

Full cusp coupling and physical kernels of the classical holonomy family

Reviewed edition, 9 October 2026: complete PK1–PK227 proofs, including PK109a, PK141a, PK143a and PK177a–PK177f, and 564 exact checks. Original rooted-face weights and actual fourth-moment bounds are evaluated. The full reflected phase gives an actual large-amplitude raw mass of $1/4$, with a different iterated weak-coupling limit and proved high-energy escape. The actual two-phase space, its complete energy form, residual interaction and convergent finite variational matrices are constructed. The actual phase density, almost-everywhere full rank, positive finite Gram matrices and complete distributional boundary identity are proved. The actual vacuum leaves an extensive electric cutoff on the original path; its full energy has a proved lower bound. Complete finite approximations to its phase Gram, energy, second-moment and residual matrices now have explicit actual-vacuum errors, including a simultaneous-path cutoff. The original-path low-energy calculation and the interacting continuum endpoint remain active research.

30 September 2026. This note constructs the full angular cusp pullback of the retained bundle and its actual classical solution family. It then maps the actual link configurations to nonzero gauge-invariant quantum vectors on the original finite lattice, computes their Gram and full Hamiltonian kernels, and constructs a nonzero vector orthogonal to the actual vacuum. Sections 8–9 fix a joint path and compute its first electric projection and the exact interacting return through the complementary space. No continuum spectral conclusion is claimed here.

The coordinate drawing is a schematic projection of the original core and its reflection at $s = 0$. It retains both cubes and their complete face count. The operator diagram is the exact identity PK54 on the reflection-odd physical space. The full hypotheses and proofs are PK38–PK54; the complete classical energy bracket is PK43f. Reproducible source: `period_physical_figure.py`.

1. Inputs, coordinates and source conventions

The original objects are the bundle $P_H \rightarrow X$, its rank-24 associated bundle $\mathcal{W}_H$, the map $\mathcal{M}_H$, and its global classical Cauchy solution $A(v; s, x)$, with $s = ct$. Complete proofs are in the Cauchy continuation, CE1–CE45 and the higher-carrier continuation, HC1–HC12. The complete period input and the original lattice Hamiltonian are retained in Spatial continuum, Sections 1–2. Its radial formulas are now extended through the entire angular coordinate.

To avoid confusing the physical support radius R with cusp depth, write the original cusp depth as T . Keep

$$t_c = e^{-2\pi T + i\phi}, \quad \sigma = \frac{\phi}{2\pi} + iT, \quad \mu = \mu(t_c), \quad \tau = \sigma + h(t_c), \quad \beta = b_{\text{per}}(t_c) - \sigma - h(t_c). \quad (\text{PK1})$$

Here h, μ, b_{per} are the original holomorphic functions, including every bounded correction. Define

$$a = \text{Re } \mu, \quad m = \text{Im } \mu, \quad u = \text{Re } \tau, \quad L = \text{Im } \tau, \quad d = \text{Im } b_{\text{per}}, \quad q = L - d, \quad c_0 = \text{Re } \beta, \\ P = \begin{pmatrix} 6a & u & 1 & 0 \\ 6m & L & 0 & 0 \\ c_0 & a & 0 & 1 \\ -q & m & 0 & 0 \end{pmatrix}, \quad D = Lq + 6m^2 > 0, \quad A_{\text{per}} = \begin{pmatrix} 6a & u \\ c_0 & a \end{pmatrix}, \quad B = \begin{pmatrix} 6m & L \\ -q & m \end{pmatrix}. \quad (\text{PK2})$$

The source's p is exactly $-q$; its δ is exactly D . This dictionary preserves the original real coordinate order $(\text{Re } z_1, \text{Im } z_1, \text{Re } z_2, \text{Im } z_2)$.

The finite-state construction is the real-label part of the heat-kernel construction of Brian C. Hall, arXiv:1707.02355v1, Sections 1.2–1.3, and its gauge average is the construction in Benjamin Bahr and

Actual fields, physical states, and the complete interaction

Exact formulas from PK13–PK54 • diagrams are schematics, not numerical samples

1. The geometric parameter reaches a physical vector

$$\begin{array}{ccc} \mathcal{W}_M = c_M^* \mathcal{W}_H & \longrightarrow & A(v; s, x) \\ [p, v] & & h_e = H_e(A)^{-1} \end{array} \longrightarrow \begin{array}{l} \Phi_{h,t} = \prod_e p_t(U_e h_e^{-1}) \\ \eta_{h,t} = \Phi_{h,t} - \Phi_{Rh,t} \end{array}$$

Bundle equivariance gives simultaneous colour conjugation; the vertex Haar projection removes it.

$\langle \psi, \eta_{h,t} \rangle = 0$ by reflection parity; $\|\eta_{h,t}\|^2 = \Gamma(h) > 0$ by an actual core face.

2. Keep both reflected core cubes and every face coefficient



$$m = \lfloor \lambda r / (2\sqrt{3}a) \rfloor, \quad M_m = 3(2m)^2(2m+1), \quad 2aK - am > \lambda R$$

$$\Gamma = e^{-6t} \sum_p |W_p(h) - W_p(Rh)|^2 + \|\eta - P_F \eta\|^2 \geq 32M_m e^{-6t} \sin^8(a/\lambda)$$

3. The first electric layer and its full interacting return

$$P = P_F|_{H_-}, \quad Q = I - P, \quad M = M_L, \quad C = Q\mathcal{W}P, \quad \mathcal{W} = \sum_p W_p$$

$$PH_0P = 3P, \quad QH_0Q \geq (9/2)Q, \quad C^*C = (M - 1)P$$

$$H - E_0 : P \rightarrow Q \quad (-bC), \quad P(H - E_0)P = dP, \quad Q(H - E_0)Q = B_- \geq 0$$

$$P(H - E_0 + s)^{-1}P = [(d + s)P - b^2C^*(B_- + s)^{-1}C]^{-1}, \quad s > 0$$

$$d = 2bM + 3\kappa - E_0 \geq (3\kappa + \sqrt{9\kappa^2 + 4b^2M})/2$$

Same-coupling path (PK43):

$$T_j = j^6, \quad M_{\text{cov},j} = j^4, \quad a_j = (100j)^{-1}, \quad L_j = j^4, \quad g_j^2 = \kappa_*/(200j), \quad \lambda_j = j^2$$

$$\mathcal{E}_{g_j}^{[\lambda_j]} = (400/(\kappa_*j)) [\text{full CE22 energy bracket}], \quad b_j = 10000j^2/\kappa_*$$

Proof: PERIOD_COUPLING_AND_PHYSICAL_KERNELS.md, PK13–PK54. The continuum spectral limit remains open.

Figure 1: Actual fields, reflected core cubes and the complete interacting return

Thomas Thiemann, arXiv:0709.4636v1, Sections 2, 3 and 4.2. The exact convention comparison is supplied in Section 5. Complex labels, their additional identifications, and semiclassical asymptotics from those papers are not imported. All arguments needed for the results here follow below. Original author TeX, reading coverage and exact hashes are recorded in SOURCE_PROVENANCE.md and the private reading ledger.

2. Angular monodromy, determinant and the complete covered metric

Put

$$N_0 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix}, \quad M_0 = I + N_0, \quad S_M = \text{diag}(1, 1, M, M), \quad M \in \mathbb{N}. \quad (\text{PK3})$$

Directly from (PK1), continuation of the logarithm by $\phi \mapsto \phi + 2\pi$ changes τ to $\tau + 1$ and β to $\beta - 1$, while leaving all three holomorphic functions unchanged. Consequently

$$P(T, \phi + 2\pi) = P(T, \phi)M_0, \quad N_0^2 = 0, \quad M_0^M = I + MN_0, \quad S_M^{-1}M_0^M S_M = M_0. \quad (\text{PK4})$$

These equations fix the orientation by the displayed logarithm change. The earlier source calls its printed matrix clockwise and later refers to a positive loop. No conclusion here relies on that verbal convention: reversing the displayed logarithm change replaces M_0 by its inverse.

On the covering base use $v_c = e^{-2\pi T/M + i\theta}$, so $t_c = v_c^M$ and $\phi = M\theta$. The covered marked torus is

$$\tilde{F}_{T,\theta,M} = \mathbb{R}^4 / P(T, M\theta)S_M\mathbb{Z}^4. \quad (\text{PK5})$$

Its marked presentation identifies

$$(T, \theta, y) \sim (T, \theta + 2\pi, M_0^{-1}y), \quad y \in \mathbb{R}^4 / \mathbb{Z}^4. \quad (\text{PK6})$$

Indeed $P(T, M\theta + 2\pi M)S_MM_0^{-1}y = P(T, M\theta)S_My$. This proves the gluing, not just invariance of a determinant. The map $[PS_My] \mapsto [PS_My]$ to the original fibre has degree M^2 and deck group $(\mathbb{Z}/M\mathbb{Z})^2$, since the lattice quotient is $\mathbb{Z}^4/S_M\mathbb{Z}^4$.

Reorder covering rows only as $(x^1, x^3; x^2, x^4)$. The row permutation has sign -1 . In this order

$$Q = \begin{pmatrix} A_{\text{per}} & MI_2 \\ B & 0 \end{pmatrix}, \quad Q^{-1} = \begin{pmatrix} 0 & B^{-1} \\ M^{-1}I_2 & -M^{-1}A_{\text{per}}B^{-1} \end{pmatrix}, \quad B^{-1} = D^{-1} \begin{pmatrix} m & -L \\ q & 6m \end{pmatrix}. \quad (\text{PK7})$$

Multiplying both ways proves the inverse. Thus in the original order

$$\det(PS_M) = -M^2D, \quad G_M = S_M^\top P^\top PS_M, \quad \sqrt{\det G_M} = M^2D, \quad (\text{PK8})$$

with inverse metric, in the unchanged marked-column order,

$$G_M^{-1} = Q^{-1}Q^{-\top} = \begin{pmatrix} B^{-1}B^{-\top} & -M^{-1}B^{-1}B^{-\top}A_{\text{per}}^\top \\ -M^{-1}A_{\text{per}}B^{-1}B^{-\top} & M^{-2}(I_2 + A_{\text{per}}B^{-1}B^{-\top}A_{\text{per}}^\top) \end{pmatrix}. \quad (\text{PK9})$$

Every angular real-part term survives in this formula. Re-marking by (PK6) gives the congruence law for G_M ; it does not identify the matrix entries in two different markings.

Here are uniform bounds with explicit roles for the constants. Fix a sufficiently small closed cusp disk on which the three holomorphic functions are bounded, and then $T \geq T_0$. There are finite constants

C_B, C_A , defined respectively as bounds on $\|B - TJ\|$ and $\|A_{\text{per}} - (\phi/(2\pi))J\|$, where $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. Enlarge T_0 so that $T_0 > C_B$; the preceding fixed disk bounds remain valid on the smaller end. On $0 \leq \theta \leq 2\pi$,

$$T - C_B \leq \sigma_{\min}(B) \leq \sigma_{\max}(B) \leq T + C_B, \quad \|A_{\text{per}}\| \leq M + C_A. \quad (\text{PK10})$$

The first inequalities follow from $|TJz| = T|z|$ and the triangle inequality. They imply

$$\begin{aligned} \sigma_{\min}(PS_M) &\geq \left[M^{-1} + \frac{1 + \|A_{\text{per}}\|/M}{T - C_B} \right]^{-1}, \\ \sigma_{\max}(PS_M) &\leq \|A_{\text{per}}\| + M + T + C_B, \\ \text{sys}(\tilde{F}_{T,\theta,M}) &\geq \min(T - C_B, M). \end{aligned} \quad (\text{PK11})$$

For the first use (PK7) and bound its three nonzero blocks separately; $\sigma_{\min} = \|Q^{-1}\|^{-1}$. The second uses the three blocks of Q . For the third, a period has the exact form $(A_{\text{per}}n + Mz, Bn)$, $n, z \in \mathbb{Z}^2$. If $n \neq 0$, its second block has length at least $T - C_B$; otherwise its first block has length at least M . No real-part correction is discarded.

Holomorphy also gives, uniformly in the original angular variable,

$$D = T^2 + (2h_0 - d_0)T + (h_0^2 - d_0h_0 + 6m_0^2) + O(Te^{-2\pi T}), \quad (\text{PK12})$$

where $h_0 = \text{Im } h(0)$, $d_0 = \text{Im } b_{\text{per}}(0)$, $m_0 = \text{Im } \mu(0)$. Substitution of $L = T + \text{Im } h(t_c)$, $q = L - \text{Im } b_{\text{per}}(t_c)$ in (PK2) proves every term.

3. Exact bundle pullback and low modes on the full end

Let $\mathcal{C} \subset X$ be the original regular cusp $0 < |t_c| < \epsilon$. Let $\mathcal{C}_M$ be its covered family (PK5)–(PK6), and $c_M : \mathcal{C}_M \rightarrow \mathcal{C} \hookrightarrow X$ the actual cover map, including the base change. Define

$$P_M = c_M^* P_H, \quad \mathcal{W}_M = c_M^* \mathcal{W}_H, \quad \mathcal{A}_M = c_M^* \mathcal{A}_H, \quad r_M = r_H c_M : \mathcal{C}_M \rightarrow F_4/\rho(\text{Sp}(1)). \quad (\text{PK13})$$

The bundle isomorphism in HC4 pulls back to $P_M \cong r_M^*(F_4 \rightarrow F_4/\rho(\text{Sp}(1)))$. Explicitly $(z, p) \mapsto (z, a(p))$, with $a(p)$ from HC5, is the map, with inverse supplied by that same fibrewise bijection. This proves the relation to the higher carrier on the whole covered end.

The original Hopf coordinate, on its open cell, remains

$$z_H = -\cot(\pi r_c/\epsilon) - \cot(\phi/2)\mathbf{i} - \cot(\pi u_1)\mathbf{j} - \cot(\pi u_2)\mathbf{k}, \quad r_c = e^{-2\pi T}, \quad \phi = M\theta, \quad u_1 = y_1, \quad u_2 = y_2. \quad (\text{PK14})$$

The last two equalities use the first two entries of S_{My} . Those entries are fixed by M_0 . Hence (PK14), its based extension, and the exact transition $g(z_H) = \bar{z}_H/|z_H|$ respect (PK6). The entire angle is retained; it is not identified with a half-angle rotation in the structure group. The nonzero global clutching class over X remains the one proved by ‘rettri:bundleclass’ in the retained source.

There is a useful strengthening about the end itself. Restrict to $r_c < \epsilon/2$. On the open cell

$$|z_H| \geq \cot(\pi r_c/\epsilon) > 0, \quad w_H = z_H^{-1} = \bar{z}_H/|z_H|^2. \quad (\text{PK15})$$

At every collapsed angular or torus seam set $w_H = 0$. This is continuous: approaching a seam makes at least one squared cotangent diverge, hence $|w_H| = |z_H|^{-1} \rightarrow 0$. It also tends uniformly to zero as $r_c \rightarrow 0$, by (PK15). Therefore the literal Hopf section

$$s_\infty(w_H) = \frac{(w_H, 1)}{\sqrt{1 + |w_H|^2}} \quad (\text{PK16})$$

is a global continuous section of P_M on this end. Its pullback transition to $s_0(z_H)$ is precisely the original $g(z_H)$. Thus the restriction to this noncompact end is trivial, even though the bundle over the whole X is nontrivial. This is proved by a section; it does not erase the global clutching class.

The smooth bundle structure constructed in HC4 can be used on the end. A smooth trivialization exists as well: view (PK16) as a continuous unit section of its associated quaternionic line; a locally finite smooth partition of unity and local approximation give a smooth section within fibre norm $1/2$ of it. Dividing that section by its displayed positive norm gives a smooth unit section. The straight interpolation with (PK16), divided by its norm, is a continuous homotopy of nowhere-zero sections. It is an isomorphism of the same bundle, not a differentiability assertion about the literal collapsed source coordinates. Formula (PK14) remains the topological comparison map.

Pullback of CE34 now gives, without a newly assumed evolution theorem,

$$\mathfrak{E}_M : \mathcal{W}_M \longrightarrow P_M \times_{\mathrm{Sp}(1)} \mathcal{S}, \quad [p, v] \longmapsto [p, A(v; s, x)]. \quad (\text{PK17})$$

Its equivariance is CE33; both pullback compositions agree pointwise. The zero subbundle and vertical ranks remain 12 and 0, 3, 6, 9. The same statement applies to the dilation family of CE38–CE45 for every $\lambda > 0$, including its support $|x| \leq \lambda R + |s|$. Smoothness is in the retained smooth bundle coordinates. It is not an unproved uniform derivative estimate at the completed cusp.

The full dual spectrum also descends. Write $k = (n, z) \in \mathbb{Z}^2 \oplus \mathbb{Z}^2$. Equations (PK7)–(PK9) give

$$\Lambda_{T, \theta, M; k} = 4\pi^2 \left[M^{-2} |z|^2 + |B^{-\top}(n - M^{-1} A_{\text{per}}^\top z)|^2 \right]. \quad (\text{PK18})$$

The corresponding character is $\exp(2\pi i \langle P^{-\top} S_M^{-1} k, x \rangle)$. The transformation $y \mapsto M_0^{-1} y$ transports its label to $k \mapsto M_0^\top k$; inserting this in the transformed metric proves spectral invariance. The original fibre characters are exactly the covered labels with k_3, k_4 divisible by M .

For completeness, an explicit full low-mode count follows from the singular-value estimate. Fix $E > 0$, set $\rho_E = \sqrt{E}/(2\pi)$, $s_* = \sigma_{\min}(PS_M)$, and suppose $\rho_E s_* > 1$. Let $N(E)$ count the nonzero labels with eigenvalue at most E . Then

$$(1 - (\rho_E s_*)^{-1})^4 \frac{E^2 M^2 D}{32\pi^2} \leq N(E) + 1 \leq (1 + (\rho_E s_*)^{-1})^4 \frac{E^2 M^2 D}{32\pi^2}. \quad (\text{PK19})$$

Indeed the ellipsoid of allowed real labels is $(PS_M)^\top \overline{B}_{\rho_E}$, of volume $M^2 D \pi^2 \rho_E^4 / 2$. Each point in a unit lattice cube is within distance one of its centre in four dimensions. The ellipsoid contains $B_{\rho_E s_*}$. Its outer unit neighbourhood is contained in its $1 + 1/(\rho_E s_*)$ dilate; its inner such dilate lies in the union of cubes whose centres it contains. Comparing volumes proves (PK19). Boundary cubes can be made half-open, so they do not change the count.

For the joint path $T = j^6$, $M = j^4$ prescribed in Section 8, (PK10)–(PK11) give $s_* \geq j^4/2$ at the explicit threshold there. Consequently (PK19) retains the full factor $E^2 j^8 D / (32\pi^2)$, with multiplicative bounds $(1 - 2/(\rho_E j^4))^4$ and $(1 + 2/(\rho_E j^4))^4$ when $\rho_E j^4 > 2$. In particular no bounded correction inside D is discarded. Also

$$\frac{4\pi^2}{(T + C_B + M + \|A_{\text{per}}\|)^2} \leq \Lambda_1 \leq \frac{4\pi^2}{(T - C_B)^2}. \quad (\text{PK20})$$

The lower bound uses $|k| \geq 1$; for the upper bound use a unit label $(n, 0)$ and (PK10). These are scalar geometric modes. The quantum Hamiltonian below is a different operator whose connection to these scales must be tested using its actual kernels.

4. Receiving the actual fields in the original lattice

The physical coordinate permutation from the source is

$$(y^0, y^1, y^2, y^3) = (x^3, x^2, x^4, x^1). \quad (\text{PK21})$$

It has determinant +1. Denote its matrix by $\mathbf{S}$. On any ball of radius b_* with $4b_* < \min(T - C_B, M)$, the quotient to (PK5) is an isometric embedding. The proof is $|x - x' + \ell| \geq |\ell| - 2b_* > 2b_*$ for every nonzero period. A physical path γ in this patch has marked coordinates

$$\eta = (PS_M)^{-1} \mathbf{S}^{-1} \gamma. \quad (\text{PK22})$$

The pulled-back one-form has components $\tilde{A}_a = \sum_\mu (SPS_M)_{\mu a} A_\mu (SPS_M \eta)$; its curvature is the exterior-square pullback. The chain rule retains each derivative and bracket. Along (PK22), the period matrix and its inverse cancel in the transport ODE, proving equality of its holonomy with the actual physical path holonomy. This cancellation is a proved map with all matrices present, not deletion of the period data. Equations (PK4)–(PK6) prove angular compatibility.

This construction uses physical patches of the Euclidean torus carrier. The physical evolution remains the Lorentzian Cauchy solution on $\mathbb{R}^{1+3}$. It does not identify physical time periodically or assert that the solution closes around a torus time cycle. Its entire curvature is pulled back on every patch where it is used.

Fix a positive integer $L \geq 1$ and the original open cubic graph with vertices $\{-L, \dots, L\}^3$, spacing $a > 0$, and a specified physical origin o , so an edge is $\gamma_{n,i}(t) = o + a(n + te_i)$. Let $H_e(A)$ be CE42's forward transport. Define the lattice centre

$$h_e = H_e(A)^{-1}. \quad (\text{PK23})$$

CE43 gives $h_e(A^q) = q_{s(e)} h_e q_{t(e)}^{-1}$, exactly the original lattice action. This inverse is essential. All reversed edges are the inverses of the same variables; they are not independent variables.

Put

$$N_L = 3(2L)(2L+1)^2, \quad M_L = 3(2L)^2(2L+1), \quad Q_L = SU(2)^{N_L}, \quad dU = \prod_e dU_e, \quad (\text{PK24})$$

where each Haar measure has total mass one. For $T_a = -i\sigma_a/2$, let $X_{e,a}$ differentiate $U_e \mapsto e^{tT_a} U_e$. The complete Hamiltonian is

$$H = \kappa \sum_e E_e + b \sum_p (2 - W_p), \quad E_e = - \sum_a X_{e,a}^2, \quad \kappa = \frac{2g^2}{a}, \quad b = \frac{1}{2g^2 a}, \quad W_p = \text{tr}(U_i(n) U_j(n+e_i) U_i(n+e_j)^{-1} U_j(n)^{-1}). \quad (\text{PK25})$$

Its domains are $H^2(Q_L)$ and form domain $H^1(Q_L)$, because the potential is smooth and $0 \leq V \leq 4bM_L$. The physical Hilbert space is the range of the orthogonal Haar projection

$$(\Pi f)(U) = \int_{SU(2)^{V_L}} f(q \cdot U) dq, \quad (q \cdot U)_e = q_s U_e q_t^{-1}. \quad (\text{PK26})$$

Haar invariance proves self-adjointness and idempotence; the image is exactly the invariant functions. Each E_e commutes with the action, and every W_p is conjugated at its base vertex, so $H\Pi = \Pi H$. The source Lie metric is $-\text{tr}(XY)/2$, for which $\Delta^c = 4 \sum_a X_a^2$. It is one quarter of CE4's matrix metric. Thus (PK25) retains the original electric coefficient; none is transferred silently into a metric or a coupling.

5. Nonzero physical vectors and their exact Gram kernel

For $t > 0$ define the heat kernel in the original T_a convention

$$p_t(U) = \sum_{j \in \frac{1}{2}\mathbb{Z}_{\geq 0}} (2j+1)e^{-tj(j+1)}\chi_j(U). \quad (\text{PK27})$$

Every derivative series converges uniformly: derivatives of a spin- j matrix are bounded by powers of j , whereas the exponential is quadratic in j . Orthogonality of matrix coefficients proves $p_t * p_s = p_{t+s}$, $\int p_t = 1$, and $\partial_t p_t = -E p_t$. The compact connected-group heat equation and the strong maximum principle give $p_t > 0$. In particular all packets below are smooth vectors in every Sobolev domain.

The exact source comparisons are as follows. Bahr–Thiemann’s real-label packet with parameter $2t$ is $p_t(Uh^{-1})$, using trace cyclicity and $\chi_j(k^{-1}) = \chi_j(k)$ on $SU(2)$. Hall uses $\partial_t \rho = \Delta^c \rho / 2 = -2E\rho$ with his displayed $SU(2)$ metric $-\text{tr}(XY)/2$; hence $\rho_{t/2} = p_t$. These factors refer to smoothing parameters, not to the coupling or Planck’s constant of (PK25).

For positive edge parameters $\mathbf{t} = (t_e)$, set

$$f_{h,\mathbf{t}}(U) = \prod_e p_{t_e}(U_e h_e^{-1}), \quad \Phi_{h,\mathbf{t}} = \Pi f_{h,\mathbf{t}} = \int dq \prod_e p_{t_e}(U_e(q \cdot h)_e^{-1}). \quad (\text{PK28})$$

The second equality follows by substituting inverse vertex variables and using conjugation invariance of p_t . It proves that Φ depends only on the gauge orbit of h . Positivity and unit integral give

$$\Phi_{h,\mathbf{t}} > 0, \quad \int \Phi_{h,\mathbf{t}} dU = 1, \quad \|\Phi_{h,\mathbf{t}}\|_2^2 \geq 1. \quad (\text{PK29})$$

Thus physical projection does not annihilate any of these actual classical configurations. In particular simultaneous colour conjugation from a change of frame in P_M disappears under Π ; (PK28) therefore gives a well-defined map on the entire associated bundle. The equal-field circles and all spectator directions of CE34a retain exactly equal vectors. No parameter labels have been made orthogonal.

The exact raw Gram kernel is

$$G(h, k; \mathbf{t}, \mathbf{s}) = \langle \Phi_{h,\mathbf{t}}, \Phi_{k,\mathbf{s}} \rangle = \int dq \prod_e p_{t_e+s_e}(h_e(q \cdot k)_e^{-1}). \quad (\text{PK30})$$

To prove it, use $\Pi^* = \Pi^2 = \Pi$, integrate the two heat kernels on each original link, and apply their convolution law. The same result comes from the two vertex integrals in (PK28), followed by their relative vertex change. It is real, strictly positive, symmetric under exchanging the two labelled packets, and positive semidefinite as a Gram kernel.

There are no further identifications for real labels at a fixed $\mathbf{t}$, apart from gauge orbits. Indeed (PK28) is the product heat operator applied to the orbit probability measure ν_h . Every Peter–Weyl coefficient is multiplied by the nonzero number $\exp(-\sum_e t_e j_e(j_e + 1))$. Equality of packets gives equality of all Fourier coefficients of the two measures. Matrix coefficients form a self-adjoint algebra separating points, so their uniform density gives equality of the measures. The support of ν_h is exactly the compact orbit of h : every nonempty open set meeting the orbit has positive Haar preimage. Equality of the measures therefore gives equality of their orbits. This argument also proves that distinct-orbit packets cannot be proportional, since their integrals are both one. More strongly, a finite collection of packets at distinct gauge orbits and fixed heat parameters is linearly independent. A vanishing linear combination, after the same injective heat operation, gives a vanishing linear combination of orbit measures. Their compact supports are disjoint. Restricting the measure to each of those Borel supports gives its coefficient zero. Thus the Gram matrix is strictly positive definite on each such finite collection.

6. Full Hamiltonian kernel, including every magnetic face

Write $x_e = (q \cdot h)_e$, $y_e = (r \cdot k)_e$. Define one-link scalar and matrix functions

$$Z_e(x, y) = p_{t_e + s_e}(xy^{-1}), \quad R_e^\pm(x, y) = \int_{SU(2)} p_{t_e}(Ux^{-1}) p_{s_e}(Uy^{-1}) U^{\pm 1} dU. \quad (\text{PK31})$$

For each oriented face with ordered distinct edges $(e_1^{\epsilon_1}, \dots, e_4^{\epsilon_4})$, put

$$B_p(h, k) = \int dq dr \left(\prod_{e \notin \partial p} Z_e(x_e, y_e) \right) \text{tr} \left(R_{e_1}^{\epsilon_1}(x_{e_1}, y_{e_1}) R_{e_2}^{\epsilon_2}(x_{e_2}, y_{e_2}) R_{e_3}^{\epsilon_3}(x_{e_3}, y_{e_3}) R_{e_4}^{\epsilon_4}(x_{e_4}, y_{e_4}) \right). \quad (\text{PK32})$$

The order in this trace is the original face order. Link integration in $\langle \Phi_h, W_p \Phi_k \rangle$ proves precisely (PK32); no commuting replacement of the four matrices is made. Differentiating the right packet in (PK28) proves the full answer

$$K(h, k; \mathbf{t}, \mathbf{s}) := \langle \Phi_{h, \mathbf{t}}, H \Phi_{k, \mathbf{s}} \rangle = -\kappa \sum_e \partial_{s_e} G + 2b M_L G - b \sum_p B_p. \quad (\text{PK33})$$

Every original boundary face, scalar Wilson term and electric coefficient is retained. All differentiations are justified by the uniformly convergent series following (PK27).

The integrals in these formulas have a completely specified convergent algebraic evaluation. Here is one that also fixes their representation conventions. Let $D^j = \text{Sym}^{2j}(U)$ in the orthonormal symmetric-tensor basis, with indices $0 \leq r, s \leq n = 2j$. For $U = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$,

$$D^j(U)_{rs} = \sqrt{\frac{\binom{n}{s}}{\binom{n}{r}}} \sum_{k=\max(0, r-n+s)}^{\min(r, s)} \binom{n-s}{r-k} \binom{s}{k} a^{n-s-r+k} c^{r-k} b^{s-k} d^k. \quad (\text{PK34})$$

This follows by expanding the two factors $(az_1 + cz_2)^{n-s} (bz_1 + dz_2)^s$, with the displayed symmetric basis factors. Thus it is an explicit finite matrix, including its signs.

For a list of these representations or their complex conjugates, let T_a^{tot} be the sum of their infinitesimal matrices in the corresponding tensor slots, and $C = -\sum_a (T_a^{\text{tot}})^2$. Let $J_{\max}$ be the sum of their spins. Their Haar matrix is

$$Q^{\text{list}} = \int \bigotimes_{\ell} D^{j_\ell}(U) dU = \begin{cases} 0, & J_{\max} \in \mathbb{Z} + \frac{1}{2}, \\ \prod_{J=1}^{J_{\max}} \left(I - \frac{C}{J(J+1)} \right), & J_{\max} \in \mathbb{Z}. \end{cases} \quad (\text{PK35})$$

Conjugated slots use the conjugated matrices in this equation; the empty product is the identity. To prove it, Haar averaging is the orthogonal projection onto invariant tensors. The finite-dimensional unitary $SU(2)$ representation decomposes orthogonally by its highest weights: starting with a largest eigenvector of iT_3^{tot} , repeated lowering gives the spin- J string with Casimir $J(J+1)$; its orthogonal complement is invariant, so induction exhausts the tensor space. Its weights have the parity of $J_{\max}$ and absolute value at most $J_{\max}$. Thus the polynomial in (PK35) is one on the spin-zero space and zero on every possible nonzero spin. For half-integral parity there is no zero spin. This proves the formula, even when some of the listed integer spins do not occur.

In particular, the entries of (PK31) are exactly

$$\begin{aligned} (R^+)_{ab}(x, y) &= \sum_{j, k} d_j d_k e^{-tj(j+1) - sk(k+1)} \sum_{m, n, p, q} D^j(x^{-1})_{nm} D^k(y^{-1})_{qp} Q_{(m, p, a), (n, q, b)}^{j, k, 1/2}, \\ (R^-)_{ab}(x, y) &= \sum_{j, k} d_j d_k e^{-tj(j+1) - sk(k+1)} \sum_{m, n, p, q} D^j(x^{-1})_{nm} D^k(y^{-1})_{qp} Q_{(m, p, b), (n, q, a)}^{j, k, 1/2}, \end{aligned} \quad d_j = 2j + 1. \quad (\text{PK36})$$

Expanding each heat trace proves the first equation. The second uses $(U^{-1})_{ab} = \overline{U}_{ba}$, hence the transposed indices shown. The remaining vertex averages in (PK30) and (PK32) are evaluated by the same (PK35) at each vertex, with the exact incoming conjugated slots. The polynomial growth in dimensions and the heat exponential give absolute convergence of these sums and all finite derivatives. Thus (PK30)–(PK36) specify actual computable kernels, including the entire nonlinear magnetic interaction, rather than a diagonal assignment to labels.

7. A nonzero vector exactly orthogonal to the actual vacuum

We now construct an excitation, not merely a positive physical packet. The original H has compact resolvent. A ground minimizer can be chosen nonnegative because $|\nabla|f|| \leq |\nabla f|$; the functions $(|f|^2 + \varepsilon^2)^{1/2}$ justify this at zeros. Elliptic regularity and the strong maximum principle give a smooth strictly positive minimizer ψ , with eigenvalue E_0 and $\int \psi^2 = 1$. The exact identity

$$\langle \psi F, (H - E_0) \psi F \rangle = \kappa \sum_{e,a} \int \psi^2 |X_{e,a} F|^2 dU \quad (\text{PK37})$$

follows by the product rule and integration by parts. Another ground vector divided by ψ has all derivatives zero, so it is constant on connected Q_L . This proves uniqueness. Gauge actions and graph reflections preserve H , positivity and norm, so they fix ψ . The bounds $0 \leq E_0 \leq 2bM_L$ follow from positivity and the unit constant trial vector, whose mean Wilson trace is zero on every face.

Let $\mathcal{R}$ be the graph reflection $n_1 \mapsto -n_1$. For reversed edges it also inverts the link; the other edges are permuted. It is an involution preserving product Haar measure, $\sum E_e$, and the sum of all face traces. With a common heat parameter $t > 0$, it sends $\Phi_{h,t}$ to $\Phi_{\mathcal{R}h,t}$. Define

$$\eta_{h,t} = \Phi_{h,t} - \Phi_{\mathcal{R}h,t}. \quad (\text{PK38})$$

It is gauge invariant and reflection odd, so $\langle \psi, \eta_{h,t} \rangle = 0$ exactly. Its complete kernels are

$$\begin{aligned} G^-(h, k) &= G(h, k) - G(h, \mathcal{R}k) - G(\mathcal{R}h, k) + G(\mathcal{R}h, \mathcal{R}k), \\ K^-(h, k) &= K(h, k) - K(h, \mathcal{R}k) - K(\mathcal{R}h, k) + K(\mathcal{R}h, \mathcal{R}k), \\ \Gamma(h) &= G^-(h, h) = 2[G(h, h) - G(h, \mathcal{R}h)]. \end{aligned} \quad (\text{PK39})$$

No norm division or vacuum overlap has been omitted.

Here is an actual field and graph for which $\Gamma > 0$. Use the equal-colour initial input, dilated by $\lambda > 0$, at $s = 0$. Choose a with $\sqrt{2}a < \lambda r$ and $0 < a/\lambda < \pi$. Let K be an integer with $aK > 2\lambda R$, choose $L > K + 2$, and set the physical origin $o = (aK, 0, 0)$. The face starting at $n = (-K, 0, 0)$ lies in the original core. There $\varphi(A_i) = 2T_i/\lambda$, so (PK23) gives its exact face word

$$h_p = e^{2aT_1/\lambda} e^{2aT_2/\lambda} e^{-2aT_1/\lambda} e^{-2aT_2/\lambda}, \quad W_p(h) = 2 - 4\sin^4(a/\lambda) < 2. \quad (\text{PK40})$$

To verify the trace, insert $e^{2aT_i/\lambda} = \cos(a/\lambda)I + 2T_i \sin(a/\lambda)$, multiply the four matrices using $T_i T_j = -\delta_{ij}I/4 + \epsilon_{ijk}T_k/2$, and take the trace. Every point of the reflected face has first coordinate at least $2aK - a > \lambda R$ after enlarging K , if necessary, by one. There $C = 0$, hence all four links are the identity and $W_p(\mathcal{R}h) = 2$. Thus the two configurations are not gauge equivalent. The real-label injectivity proved after (PK30) gives $\eta_{h,t} \neq 0$. This proves a nonzero, smooth, finite-energy physical vector orthogonal to the actual interacting finite-lattice vacuum.

There is also a quantitative raw-norm lower bound. A face Wilson trace is an electric eigenfunction with eigenvalue $4 \cdot (3/4) = 3$; each of its four links occurs in the fundamental representation. Equations

(PK27)–(PK28) therefore give

$$\int W_p(U) \Phi_{h,t}(U) dU = e^{-3t} W_p(h).$$

Its Haar squared norm is one: integrating three links reduces its holonomy to a single Haar element, and character orthogonality gives $\int \chi_{1/2}^2 = 1$. Cauchy–Schwarz applied to (PK38) proves

$$\Gamma(h) \geq e^{-6t} |W_p(h) - W_p(\mathcal{R}h)|^2 = 16e^{-6t} \sin^8(a/\lambda) > 0. \quad (\text{PK41})$$

This bound retains the decreasing amplitude; it is not a uniform positive lower bound in a limit with $a/\lambda \rightarrow 0$.

For an explicit finite energy control put $D_h = -\sum_e \partial_{s_e} G(h, h; \mathbf{t}, \mathbf{s})|_{\mathbf{s}=\mathbf{t}}$. This is $\sum_{e,a} \|X_{e,a} \Phi_{h,t}\|_2^2 \geq 0$. Reflection invariance and $|u - v|^2 \leq 2|u|^2 + 2|v|^2$ give

$$0 < \frac{K^-(h, h) - E_0 \Gamma(h)}{\Gamma(h)} \leq \frac{4\kappa D_h}{\Gamma(h)} + 4bM_L. \quad (\text{PK42})$$

The strict inequality follows from uniqueness of the ground state and $\eta \neq 0$; all vectors are in the operator domain. The upper bound is finite and explicit in the kernels. It is not claimed to tend to zero.

8. One fixed joint path and the next actual spectral calculation

Fix $\kappa_* > 0$, $t_* > 0$, the original cutoff χ, r, R , and the original physical unit $\ell_* = 1$, the length of its real period vector. Use the same g_j in the classical physical energy and the quantum Hamiltonian. Before estimating any spectral limit, prescribe

$$\begin{aligned} T_j &= j^6, \quad M_j = j^4, \quad v_{c,j} = e^{-2\pi j^2}, \quad a_j = \frac{1}{100j}, \quad L_j = j^4, \quad g_j^2 = \frac{\kappa_*}{200j}, \quad t_j = t_*, \quad \lambda_j = j^2, \\ K_j &= \left\lceil \frac{3\lambda_j R}{a_j} \right\rceil + 1, \quad o_j = (a_j K_j, 0, 0), \quad h_j = H(A^{[\lambda_j]}(0))^{-1}, \quad \eta_j = \Phi_{h_j, t_*} - \Phi_{\mathcal{R}h_j, t_*}. \end{aligned} \quad (\text{PK43})$$

The base identity remains exact: $v_{c,j}^{M_j} = e^{-2\pi j^6}$. The full graph and Hamiltonian constants are

$$\begin{aligned} N_{L_j} &= 3(2j^4)(2j^4 + 1)^2, \quad M_{L_j} = 3(2j^4)^2(2j^4 + 1), \\ \kappa_j &= \kappa_*, \quad b_j = \frac{10000j^2}{\kappa_*}, \quad \xi_j = \frac{10000j^2}{\kappa_*^2}, \quad \frac{a_j}{\lambda_j} = \frac{1}{100j^3}, \quad K_j = \lceil 300Rj^3 \rceil + 1. \end{aligned} \quad (\text{PK43a})$$

In particular every original edge and face remains in (PK25).

Here is one sufficient threshold, including the small-core condition. Increase the already fixed T_0 , if needed, so (PK10) holds and $e^{-2\pi T_0} < \epsilon/2$. Take every integer j satisfying

$$j \geq 1 + \left\lceil \max \left\{ 2, T_0^{1/6}, (2C_B)^{1/6}, \sqrt{4 + 2C_A}, \sqrt{32(3R + 1)}, 100(4R + 2), \left(\frac{\sqrt{3}}{25r} \right)^{1/3} \right\} \right\rceil. \quad (\text{PK43b})$$

The constants here are those of the original cutoff and full period matrix. Since $j^6 - C_B \geq j^6/2 \geq j^4$, (PK11) gives $\text{sys} \geq j^4$. Its inverse-matrix estimate also gives

$$\sigma_{\min}(PS_{M_j}) \geq [j^{-4} + 2j^{-6}(2 + C_A j^{-4})]^{-1} \geq \frac{j^4}{2}. \quad (\text{PK43c})$$

The last inequality follows from $j^2 \geq 4 + 2C_A$ and $j \geq 1$. All bounds hold uniformly for $0 \leq \theta \leq 2\pi$.

The spatial box has half-width $a_j L_j = j^3/100$, and

$$3Rj^2 < a_j K_j < 3Rj^2 + 2a_j.$$

Consequently it exhausts $\mathbb{R}^3$. For the entire physical time slab $|s| \leq j^2$, its image in $\mathbb{R}^4$ is contained in the ball of radius

$$B_j = \frac{\sqrt{3}j^3}{100} + (3R+1)j^2 + \frac{1}{50j} < \frac{j^4}{8}. \quad (\text{PK43d})$$

Indeed each of the three terms is at most $j^4/32$ under (PK43b): for the first and third it suffices that $j \geq 1$, while the middle term uses $j^2 \geq 32(3R+1)$. Thus the slightly larger open ball of radius $j^4/8$ embeds isometrically in the covered torus by (PK22). This leaves a collar around the entire slab.

The complete evolved support also lies inside this box on the stated slab. CE39 gives its radius at most $(R+1)j^2$, whereas

$$(R+1)j^2 + a_j K_j < (4R+1)j^2 + \frac{1}{50j} < \frac{j^3}{100}. \quad (\text{PK43e})$$

The last inequality follows from $j \geq 100(4R+2)$. This verifies the boundary condition with the actual translated origin, including the negative x^1 boundary. Moreover $L_j > K_j + 2$, since $j > 300R + 4$, and $\sqrt{2}a_j < \lambda_j r$. The reflected face has first coordinate at least $2a_j K_j - a_j > 6Rj^2 + a_j > \lambda_j R$. Hence the nonzero-vector construction in Section 7 applies at every integer in (PK43b).

The complete classical energy comparison, with the original coefficients of CE18–CE22, is

$$\begin{aligned} \mathcal{E}_{g_j}^{[\lambda_j]}(v) &= \frac{2}{g_j^2 \lambda_j} \left[\sum_{a,b} \langle c_a, c_b \rangle I_{ab} + 2 \sum_{a, b < c} \langle c_a, [c_b, c_c] \rangle J_{a,bc} + \sum_{a < b, c < d} \langle [c_a, c_b], [c_c, c_d] \rangle K_{ab,cd} \right] \\ &= \frac{400}{\kappa_* j} \left[\sum_{a,b} \langle c_a, c_b \rangle I_{ab} + 2 \sum_{a, b < c} \langle c_a, [c_b, c_c] \rangle J_{a,bc} + \sum_{a < b, c < d} \langle [c_a, c_b], [c_c, c_d] \rangle K_{ab,cd} \right] \\ &= \frac{200g_{\text{ref}}^2}{\kappa_* j} \mathcal{E}_{g_{\text{ref}}}(v) \end{aligned} \quad (\text{PK43f})$$

for any fixed reference $g_{\text{ref}} > 0$. The first equality is CE39 together with the explicit g^{-2} in CE22. Thus no coupling change has been hidden in a classical norm. This energy tends to zero for each fixed original v , and its zero inputs remain exactly the rank-12 zero subbundle. At the equal-colour input, CE23 retains the lower bound

$$\mathcal{E}_{g_j}^{[\lambda_j]} \geq \frac{6400\pi r^3}{\kappa_* j} > 0. \quad (\text{PK43g})$$

The annulus contribution remains in (PK43f); (PK43g) is only its core lower bound. The superseded choice $\lambda_j = \sqrt{j}$, with the same g_j , would instead give $200g_{\text{ref}}^2 \sqrt{j} \mathcal{E}_{g_{\text{ref}}}/\kappa_*$. It increases for every nonzero input. This is the exact defect corrected by (PK43), before any quantum spectral asymptotics are inferred.

The exact finite raw spectral measure is now a definite object:

$$\nu_j(I) = \langle \eta_j, \mathbf{1}_I(H_j - E_{0,j})\eta_j \rangle, \quad \nu_j(\{0\}) = 0, \quad \nu_j([0, \infty)) = \Gamma_j > 0. \quad (\text{PK44})$$

The finite-volume heat kernel or spectral resolution of the actual operator (PK25) computes its Laplace transform. Equivalently expand the bounded magnetic perturbation around its electric heat semigroup:

$$e^{-uH_j} = \sum_{n=0}^{\infty} (-1)^n \int_{0 < s_1 < \dots < s_n < u} e^{-(u-s_n)\kappa_* H_0} V_j e^{-(s_n-s_{n-1})\kappa_* H_0} \dots V_j e^{-s_1 \kappa_* H_0} d^n s. \quad (\text{PK45})$$

The $n = 0$ term is the electric semigroup. The operator norm of the n -th term is at most $(4b_j M_{L_j} u)^n / n!$; hence the series converges in operator norm for every fixed j, u . Its matrix elements are computed by the same heat kernels and face insertions in Section 6, retaining every face word. Multiplication by $e^{uE_{0,j}}$ gives $\int e^{-u\omega} d\nu_j(\omega)$. This specifies a convergent calculation of the full nonlinear dynamics on the constructed vectors.

The next calculation is to estimate these actual raw masses, energy moments and Laplace transforms uniformly along (PK43), including Γ_j , whose lower bounds in (PK41) and (PK49) decrease. Fixed-scale scalar low modes (PK18)–(PK20) and decreasing classical energy do not provide that estimate. In particular the present finite-state construction does not establish persistence of positive spectral mass near zero, continuum vacuum uniqueness, reconstruction, or the theory-identification map. Those remain the active programme calculations on this specified path.

9. The complete first electric layer and its interacting return

The original cubic graph supplies a finite subspace on which the following computations are exact. Write $M = M_L$ in this section only; the covering integer remains M_{cov} when both occur in one formula. Keep $\mathcal{W} = \sum_p W_p$, $H_0 = \sum_e E_e$, and the original identity

$$H = \kappa H_0 + 2bMI - b\mathcal{W}. \quad (\text{PK46})$$

Here $\mathcal{W}$ means multiplication by the complete face sum.

Theorem 9.1 (first physical electric layer). On the original physical Hilbert space, the first positive eigenvalue of H_0 is 3. Its eigenspace is exactly $\mathcal{F} = \text{span}\{W_p : p \text{ an elementary face}\}$. The displayed M functions form an orthonormal basis of $\mathcal{F}$. On the orthogonal complement of the constants and $\mathcal{F}$, the electric operator has lowest eigenvalue exactly $9/2$.

Proof. The integral of a matrix entry of U or U^{-1} against Haar measure is zero: replacing U by $-U$ reverses its sign. Hence $\int W_p = 0$. Integrating any one link in a face reduces its squared trace to $\int_{SU(2)} (\text{tr } U)^2 dU = 1$. One can see this last constant without a character convention: $SU(2)$ is the unit quaternion three-sphere, $\text{tr } \Psi(a) = 2a_0$, its Haar probability is the rotation-invariant sphere probability, and each of the four squared coordinates has mean $1/4$. If $p \neq q$, there is an edge of p absent from q ; integration in that edge gives $\langle W_p, W_q \rangle = 0$. There are exactly M original faces. For every edge of p , $E_e W_p = (3/4)W_p$, because $-\sum_a T_a^2 = 3I/4$. The same calculation holds for an inverse link. Thus $H_0 W_p = 3W_p$.

For the completeness and lower spectral bounds, decompose the product matrix coefficients into edge spins j_e . On this finite spin block, H_0 is the scalar $\sum_e j_e(j_e + 1)$. These blocks span a dense subspace: polynomials in fundamental matrix entries and their conjugates separate points on the compact product group, their algebra contains the constants, and uniform density followed by the finite-dimensional highest-weight decomposition used in (PK35) gives the claimed matrix coefficient density. Averaging at a vertex acts only on its incident representation slots. A vertex incident to exactly one nonzero spin has zero invariant part, because a nontrivial irreducible representation has no fixed vector. Therefore the support graph of any nonconstant physical spin block has minimum degree at least two.

A nonempty finite graph of minimum degree at least two contains a cycle: extend a nonbacktracking path until a vertex repeats. The cubic graph is simple and bipartite, so this cycle has at least four edges. Each nonzero spin contributes at least $3/4$, giving the lower bound 3. Equality requires exactly four spin-1/2 edges. Four such edges form a unit coordinate square; at its degree-two vertices, the invariant pairing is one-dimensional. Contracting those pairings is precisely its Wilson trace. The dimension statement also follows directly from Schur's lemma: an intertwiner between the two irreducible incident

slots is scalar when their spins agree and zero otherwise. This proves that no other vector occurs at eigenvalue 3.

There is no five-edge nonempty support graph of minimum degree at least two in the cubic graph. It would be connected, since two components would require at least eight edges. With at most four vertices a simple bipartite graph has at most four edges; with five vertices and five edges, minimum degree two forces every degree to be two, giving an odd cycle. A four-edge support is a square, whose successive degree-two invariant pairings force one common spin j ; after $j = 1/2$, its energy is at least $4 \cdot 1(1+1) = 8$. Every support with at least six edges has energy at least $6 \cdot 3/4 = 9/2$. This proves the complementary bound. It is attained: the rectangle with corners $(-1, -1, 0), (1, -1, 0), (1, 0, 0), (-1, 0, 0)$ lies in every box $L \geq 1$. Its ordered boundary uses six distinct links. The fundamental trace of that holonomy has Haar squared norm one by integrating one boundary link, and each of its six link Casimirs contributes $3/4$. It is therefore a nonzero eigenvector at $9/2$, orthogonal to the constants and the eigenvalue-3 face space by self-adjointness. The arguments extend from the dense finite spin sums to the form domain by their orthogonal spectral expansion. $\square$

Let $P_{\mathcal{F}}$ be this electric spectral projection. Applying (PK28) to each gauge-invariant electric eigenfunction gives the full projection, with every original face retained:

$$\begin{aligned} P_{\mathcal{F}}\eta_{h,t} &= e^{-3t} \sum_p (W_p(h) - W_p(\mathcal{R}h)) W_p, \\ S(h,t) &= e^{-6t} \sum_p |W_p(h) - W_p(\mathcal{R}h)|^2, \\ \Gamma(h) &= S(h,t) + \|\eta_{h,t} - P_{\mathcal{F}}\eta_{h,t}\|_2^2, \\ \langle \eta_{h,t}, H_0 \eta_{h,t} \rangle &\geq 3S(h,t) + \frac{9}{2}(\Gamma(h) - S(h,t)). \end{aligned} \tag{PK47}$$

The constant coefficient of η is zero since its two integrals equal one. This is why Theorem 9.1 applies to the remainder. These are electric spectral statements; the magnetic terms in (PK46) are still present.

Here is the resulting strengthened lower bound for the actual field. Set

$$m = \left\lfloor \frac{\lambda r}{2\sqrt{3}a} \right\rfloor, \quad M_m = 3(2m)^2(2m+1).$$

Use all faces of the coordinate cube $(-K, 0, 0) + \{-m, \dots, m\}^3$, assuming $m \geq 1$ and $L > K + m$. Every point of this cube has physical norm at most $\sqrt{3}am \leq \lambda r/2$, so every one of its M_m face traces is the value (PK40). Their reflected cube lies outside the original support: its first physical coordinate is at least $2aK - am > \lambda R$. The two face sets are disjoint. Their coefficients in (PK47) have equal magnitudes and opposite signs, which proves

$$\Gamma(h) \geq 32M_m e^{-6t} \sin^8(a/\lambda). \tag{PK48}$$

The single-face-plus-reflection case already improves (PK41)'s constant from 16 to 32. Equation (PK48) retains every face of the selected core cube; all remaining face coefficients and the orthogonal remainder remain in the exact identity (PK47).

On (PK43), $m_j = \lfloor 50rj^3/\sqrt{3} \rfloor$, and (PK43b) gives $m_j \geq 25rj^3/\sqrt{3} \geq 1$ and $L_j > K_j + m_j$. For the second assertion, $m_j \leq 50rj^3/\sqrt{3} < 30Rj^3$, whereas $K_j < 300Rj^3 + 2$ and $j \geq 100(4R+2) > 330R+2$. Concavity of sine on $[0, \pi/2]$, or its second derivative, gives $\sin x \geq 2x/\pi$ on that interval. Consequently

the completely specified lower bound along the corrected path is

$$\begin{aligned}\Gamma_j &\geq 32 [3(2m_j)^2(2m_j + 1)]e^{-6t_*} \sin^8\left(\frac{1}{100j^3}\right) \\ &\geq 32 \cdot 24 \left(\frac{25r}{\sqrt{3}}\right)^3 \left(\frac{2}{100\pi}\right)^8 e^{-6t_*} j^{-15} > 0.\end{aligned}\tag{PK49}$$

This is a lower bound for the raw norm, not an assertion that its actual asymptotic size is j^{-15} , and not a limiting quantum spectral mass.

To calculate the interaction with this electric layer, we need a small geometric fact. A nonempty collection of at most five distinct elementary faces in the cubic lattice has an edge incident to an odd number of the selected faces. Here is a proof retaining the ambient graph. For a finite collection with every edge even, define the occupation of a unit cube, modulo two, by the number of selected horizontal faces crossed by the vertical ray from its centre to positive infinity. Across a horizontal face the occupation changes exactly when that face is selected. Across a vertical face, sum the edge-even equations in the finite vertical strip between the two rays; internal terms cancel in pairs and give that same change law. Above all selected faces every occupation is zero. Below all selected faces, the vertical-face change law makes the occupations equal in every horizontal column; a column outside the finite horizontal support has occupation zero. Thus the occupations also vanish below all selected faces. Only finitely many columns can meet the selected horizontal faces, so the occupied cube set is finite. Its boundary is exactly the selected face collection. If it is nonempty, choose a cube at the maximum and minimum in each of the three coordinate directions. Its outward face is a boundary face. These six faces are distinct by their directions and supporting coordinate planes. Thus the boundary contains at least six faces, proving the assertion. The open finite box embeds in this infinite lattice, so the fact applies to its face subsets without a periodic-boundary assumption.

Replacing one link U_e by $-U_e$ changes the sign of each trace using that edge and preserves Haar measure. It follows that a product of at most five Wilson traces has zero integral whenever the collection of faces with odd multiplicity is nonempty. Thus $\int W_p W_r W_q = 0$ and every product of five traces has zero integral. For fourth moments, the only nonzero cases are pairs of equal faces or four copies of one face. Distinct squared traces have integral one: first integrate an edge unique to one of the two faces, making its squared trace equal to one, then integrate the other. A single fourth power has integral two. For this last constant, the spin decomposition $\chi_{1/2}^2 = 1 + \chi_1$ follows by separating symmetric and antisymmetric tensors of two fundamental vectors, and character orthogonality gives $\int (1 + \chi_1)^2 = 2$.

These identities determine three complete finite matrices in the original orthonormal face basis. Let $\mathbf{1}$ be the M -component column of ones. Then

$$\begin{aligned}P_{\mathcal{F}} W P_{\mathcal{F}} &= 0, \\ P_{\mathcal{F}} W^2 P_{\mathcal{F}} &= (M - 1)I_M + 2\mathbf{1}\mathbf{1}^{\top}, \\ P_{\mathcal{F}} W H_0 W P_{\mathcal{F}} &= (6M - 4)I_M + 6\mathbf{1}\mathbf{1}^{\top}, \\ P_{\mathcal{F}} W^3 P_{\mathcal{F}} &= 0.\end{aligned}\tag{PK50}$$

For the second matrix its diagonal is $2 + (M - 1) = M + 1$ and its off-diagonal entries are 2, from the two pairings of two different faces. For the third, first observe

$$\langle W_p^2, H_0 W_p^2 \rangle = 8, \quad \langle W_p^2, H_0 W_q^2 \rangle = 0 \ (p \neq q), \quad \langle W_p W_q, H_0 W_p W_q \rangle = 6 \ (p \neq q).$$

The first two follow from $W_p^2 = 1 + \chi_1(U_p)$, whose nonconstant part has electric energy $4 \cdot 2 = 8$; distinct such face characters are orthogonal by their unique edges. For the third, expand the gradient of

the product. Each individual face has Dirichlet energy 3; integrating an edge unique to the other face removes its squared trace with factor one. Every cross term vanishes: integrate an edge unique to one face to obtain $\int W_p X_{e,a} W_p = \frac{1}{2} X_{e,a} \int W_p^2 = 0$. This proves the total $3 + 3 = 6$, including adjacent faces sharing an edge. Consequently the third matrix has diagonal $8 + 6(M - 1) = 6M + 2$ and off-diagonal 6. Haar link sign changes commute with H_0 , so the same parity argument eliminates all other terms. The first and fourth matrices are respectively the triple and quintuple moments proved above.

Compression of the *full* Hamiltonian to constants plus faces is therefore

$$\begin{pmatrix} 2bM & -b\mathbf{1}^\top \\ -b\mathbf{1} & (2bM + 3\kappa)I_M \end{pmatrix}. \quad (\text{PK51})$$

Restricting this matrix to the constant vector and the direction $\sum_p W_p$, with its squared norm M retained, gives the ground variational upper bound

$$E_0 \leq 2bM + \frac{3\kappa - \sqrt{9\kappa^2 + 4b^2M}}{2}. \quad (\text{PK52})$$

This is an upper bound for the actual interacting ground energy; it is not an identification of the compressed eigenvalue with E_0 .

Now use the reflection-odd subspace $\mathcal{H}_-$. Put $P_- = P_{\mathcal{F}}|_{\mathcal{H}_-}$ and $Q_- = I_{\mathcal{H}_-} - P_-$. Every face coefficient vector in this subspace has sum zero. Equations (PK50)–(PK52) give

$$\begin{aligned} P_- H P_- &= (2bM + 3\kappa)P_-, & d &:= 2bM + 3\kappa - E_0 \geq \frac{3\kappa + \sqrt{9\kappa^2 + 4b^2M}}{2}, \\ C &:= Q_- \mathcal{W} P_- = \mathcal{W} P_-, & C^* C &= (M - 1)P_-, \\ C^* H_0 C &= (6M - 4)P_-, & C^* \mathcal{W} C &= 0. \end{aligned} \quad (\text{PK53})$$

The equality for C uses both reflection invariance and $P_- \mathcal{W} P_- = 0$. Every nonzero vector confined to this first odd electric layer has excitation-energy mean exactly d . On (PK43) its proved lower bound in (PK53) grows at least as $b_j \sqrt{M_{L_j}}$, rather than tending to zero.

This calculation identifies and evaluates the next return through the complement instead of stopping at that limitation. Let $s > 0$, and define the self-adjoint operator

$$B_- = Q_-(H - E_0)Q_-$$

on $Q_- \mathcal{H}_-$ by its closed quadratic form. Since $H - E_0 \geq 0$, $B_- \geq 0$, and $B_- + s$ is invertible without a gap assumption. All vectors in the range of C are smooth finite polynomials. The exact block resolvent and its first-return bounds are

$$\begin{aligned} P_-(H - E_0 + s)^{-1}P_- &= [(d + s)P_- - b^2 C^*(B_- + s)^{-1}C]_{P_- \mathcal{H}_-}^{-1}, \\ \frac{(M - 1)^2}{\kappa(6M - 4) + (2bM - E_0 + s)(M - 1)}P_- &\leq C^*(B_- + s)^{-1}C \leq \frac{M - 1}{s}P_-. \end{aligned} \quad (\text{PK54})$$

To prove the first identity, solve the Q_- component of the block equation for $H - E_0 + s$; its off-diagonal block is $-bC$. Substitution into the P_- equation gives the displayed Schur complement. It is strictly positive: minimizing the full positive form over the Q_- variable gives a value at least $s\|f\|^2$ for each $f \in P_- \mathcal{H}_-$. The finite inverse therefore exists. The upper bound follows from $B_- \geq 0$ and $C^*C = (M - 1)P_-$. For the lower bound, Cauchy–Schwarz applied to $(B_- + s)^{1/2}Cf$ and $(B_- + s)^{-1/2}Cf$, together with (PK53), gives

$$(M - 1)^2\|f\|^4 \leq [\kappa(6M - 4) + (2bM - E_0 + s)(M - 1)]\|f\|^2\langle Cf, (B_- + s)^{-1}Cf \rangle.$$

This proves (PK54), retaining the actual vacuum shift and both couplings. No artificial orthogonality between geometric parameters or omitted magnetic interaction enters the calculation.

The full packet $\eta_{h,t}$ also has its complementary part in (PK47). Its actual spectral measure remains (PK44), evaluated by (PK33), (PK45) and the full complementary operator in (PK54). The first electric projection alone cannot supply a decreasing excitation-energy mean on the chosen path. The exact complementary return above is the next receiving operator; further uniform estimates of that operator and of the packet's complementary coefficients remain active calculations.

9.2. The complete second complementary moment

Continue with the original open box and every face in (PK24). In this subsection write $P = P_-$, $Q = Q_-$, $B = B_-$, and $\mathcal{W} = \sum_p W_p$, with exactly the definitions of (PK53)–(PK54). No change of spatial scale, coupling, inner product or Hamiltonian is made. Put

$$m = M - 1, \quad \alpha = 2bM - E_0, \quad \mu = \kappa(6M - 4) + \alpha(M - 1). \quad (\text{PK55})$$

Thus $C^*C = mP$ and $C^*BC = \mu P$. The latter identity follows from (PK53), since C has range in $Q\mathcal{H}_-$. The ground-energy bound (PK52) gives $\alpha \geq (\sqrt{9\kappa^2 + 4b^2M} - 3\kappa)/2 > 0$. Consequently $m > 0$ and $\mu > 0$.

For two distinct original faces, define $A_{pq} = 1$ when they share a link and $A_{pq} = 0$ otherwise; set $A_{pp} = 0$. Let $D_{pp} = \sum_q A_{pq}$, $D_{pq} = 0$ for $p \neq q$. Let N_{pq} , for $p \neq q$, count elementary cubes in the original box whose boundaries contain both p and q , and set $N_{pp} = 0$. All three matrices act on the full original face coefficient space $\mathbb{C}^M$, identified isometrically with $\mathcal{F}$ by $(z_p) \mapsto \sum_p z_p W_p$. Their definitions include faces on the box boundary. Reflection permutes faces, links and cubes, so these matrices commute with the reflection and preserve $P\mathcal{H}_-$.

Theorem 9.2. In the original orthonormal face basis,

$$\begin{aligned} P_{\mathcal{F}} \mathcal{W} H_0^2 \mathcal{W} P_{\mathcal{F}} &= (36M - 8)I_M + 36\mathbf{1}\mathbf{1}^\top + \frac{3}{4}(D + A), \\ P_{\mathcal{F}} \mathcal{W}^4 P_{\mathcal{F}} &= (3M^2 - 7M + 5)I_M + (12M - 8)\mathbf{1}\mathbf{1}^\top + \frac{3}{2}N. \end{aligned} \quad (\text{PK56})$$

The full complementary second moment is

$$\begin{aligned} K_2 := C^*B^2C &= \kappa^2 P \left[(36M - 8)I_M + \frac{3}{4}(D + A) \right] P \\ &+ [2\kappa\alpha(6M - 4) + \alpha^2(M - 1) + b^2(2M^2 - 5M + 4)] P + \frac{3}{2}b^2 PNP. \end{aligned} \quad (\text{PK57})$$

Here B is the entire complementary operator, with the actual vacuum energy E_0 , not a finite compression substituted for that energy.

Proof of the geometry needed for (PK56). The vertical-ray occupation construction before (PK50) maps every finite edge-even face set S to a finite cube set K with boundary S , over $\mathbb{Z}/2\mathbb{Z}$. To make the change law explicit, for a cube with lower corner $n = (n_1, n_2, n_3)$ let its occupation be the parity of selected horizontal faces above its centre. The difference of occupations above and below a horizontal face is that face's indicator. For a vertical face between two horizontally adjacent columns, sum the edge-even equations on the common vertical strip from its lower edge to a height beyond S . Each internal vertical-face indicator occurs twice. The surviving terms are the two horizontal-ray counts and the indicator of that vertical face. Thus the occupation difference across it is again precisely its indicator. At sufficiently high height all occupations are zero. At sufficiently low height these change laws make occupations equal

across all horizontal columns, and an exterior column has value zero. Occupied cubes are therefore finite in both height and horizontal extent, and their boundary is exactly S . A cube outside the open box has occupation zero: its column is exterior, or it is above or below all box faces. Thus the filling respects the original box when S lies there.

In integer coordinates, write $s_{ij}(n) \in \{0, 1\}$ for the indicator of the face with lower corner n in coordinate directions i, j , and let e_1, e_2, e_3 be the integer coordinate vectors. The construction and all three boundary components are exactly

$$\begin{aligned} k(n) &= \sum_{t=n_3+1}^{\infty} s_{12}(n_1, n_2, t) \pmod{2}, \\ s_{12}(n) &= k(n - e_3) + k(n), \\ s_{23}(n) &= k(n - e_1) + k(n), \\ s_{13}(n) &= k(n - e_2) + k(n). \end{aligned}$$

The last two equations follow by summing, respectively, the edge equations

$$\begin{aligned} s_{12}(n) + s_{12}(n - e_1) + s_{23}(n) + s_{23}(n - e_3) &= 0, \\ s_{12}(n) + s_{12}(n - e_2) + s_{13}(n) + s_{13}(n - e_3) &= 0 \end{aligned}$$

over heights $t \geq n_3 + 1$. The vertical-face terms telescope to the face at height n_3 ; the face at infinity is zero by finite support. All additions in these five boundary equations are modulo two.

Let $\pi_{23}K, \pi_{13}K, \pi_{12}K$ be the sets of unit cells obtained by projecting the occupied cubes in the three coordinate directions. Every nonempty line of cubes parallel to the first axis has two end faces normal to that axis; these end faces are distinct for different projected cells. The same statement holds for the second and third axes. Hence

$$|\partial K| \geq 2(|\pi_{23}K| + |\pi_{13}K| + |\pi_{12}K|) \geq 6.$$

If $|\partial K| = 6$, each of these three projections consists of one cell. Any two cubes of K then have the same pair of second/third coordinates, the same first/third coordinates and the same first/second coordinates. They coincide. Conversely one cube has six boundary faces. We have proved that every six-face nonempty edge-even set is exactly one cube boundary, including at the box boundary.

There are two further elementary geometric facts used below. Distinct unit squares of the cubic lattice share at most one edge. Therefore in a set of at most three distinct faces, each face has an edge unused by the other faces. Integrating such an edge makes any power of its trace equal to the corresponding single-group Haar moment, independently of all remaining link variables. Repeating proves factorization of trace powers for these sets; no independence of larger face collections is assumed.

Haar coefficients. For $U = q_0 I - i_{\mathbb{C}} \sum_{a=1}^3 q_a \sigma_a$, with $\sum_{a=0}^3 q_a^2 = 1$, Haar probability is uniform measure on S^3 and $\text{tr } U = 2q_0$. Its even coordinate moment is

$$\int q_0^{2r} dU = \frac{1 \cdot 3 \cdots (2r-1)}{4 \cdot 6 \cdots (2r+2)}.$$

For completeness, integrate a rotationally invariant Gaussian in $\mathbb{R}^4$ in Cartesian and polar coordinates. Integration by parts gives the one-coordinate Gaussian recursion $\int_{\mathbb{R}} t^{2r} e^{-t^2/2} dt = (2r-1) \int_{\mathbb{R}} t^{2r-2} e^{-t^2/2} dt$. Radial integration gives the recursion by $2r+2$ for the corresponding four-dimensional radial moment. Dividing the Cartesian moment by that radial moment yields the displayed ratio, including the initial moment one. Thus the trace moments at powers 2, 4, 6 are respectively 1, 2, 5.

The integral of one trace on each of the six cube faces is exactly $1/16$. Orient the six faces outward. The two traversals of each of the twelve edges then have opposite orientations. Reversing any original face orientation does not change its trace, since for $SU(2)$ the trace of the inverse equals the trace. The entry identity is

$$\int U_{ij}(U^{-1})_{kl} dU = \frac{1}{2} \delta_{il} \delta_{jk}.$$

One can prove it directly from the displayed quaternion matrix and the second coordinate moments $\int q_a q_b = \delta_{ab}/4$. Expand all six traces into matrix entries and integrate every edge. Each contributes $1/2$ and identifies the corner indices at its two ends. At each cube vertex its three incident face-corner indices are identified cyclically; no equality joins distinct vertices. There remain eight free indices, each taking two values. The integral is therefore $2^8/2^{12} = 1/16$. This also proves positivity and the orientation sign without a surface approximation.

For a diagonal entry of the second matrix in (PK56), the two external traces are both W_p . The four internal choices have nonzero integrals only when every face has even multiplicity. A nonempty odd-multiplicity set would have at most four faces and is excluded by the proved geometry. The four possibilities, with their multiplicities retained, give

$$5 + 2(M-1) + 6 \cdot 2(M-1) + 6 \binom{M-1}{2} = 3M^2 + 5M - 3.$$

They are respectively four further copies of p ; four copies of another face; two further copies of p and two of another face; and two each of two other faces. Each integral uses at most three distinct faces, so the factorization proved above applies.

For distinct external faces p, q , the even-multiplicity cases have internal multiplicities $(3p, 1q)$, $(1p, 3q)$, or $(1p, 1q, 2r)$ with $r \neq p, q$. Their sum is

$$4 \cdot 2 + 4 \cdot 2 + 12(M-2) = 12M - 8.$$

If the odd-multiplicity set is nonempty, it must contain six distinct faces, each appearing once in the full product, and must be a cube boundary. For each cube containing p, q , the remaining four faces have $4! = 24$ orders. The additional contribution is $24/16 = 3/2$ per such cube. This proves the magnetic matrix in (PK56), including every entry of N .

Electric coefficients. The single-face identity $W_p^2 = 1 + \chi_1(U_p)$ gives

$$H_0 W_p^2 = 8\chi_1(U_p), \quad \|H_0 W_p^2\|_2^2 = 64, \quad \langle H_0 W_p^2, H_0 W_q^2 \rangle = 0 \quad (p \neq q).$$

The last assertion follows by integrating an edge unique to one of the two faces: a spin-one character has zero average. The eigenvalue eight retains all four link Casimirs $1(1+1) = 2$.

For faces $p \neq q$ with no common link, $W_p W_q$ is an electric eigenfunction with eigenvalue $8 \cdot 3/4 = 6$, and its squared norm is one. Its second moment is therefore 36. For adjacent faces let U be the shared link, and write the two traces, after cyclic permutation and permitted inversion, as $\text{tr}(AU)$ and $\text{tr}(U^{-1}B)$. The other six links are distinct. Haar projection onto constants in the shared link is

$$F_0(A, B) = \int \text{tr}(AU) \text{tr}(U^{-1}B) dU = \frac{1}{2} \text{tr}(AB).$$

Its squared norm over the other six links is $1/4$. The full product has squared norm one by the two-face integration argument. The complementary part F_1 consequently has squared norm $3/4$. Its shared-link representation is spin one: the tensor product of the fundamental representation and its dual splits into the scalar identity matrix and the three traceless matrices. Conjugation on the traceless matrices is

the spin-one representation. Its Casimir for the original $T_a = -i_C \sigma_a / 2$ is two, as follows by applying $-\sum_a [T_a, [T_a, V]]$ to each $V = T_b$. On the scalar part the shared-link Casimir is zero. Each of the other six links contributes $3/4$ to both parts. Hence

$$H_0 F_0 = \frac{9}{2} F_0, \quad H_0 F_1 = \frac{13}{2} F_1, \quad \|H_0(W_p W_q)\|_2^2 = \frac{1}{4} \left(\frac{9}{2}\right)^2 + \frac{3}{4} \left(\frac{13}{2}\right)^2 = \frac{147}{4}.$$

This is $36 + 3/4$, not 36. Keeping the shared-link tensor sector is essential for the second moment even though its first moment is six.

For an arbitrary matrix entry, expand $\langle H_0(WW_p), H_0(WW_q) \rangle$. Changing one link to its negative commutes with H_0 ; therefore a term vanishes if its four face factors have a nonempty odd-multiplicity set. The diagonal terms are 64 for the repeated face, $147/4$ for every adjacent face and 36 for every other face. Their sum is $36M + 28 + (3/4)D_{pp}$. For $p \neq q$, one remaining pairing gives the squared moment of $W_p W_q$; the other gives $\langle H_0 W_p^2, H_0 W_q^2 \rangle = 0$. Thus the off-diagonal entry is $36 + (3/4)A_{pq}$. This proves the first matrix in (PK56).

Passage to the entire complement. All face polynomials, and their images under any finite word in H_0 and $\mathcal{W}$, are smooth on the compact product of link groups. They lie in the domain of every required power of the elliptic operator. The projections P and Q preserve this domain, since P has smooth finite-dimensional range. Thus the following expressions are operator identities between finite-dimensional maps, with no form-domain substitution:

$$\begin{aligned} C^* B^2 C &= C^* (H - E_0) Q (H - E_0) C \\ &= C^* (H - E_0)^2 C - C^* (H - E_0) P (H - E_0) C. \end{aligned}$$

Since $PH_0 = 3P$, $PC = 0$, and $PWC = mP$, $P(H - E_0)C = -bmP$. The second term is exactly $b^2 m^2 P$. Expand the first term using $H - E_0 = \kappa H_0 + \alpha - b\mathcal{W}$:

$$\begin{aligned} C^* (H - E_0)^2 C &= \kappa^2 C^* H_0^2 C + 2\kappa\alpha C^* H_0 C + \alpha^2 C^* C + b^2 C^* \mathcal{W}^2 C \\ &\quad - \kappa b C^* (H_0 \mathcal{W} + \mathcal{W} H_0) C - 2\alpha b C^* \mathcal{W} C. \end{aligned}$$

The last term vanishes by (PK53). Each mixed electric/magnetic term on the preceding line has five total face factors, and H_0 preserves each edge parity. Five factors cannot have all face multiplicities even; the nonempty odd set has at most five faces and is excluded. Hence both mixed terms vanish. Insert (PK56), (PK53), and the exact $b^2 m^2 P$ subtraction. The all-ones matrix annihilates the odd subspace. The remaining magnetic scalar is $b^2[(3M^2 - 7M + 5) - (M - 1)^2] = b^2(2M^2 - 5M + 4)$. This proves (PK57). $\square$

9.3. A second exact return and a stronger resolvent bound

The calculated moment supplies the next connecting map, not only a finite matrix comparison. Define projections and operators on the original complementary Hilbert space by

$$\begin{aligned} \Pi &= \frac{1}{m} C C^*, \quad Q_2 = Q - \Pi, \quad T = Q_2 B C, \quad B_2 = Q_2 B Q_2, \\ V &:= T^* T = K_2 - \frac{\mu^2}{m} P. \end{aligned} \tag{PK58}$$

The operator B_2 is defined by restriction of the closed nonnegative form to $Q_2 \mathcal{H}_-$; it is self-adjoint and nonnegative. Indeed $C^* C = mP$ proves $\Pi^* = \Pi$, $\Pi^2 = \Pi$ and that its range is exactly $\text{ran } C$. This finite-dimensional range consists of smooth vectors. Its orthogonal complement preserves the form

domain; restriction is therefore dense and closed. In the last identity of (PK58), subtract $C^*B\Pi BC = (C^*BC)^2/m = (\mu^2/m)P$.

The full expression, retaining both its previous form and its exact evaluated comparison, is

$$\begin{aligned} V &= K_2 - \frac{[\kappa(6M-4) + \alpha(M-1)]^2}{M-1}P \\ &= \kappa^2 P \left[\left(4 - \frac{4}{M-1}\right) I_M + \frac{3}{4}(D+A) \right] P \\ &\quad + b^2 P \left[(2M^2 - 5M + 4)I_M + \frac{3}{2}N \right] P. \end{aligned} \tag{PK59}$$

The cancellation of α follows by expanding its entire square: the $\alpha^2(M-1)$ and $2\kappa\alpha(6M-4)$ terms cancel exactly. Also $(6M-4)^2/(M-1) = 36M-12+4/(M-1)$, giving the displayed electric scalar. No vacuum energy has been removed from B , B_2 , or μ .

These matrices give explicit nonzero bounds for the next map. For any face vector z ,

$$z^*(D+A)z = \sum_{\{p,q\}:A_{pq}=1} |z_p + z_q|^2.$$

Every face has four edges, and each edge belongs to at most four faces. Different faces share at most one edge, so $D_{pp} \leq 12$. Thus $0 \leq D+A \leq 24I_M$. Let $J_{pc} = 1$ if face p belongs to cube c , and zero otherwise. With $k_p = \sum_c J_{pc} \in \{1, 2\}$, the exact identity is $N = JJ^T - \text{diag}(k_p)$. Therefore $N \geq -2I_M$. Its row sum is $5k_p \leq 10$, and $|z^*Nz| \leq \sum_{p<q} N_{pq}(|z_p|^2 + |z_q|^2) \leq 10\|z\|^2$, so $N \leq 10I_M$. Consequently

$$\begin{aligned} \left[\kappa^2 \left(4 - \frac{4}{m}\right) + b^2(2M^2 - 5M + 1) \right] P &\leq V \\ &\leq \left[\kappa^2 \left(22 - \frac{4}{m}\right) + b^2(2M^2 - 5M + 19) \right] P. \end{aligned} \tag{PK60}$$

Here $L \geq 1$ gives $M = 3(2L)^2(2L+1) \geq 36$, so the lower bound is strictly positive. In particular T is injective on the actual odd face space: the next return is not zero.

For every $s > 0$, the exact second return is

$$C^*(B+s)^{-1}C = m^2 [(\mu + sm)P - T^*(B_2 + s)^{-1}T]_{P\mathcal{H}_-}^{-1}. \tag{PK61}$$

To prove it without changing the metric, solve $(B+s)(Cf+y) = Cu$, where $y \in Q_2\mathcal{H}_-$. The Q_2 equation gives $y = -(B_2+s)^{-1}Tf$. Applying C^* to the other equation gives $[(\mu + sm)P - T^*(B_2 + s)^{-1}T]f = mu$. Finally $C^*(Cf+y) = mf$, producing both factors m in (PK61). Its denominator is strictly positive: minimizing the quadratic form of $B+s$ over y leaves at least $sm\|f\|^2$. These arguments extend from smooth vectors by the closed form and resolvent; the finite off-diagonal map T is bounded.

There is also an upper bound using the complete second moment:

$$\frac{m^2}{\mu + sm}P \leq C^*(B+s)^{-1}C \leq \frac{m}{s}P - \frac{\mu^2}{s}(K_2 + s\mu P)^{-1}. \tag{PK62}$$

The first inequality is (PK54). For the second put

$$X = [B(B+s)]^{1/2}C, \quad Y = [B(B+s)^{-1}]^{1/2}C.$$

These maps are defined since C has range in the domain of B . The spectral functional calculus gives

$$X^*X = K_2 + s\mu P, \quad X^*Y = \mu P, \quad Y^*Y = mP - sC^*(B+s)^{-1}C.$$

The first matrix is positive definite because $s\mu > 0$. Expanding $[Y - X(X^*X)^{-1}X^*Y]^*[Y - X(X^*X)^{-1}X^*Y] \geq 0$ gives $Y^*Y \geq \mu^2(K_2 + s\mu P)^{-1}$. Rearrangement proves (PK62). Its positive subtracted matrix makes the upper bound strictly stronger than mP/s for every finite box and every $s > 0$.

Equations (PK57)–(PK62) retain the exact relation of the full complementary operator to the original first electric layer. They do not substitute a finite-level spectrum for the spectrum of H . For the actual corrected path (PK43), $M = 3(2j^4)^2(2j^4 + 1)$ and $b = 10000j^2/\kappa_*$; the nonzero term $b^2(2M^2 - 5M + 1)$ in (PK60) explicitly prevents treating the next map as a uniformly small perturbation. The next calculation is the entire $T^*(B_2 + s)^{-1}T$, together with the complementary coefficients of the actual heat packet in (PK47). Its small-energy measure remains the quantity (PK44); neither a shrinking classical energy nor these finite moments replace that calculation.

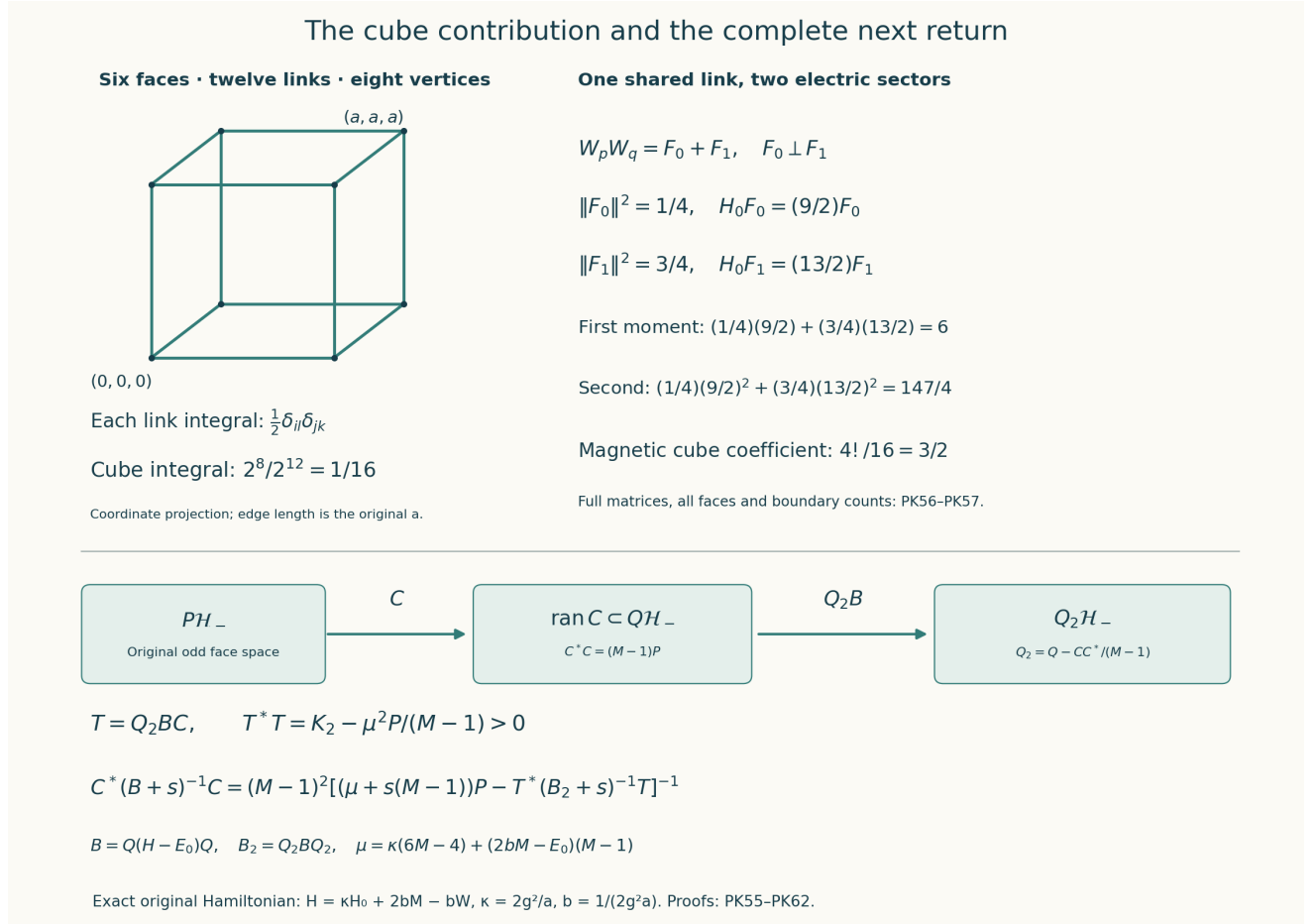


Figure 2: Original cube integration and the complete next return

The figure shows the coordinate projection of an actual side-a lattice cube, the two shared-link electric sectors, and the exact composition $T = Q_2 B C$. Complete proofs and all factors are PK55–PK62. Its reproducible source retains the original operators. The finite Haar calculation is derived here; no novelty claim is made for character integration or Schur inversion.

9.4. The full packet through both complementary returns

Keep the original finite open box, Haar measure, physical Hilbert space, Hamiltonian $H = \kappa H_0 + 2bM - bW$, and actual ground energy E_0 of (PK24)–(PK37). All inverses in this section have a positive energy parameter $s > 0$. The inner product is conjugate linear in its first argument. On the reflection-odd physical space put $A = H - E_0$, $p = P\eta_{h,t}$, $q = Q\eta_{h,t}$, with the original first-face projections P, Q of (PK51)–(PK54). In particular p is the explicit face sum (PK47); q is the entire remainder. The complete raw Stieltjes transform is

$$\begin{aligned} F_{h,t}(s) &:= \langle \eta_{h,t}, (A + s)^{-1} \eta_{h,t} \rangle = \int_{(0,\infty)} \frac{d\nu_{h,t}(\omega)}{\omega + s}, \\ D_s &= B + s, \quad S_s = (d + s)P - b^2 C^* D_s^{-1} C, \\ F_{h,t}(s) &= \langle q, D_s^{-1} q \rangle + \langle p + bC^* D_s^{-1} q, S_s^{-1} (p + bC^* D_s^{-1} q) \rangle. \end{aligned} \tag{PK63}$$

Here $d = 2bM + 3\kappa - E_0$, $C = QWP$, and $B = QAQ$ remain the operators already proved in (PK54).

Indeed the block equations for $(A + s)(x + y) = p + q$, with $x = Px$, $y = Qy$, are $(d + s)x - bC^*y = p$ and $D_s y - bCx = q$. Thus $y = D_s^{-1}(q + bCx)$ and $S_s x = p + bC^* D_s^{-1} q$. Taking the inner product with $p + q$ gives (PK63), including both cross terms. The closed form of $A + s$ is at least sI . Minimizing it over y proves $S_s \geq sP$; also $D_s \geq sQ$. Therefore every inverse used here is defined and bounded. The off-diagonal map has finite-dimensional smooth range, so the calculation on smooth vectors extends to the full form domain and the resolvent. No packet norm has been divided out.

The next return also acts on the entire complementary packet. Retain $m = M - 1$, $\Pi = CC^*/m$, $Q_2 = Q - \Pi$, $T = Q_2 BC$, $B_2 = Q_2 B Q_2$, and $\mu = \kappa(6M - 4) + (2bM - E_0)m$. Define

$$\begin{aligned} w &= m^{-1} C^* q = m^{-1} P W \eta_{h,t}, \quad z = Q_2 q, \quad q = Cw + z, \\ D_{2,s} &= B_2 + s, \quad K_s = (\mu + sm)P - T^* D_{2,s}^{-1} T, \quad r_s = mw - T^* D_{2,s}^{-1} z, \\ C^* D_s^{-1} q &= m K_s^{-1} r_s, \quad \langle q, D_s^{-1} q \rangle = \langle z, D_{2,s}^{-1} z \rangle + \langle r_s, K_s^{-1} r_s \rangle, \\ S_s &= (d + s)P - b^2 m^2 K_s^{-1}, \\ F_{h,t}(s) &= \langle z, D_{2,s}^{-1} z \rangle + \langle r_s, K_s^{-1} r_s \rangle + \langle p + bm K_s^{-1} r_s, S_s^{-1} (p + bm K_s^{-1} r_s) \rangle. \end{aligned} \tag{PK64}$$

The equality for w uses $C^* C = mP$ and $PWP = 0$; it retains the original non-unit Gram of the map C . To prove the other formulas, write $D_s^{-1} q = Cf + y$, $Q_2 y = y$. The two equations are $(\mu + sm)f + T^* y = mw$ and $Tf + D_{2,s} y = z$. They give $K_s f = r_s$ and $y = D_{2,s}^{-1}(z - Tf)$. Taking the inner product with $Cw + z$ proves the middle line. The same minimization as in (PK61) gives $K_s \geq smP$. Substitution in (PK63) proves the final line. In particular omitting z , either of its return terms, or the change from p to $p + bm K_s^{-1} r_s$ does not compute the original measure.

9.5. A complete electric cutoff with a vacuum-energy interval

We next bound the entire omitted operator space, while retaining all magnetic faces. This supplies a finite calculation with an explicit error for the same $F_{h,t}$, without evaluating successive moments indefinitely.

For a real number $\Lambda \geq 3$, let

$$\begin{aligned} P_\Lambda &= \mathbf{1}_{[0,\Lambda]}(H_0) \quad \text{on the full physical Hilbert space,} \quad Q_\Lambda = I - P_\Lambda, \\ H_\Lambda &= P_\Lambda H P_\Lambda \Big|_{\text{Ran } P_\Lambda}, \quad u_\Lambda = \min \sigma(H_\Lambda), \quad \ell = 2bM. \end{aligned} \tag{PK65}$$

The ground calculation uses both reflection parities. Its vacuum is not replaced by the bottom of the odd compression.

Here is the exact finite construction of this space and matrix. For each assignment of link spins $j_e \in \{0, \frac{1}{2}, 1, \dots\}$ with $\lambda_j = \sum_e j_e(j_e + 1) \leq \Lambda$, take all products $\prod_e \sqrt{2j_e + 1} D^{j_e}(U_e)_{r_e s_e}$. They are orthonormal in the full product Haar space by matrix-entry orthogonality. Their union spans the electric cutoff: on each link the Casimir is $j_e(j_e + 1)$, and the matrix coefficients give the complete Peter–Weyl expansion used in (PK27)–(PK36). There are finitely many assignments and finitely many entries per assignment. Apply the vertex Haar projections (PK35). Remove the zero eigenvectors of their finite Gram matrix and orthonormalize its positive eigenspaces. Since gauge averaging commutes with H_0 , this gives an orthonormal physical basis v_α , with $H_0 v_\alpha = \lambda_\alpha v_\alpha$. Reflection commutes with the same operators; diagonalize its involution within each electric eigenspace to retain both parities explicitly. Every entry is then

$$\begin{aligned} (H_\Lambda)_{\alpha\beta} &= \kappa \lambda_\beta \delta_{\alpha\beta} + 2bM \delta_{\alpha\beta} - b \sum_{p \in \mathcal{F}_L} \int \overline{v_\alpha(U)} W_p(U) v_\beta(U) dU, \\ a_\alpha(h, t) &:= \langle v_\alpha, \eta_{h,t} \rangle = e^{-t\lambda_\alpha} (\overline{v_\alpha(h)} - \overline{v_\alpha(\mathcal{R}h)}), \\ \eta_\Lambda &= \mathbf{P}_\Lambda \eta_{h,t} = \sum_\alpha a_\alpha(h, t) v_\alpha, \quad \Gamma_\Lambda = \sum_\alpha |a_\alpha(h, t)|^2. \end{aligned} \tag{PK66}$$

The link integrals are finite products of the exact entry projectors (PK35), after the ordered face word has been inserted; the entire face sum is present. The coefficient formula follows by applying the heat semigroup to v_α and evaluating at the label, then averaging over its gauge orbit. It is zero for even v_α . The same formula, over all electric eigenvalues, constructs the full packet and hence q, w, z in (PK64). Here h remains the actual inverse holonomy of (PK23), including its original compact-support field and parameters. This is a finite algebraic matrix together with exact evaluations at those holonomies; no numerical holonomy evaluation is claimed.

Positivity of the original Wilson potential and its full bound give

$$H \geq \kappa H_0, \quad 0 \leq b \sum_p (2 - W_p) \leq 4bM, \quad \|\mathbf{Q}_\Lambda H \mathbf{P}_\Lambda\| = b \|\mathbf{Q}_\Lambda \mathcal{W} \mathbf{P}_\Lambda\| \leq \ell. \tag{PK67}$$

The last equality holds because both H_0 and the scalar $2bM$ commute with the cutoff. The inequality uses $|W_p| \leq 2$ for each original face. The high compression, defined by its closed quadratic form, is at least $\kappa \Lambda \mathbf{Q}_\Lambda$. The cutoff contains the unit constant, whose energy is $2bM$. Consequently $0 \leq E_0 \leq u_\Lambda \leq 2bM$.

For $\kappa \Lambda > 2bM$, define

$$\begin{aligned} e_\Lambda &= \max \left\{ 0, \frac{u_\Lambda + \kappa \Lambda - \sqrt{(\kappa \Lambda - u_\Lambda)^2 + 4\ell^2}}{2} \right\}, \\ w_\Lambda &= u_\Lambda - e_\Lambda, \quad D_\Lambda = \kappa \Lambda - 2bM > 0. \end{aligned} \tag{PK68}$$

Then the true vacuum energy satisfies the proved interval

$$0 \leq e_\Lambda \leq E_0 \leq u_\Lambda \leq 2bM, \quad 0 \leq w_\Lambda \leq \frac{2\ell^2}{\sqrt{(\kappa \Lambda - u_\Lambda)^2 + 4\ell^2} + \kappa \Lambda - u_\Lambda} \leq \frac{\ell^2}{D_\Lambda}. \tag{PK69}$$

For a unit form-domain vector $x+y$, $x = \mathbf{P}_\Lambda x$, $y = \mathbf{Q}_\Lambda y$, its energy is at least $u_\Lambda \|x\|^2 - 2\ell \|x\| \|y\| + \kappa \Lambda \|y\|^2$. The smallest eigenvalue of the real two-by-two matrix with diagonal $u_\Lambda, \kappa \Lambda$ and off-diagonal $-\ell$ is the root in (PK68). Minimizing proves the lower bound; the independent inequality $H \geq 0$ supplies the maximum with zero. Rayleigh minimization in the cutoff gives the upper bound. Subtracting that root

from u_Λ and rationalizing gives the middle expression of (PK69); if the root is negative, taking the maximum with zero can only decrease the width. Finally $\kappa\Lambda - u_\Lambda \geq D_\Lambda > 0$ proves the last bound. Thus no unknown vacuum shift is assumed in the finite calculation.

9.6. The raw packet, its tail, and a resolvent error

For fixed original h, t, L, a, g set $\Gamma = \|\eta_{h,t}\|^2$ and $\tau_\Lambda = \|\mathbf{Q}_\Lambda \eta_{h,t}\|^2$. The exact identity is $\Gamma = \Gamma_\Lambda + \tau_\Lambda$. The common heat parameter gives $\eta_{h,t} = e^{-(t/2)H_0} \eta_{h,t/2}$, including the physical projection and reflection. Therefore

$$\begin{aligned} 0 \leq \tau_\Lambda &\leq e^{-t\Lambda} \|\eta_{h,t/2}\|^2 \leq 4e^{-t\Lambda} Z_t^{N_L}, \\ Z_t = p_t(I) &= \sum_{j \in \{0, 1/2, \dots\}} (2j+1)^2 e^{-tj(j+1)} = \sum_{n=1}^{\infty} n^2 e^{-t(n^2-1)/4}, \\ Z_t &\leq \mathcal{Z}_t := \frac{1 + e^{-t/4}}{(1 - e^{-t/4})^3}. \end{aligned} \tag{PK70}$$

The first inequality is spectral calculus on electric eigenvalues strictly greater than Λ . Before physical projection, the squared norm of a product heat packet of time $t/2$ is $p_t(I)^{N_L}$, by Haar invariance and the convolution law. Orthogonal gauge projection decreases norm; the difference of two such packets has squared norm at most four times this number. Finally, with $q_t = e^{-t/4}$, $n^2 \geq n$ gives $Z_t \leq e^{t/4} \sum_{n \geq 1} n^2 q_t^n = e^{t/4} q_t(1 + q_t)/(1 - q_t)^3$. This proves the stated bound and retains the original heat convention.

The finite number to compute is

$$\begin{aligned} f_\Lambda(s) &= \langle \eta_\Lambda, (H_\Lambda - e_\Lambda + s)^{-1} \eta_\Lambda \rangle, \quad \beta_\Lambda(s) = \kappa\Lambda - e_\Lambda + s, \\ |F_{h,t}(s) - f_\Lambda(s)| &\leq \frac{\Gamma_\Lambda \ell^2}{\beta_\Lambda(s) s^2} + \frac{2\sqrt{\Gamma_\Lambda \tau_\Lambda} + \tau_\Lambda}{s} + \frac{\Gamma w_\Lambda}{s^2} =: \mathcal{E}_\Lambda(s). \end{aligned} \tag{PK71}$$

Here Γ , E_0 , the packet, and H have their original meanings. The new e_Λ is explicitly an endpoint used for an approximation with the displayed error, never a replacement definition of the vacuum energy.

To prove (PK71), first use the full resolvent $R_e = (H - e_\Lambda + s)^{-1}$. Its norm is at most $1/s$ by (PK69). Its high block before inversion, D_e , is at least $\beta_\Lambda(s) \mathbf{Q}_\Lambda$. Writing $J = \mathbf{Q}_\Lambda H \mathbf{P}_\Lambda$ and $L_e = H_\Lambda - e_\Lambda + s$, block elimination gives

$$\mathbf{P}_\Lambda R_e \mathbf{P}_\Lambda = (L_e - J^* D_e^{-1} J)^{-1}, \quad 0 \leq J^* D_e^{-1} J \leq \frac{\ell^2}{\beta_\Lambda(s)} \mathbf{P}_\Lambda.$$

Both inverses, L_e^{-1} and the displayed compressed resolvent, have norm at most $1/s$. The latter fact follows either by compression of R_e or by minimizing the full positive form over its high component. The identity $(L_e - R)^{-1} - L_e^{-1} = (L_e - R)^{-1} R L_e^{-1}$ therefore bounds their difference by $\ell^2/(\beta_\Lambda(s) s^2)$. Multiplying by the actual low-vector norm gives the first term of (PK71). Expanding $\eta_{h,t} = \eta_\Lambda + \mathbf{Q}_\Lambda \eta_{h,t}$ in R_e and applying Cauchy-Schwarz gives its second term. Lastly the resolvent identity for the two scalar shifts gives

$$(H - E_0 + s)^{-1} - R_e = (E_0 - e_\Lambda)(H - E_0 + s)^{-1} R_e.$$

Both factors have norm at most $1/s$, and $0 \leq E_0 - e_\Lambda \leq w_\Lambda$. This gives the third term and completes the proof, including all infinite complementary components.

A fully finite upper error is also available: replace τ_Λ by $T_\Lambda = 4e^{-t\Lambda} \mathcal{Z}_t^{N_L}$ and Γ by $\Gamma_\Lambda + T_\Lambda$ on the right of (PK71). Every term is increasing in these nonnegative variables, so the resulting number is a proved upper bound, without a separate evaluation of Γ .

9.7. An explicit cutoff on the original regulator path

Take the same integers j satisfying (PK43b), the same labels h_j , and the same $t_* > 0, \kappa_* > 0, r, R$. Define, without dropping any factors,

$$\begin{aligned} N_j &= 3(2j^4)(2j^4 + 1)^2, & M_j^{\text{face}} &= 3(2j^4)^2(2j^4 + 1), & b_j &= \frac{10000j^2}{\kappa_*}, \\ m_j^{\text{core}} &= \left\lfloor \frac{50rj^3}{\sqrt{3}} \right\rfloor, & \gamma_j &= 32 [3(2m_j^{\text{core}})^2(2m_j^{\text{core}} + 1)]e^{-6t_*} \sin^8 \left(\frac{1}{100j^3} \right), \\ s_j &= \frac{\kappa_*}{j}, & \delta_j &= \frac{1}{\kappa_* j}, & \rho_j &= \frac{1}{6j^2}, \\ \Lambda_j &= 1 + \left[\max \left\{ 3, \frac{2b_j M_j^{\text{face}}}{\kappa_*} + \frac{16b_j^2 (M_j^{\text{face}})^2 j^3}{\kappa_*^2}, \frac{\log 4 + N_j \log \mathcal{Z}_{t_*} - \log \gamma_j + 2 \log(6j^2)}{t_*} \right\} \right]. \end{aligned} \quad (\text{PK72})$$

The superscripts distinguish the face count and core size from the covering degree $M_{\text{cov}} = j^4$ and the return Gram $m = M_L - 1$. The proof of (PK48)–(PK49) gives $0 < \gamma_j \leq \Gamma_j$. All entries of (PK72) are defined at the stated threshold.

For this explicit finite cutoff the actual transform obeys

$$|F_{h_j, t_*}(s_j) - f_{\Lambda_j}(s_j)| \leq \frac{\Gamma_j}{\kappa_* j}. \quad (\text{PK73})$$

To verify the constants, put $\ell_j = 2b_j M_j^{\text{face}}$ and $D_j = \kappa_* \Lambda_j - 2b_j M_j^{\text{face}}$. Formula (PK72) gives

$$D_j \geq \frac{4\ell_j^2}{\delta_j s_j^2}, \quad \tau_{\Lambda_j} \leq \gamma_j \rho_j^2 \leq \Gamma_j \rho_j^2, \quad \beta_{\Lambda_j}(s_j) \geq D_j.$$

Using $\Gamma_{\Lambda_j} \leq \Gamma_j$ and (PK69), the first and third terms of (PK71) together are at most $2\Gamma_j \ell_j^2 / (D_j s_j^2) \leq \delta_j \Gamma_j / 2$. The second term is at most $\Gamma_j (2\rho_j + \rho_j^2) / s_j \leq 3\Gamma_j \rho_j / s_j = \delta_j \Gamma_j / 2$. Adding proves (PK73). This is an error measured against the original raw norm; the measure and vectors have not been redefined. The cutoff grows substantially because the full $b_j^2 (M_j^{\text{face}})^2$ contribution is retained. No feasible matrix size or evaluated small-energy mass at these cutoffs is asserted.

The transform error has an exact consequence for the original spectral mass. For every $\epsilon, s > 0$, writing $n(\epsilon) = \nu_{h, t}((0, \epsilon])$, one has

$$\max \left\{ 0, \frac{s(\epsilon + s)}{\epsilon} F_{h, t}(s) - \frac{s\Gamma}{\epsilon} \right\} \leq n(\epsilon) \leq \min \{ \Gamma, (\epsilon + s) F_{h, t}(s) \}. \quad (\text{PK74})$$

Indeed $1/(\omega + s) \geq 1/(\epsilon + s)$ on $(0, \epsilon]$, which proves the upper bound. On this interval it is at most $1/s$, and on (ϵ, ∞) it is at most $1/(\epsilon + s)$. Thus $F(s) \leq n(\epsilon)/s + (\Gamma - n(\epsilon))/(\epsilon + s)$; rearranging proves the lower bound. There is no atom at zero by (PK44). Consequently a fully finite interval for $n(\epsilon)$ is obtained by using $f_{\Lambda} - \mathcal{E}_{\Lambda}$ in the lower expression and $f_{\Lambda} + \mathcal{E}_{\Lambda}$ in the upper one. If Γ is not evaluated, its finite upper bound $\Gamma_{\Lambda} + T_{\Lambda}$ may be used in the negative lower-bound term and in the upper cap.

In particular put $\epsilon = s = s_j$, $f_j = f_{\Lambda_j}(s_j)$. The exact path estimate (PK73) yields

$$\max \{ 0, 2s_j f_j - \Gamma_j - 2\Gamma_j / j^2 \} \leq \nu_j((0, s_j]) \leq \min \{ \Gamma_j, 2s_j f_j + 2\Gamma_j / j^2 \}. \quad (\text{PK75})$$

These inequalities are proved for the constructed original vectors. They do not assert that their lower endpoint is positive. They turn the remaining low-energy-weight question into a specific finite expression with a proved error, while (PK63)–(PK64) identify exactly which full packet components it receives. The next calculation is to estimate or evaluate that expression along the path, and to pursue the continuum state and reconstruction using the resulting spectral information.

9.8. The complete first cutoff on the original smallest box

The finite construction can already be evaluated exactly at its first nonconstant cutoff. For $3 \leq \Lambda < 9/2$, Theorem 9.1 says $\text{Ran } \mathbf{P}_\Lambda = \text{span}\{1, W_p : p \in \mathcal{F}_L\}$. These $M+1$ vectors are orthonormal. The constant electric eigenvalue is zero, every face eigenvalue is 3, and (PK50) gives $P_{\mathcal{F}} \mathcal{W} P_{\mathcal{F}} = 0$. Also $\langle 1, \mathcal{W} W_p \rangle = 1$ and $\langle 1, \mathcal{W} 1 \rangle = 0$. Hence the entire cutoff matrix is

$$H_\Lambda = \begin{pmatrix} 2bM & -b\mathbf{1}^T \\ -b\mathbf{1} & (2bM + 3\kappa)I_M \end{pmatrix}, \quad u_\Lambda = 2bM + \frac{3\kappa - \sqrt{9\kappa^2 + 4b^2M}}{2}. \quad (\text{PK76})$$

On the $M-1$ face-coefficient vectors orthogonal to $\mathbf{1}$, the eigenvalue is $2bM + 3\kappa$. On the remaining two-dimensional space, use the unit constant and the unit face sum $M^{-1/2} \sum_p W_p$: the off-diagonal entry is $-b\sqrt{M}$. Its two eigenvalues are the two roots displayed by (PK76), with minus and plus signs. This proves the asserted minimum and recovers the earlier trial bound as the exact minimum of this whole cutoff.

For the actual box $L = 1$, choose $a = 1$, $g = 2$, and $\Lambda = 3$. Then $N_L = 54$, $M = 36$, $\kappa = 8$, $b = 1/8$, $\ell = 9$, and $\kappa\Lambda = 24 > 2bM = 9$. The full physical cutoff has dimension 37, and

$$u_\Lambda = 21 - \frac{3\sqrt{257}}{4}, \quad e_\Lambda = \frac{u_\Lambda + 24 - \sqrt{(24 - u_\Lambda)^2 + 324}}{2} > 0. \quad (\text{PK77})$$

Positivity follows from $24u_\Lambda > 81$: indeed $\sqrt{257} < 17$ gives $u_\Lambda > 33/4$. Thus (PK69) gives the explicit genuine vacuum interval $e_\Lambda \leq E_0 \leq u_\Lambda$ in this original interacting box. This interval is not asserted to have sharp endpoints.

Reflection fixes exactly the four faces in the $n_1 = 0$ coordinate plane parallel to the second and third axes; the other 32 faces form 16 pairs. Therefore the odd cutoff has dimension 16 and its entire Hamiltonian is $33I_{16}$. For every original real label h and heat time $t > 0$ on this box,

$$f_\Lambda(s) = \frac{e^{-6t} \sum_{p \in \mathcal{F}_1} |W_p(h) - W_p(\mathcal{R}h)|^2}{33 - e_\Lambda + s}. \quad (\text{PK78})$$

This is an evaluated finite resolvent with the full original face coefficients; the cutoff error is still (PK71), with $\beta_\Lambda(s) = 24 - e_\Lambda + s$ and $T_\Lambda = 4e^{-3t} \mathcal{Z}_t^{54}$. It is a calculation at these stated box and coupling parameters, not an evaluation at the much larger cutoffs (PK72).

The block-inversion method is classical; compare Geneviève Dusson, Israel Michael Sigal and Benjamin Stamm, The Feshbach–Schur map and perturbation theory, arXiv:2105.02058v1, original TeX labels thm:isospF, Fesh, QP and Ulam-def. The original source was read at lines 398–473. Its complementary inverse and reconstruction map agree with the block equations above when its operator is $H - E_0 + s$ and its projection is the indicated electric projection. All invertibility and error estimates used here are proved above for the actual lattice operators. No perturbative smallness hypothesis or novelty of Schur inversion is asserted.

The diagram retains the exact maps in PK63–PK64 and the three contributions in PK71. Its numerical endpoints illustrate the exact radicals PK77; the displayed 33 is the odd cutoff eigenvalue in PK78. Reproducible figure source.

9.9. The actual first and second interacting packet moments

Retain the original open cube, all $N = N_L$ links and $M = M_L$ faces, the gauge projection (PK26), and

$$H = \kappa H_0 + 2bM - b\mathcal{W}, \quad H_0 = \sum_e E_e, \quad \mathcal{W} = \sum_p W_p.$$

The original packet, the complete complement, and a proved error

$H = \kappa H_0 + 2bM - bW \mid F(s) = \langle \eta, (H - E_0 + s)^{-1} \eta \rangle \mid$ no packet rescaling

Every packet component remains

$$p = P\eta$$

First electric faces: eigenvalue 3

$$Cw = \Pi Q\eta$$

$$C^*C = mP, \quad w = m^{-1}C^*Q\eta$$

$$z = Q_2Q\eta$$

Entire remaining odd physical space

$$\eta = p + Cw + z \quad (\text{PK63-PK64})$$

Complete cutoff error: three contributions

1 Omitted operator space

$$\Gamma_\Lambda l^2 / (\beta_\Lambda(s) s^2)$$

$$l = 2bM, \quad \beta_\Lambda(s) = \kappa\Lambda - e_\Lambda + s$$

2 Omitted packet and its cross terms

$$(2\sqrt{\Gamma_\Lambda \tau_\Lambda} + \tau_\Lambda)/s$$

$$\tau_\Lambda \leq 4e^{-t\Lambda} Z_t^{N_\Lambda}$$

3 Actual vacuum-energy uncertainty

$$\Gamma(u_\Lambda - e_\Lambda)/s^2$$

$$e_\Lambda \leq E_0 \leq u_\Lambda$$

$$\text{Sum bounds } |F(s) - f_\Lambda(s)| \quad (\text{PK71})$$

Original box $L = 1$, $a = 1$, $g = 2$, electric cutoff $\Lambda = 3$

$$e_\Lambda \leq E_0 \leq u_\Lambda$$



The full cutoff has dimension 37. The interval bounds the actual vacuum; 33 is the odd cutoff eigenvalue, not an asserted full-spectrum value.

Proof: PK63-PK78, especially (PK69), (PK71), (PK77) and (PK78). Displayed decimals illustrate the exact radical endpoints. No continuum mass is plotted.

Figure 3: Every packet component, all three error terms, and the actual smallest-box vacuum interval

All kernels below have the original, un-divided packet norms. Put $D_{\mathbf{t}} = \sum_e \partial_{t_e}$ and $D_{\mathbf{s}} = \sum_e \partial_{s_e}$. The new two-face kernel is

$$C_{pq}(h, k; \mathbf{t}, \mathbf{s}) = \langle \Phi_{h, \mathbf{t}}, W_p W_q \Phi_{k, \mathbf{s}} \rangle. \quad (\text{PK79})$$

Here the sum over p, q is ordered and includes $p = q$, adjacent faces, disjoint faces and every boundary face.

For completeness, (PK79) has the following exact link contraction. Treat p and q as two labelled occurrences $A = 1, 2$, even when their underlying face is the same. Write their four ordered edges as $(e_{A, \ell}^{\epsilon_{A, \ell}})_{\ell=1}^4$. Introduce indices $i_{A, \ell} \in \{1, 2\}$ with $i_{A, 5} = i_{A, 1}$. For a link e , let O_e be its list of occurrences (A, ℓ) , in lexicographic order. With the original $x_e = (q_0 \cdot h)_e$ and $y_e = (r_0 \cdot k)_e$, define

$$S_e^{O_e}(x_e, y_e; \mathbf{i}) = \int_{SU(2)} p_{t_e}(U x_e^{-1}) p_{s_e}(U y_e^{-1}) \prod_{(A, \ell) \in O_e} (U^{\epsilon_{A, \ell}})_{i_{A, \ell}, i_{A, \ell+1}} dU, \quad (\text{PK80})$$

$$C_{pq} = \int dq_0 dr_0 \sum_{\{i_{A, \ell}=1, 2\}} \prod_e S_e^{O_e}(x_e, y_e; \mathbf{i}).$$

The empty occurrence list gives $S_e^{\emptyset} = Z_e$. Each nonempty list has one or two slots. Shared links are integrated once with both slots; two independent link integrals would give a different answer. Formula (PK80) follows by expanding the two original cyclic traces into their matrix entries and then integrating each link. The scalar entries commute, while their index contractions preserve the original matrix order.

There is also a fully specified representation expansion for every local integral. For an occurrence with matrix indices (a_o, b_o) , put

$$(\rho_o, \alpha_o, \beta_o) = \begin{cases} (1/2, a_o, b_o), & \epsilon_o = 1, \\ (1/2, b_o, a_o), & \epsilon_o = -1. \end{cases}$$

Using exactly the matrices and Haar polynomial of (PK34)–(PK35),

$$S_e^{O_e} = \sum_{j, k} d_j d_k e^{-t_e j(j+1) - s_e k(k+1)} \sum_{m, n, p', q'} D^j(x_e^{-1})_{nm} D^k(y_e^{-1})_{q'p'} Q_{(m, p', \alpha_1, \dots), (n, q', \beta_1, \dots)}^{j, k, \rho_1, \dots, \rho_{|O_e|}}. \quad (\text{PK81})$$

The two minus slots, when present, each retain their own transposition. The remaining vertex integrals use (PK35) with the actual incident slots. Polynomial growth of all finite derivatives and the positive heat parameters prove absolute convergence. Thus (PK80) is an actual convergent calculation on the original graph.

Define $\mathcal{L}(h, k) = \langle H \Phi_h, H \Phi_k \rangle$, keeping the two heat-parameter lists independent until after differentiation. Direct expansion gives

$$\begin{aligned} \mathcal{L} &= \kappa^2 D_{\mathbf{t}} D_{\mathbf{s}} G - 2\kappa b M(D_{\mathbf{t}} + D_{\mathbf{s}})G \\ &\quad + \kappa b(D_{\mathbf{t}} + D_{\mathbf{s}}) \sum_p B_p + 4b^2 M^2 G - 4b^2 M \sum_p B_p + b^2 \sum_{p, q} C_{pq}. \end{aligned} \quad (\text{PK82})$$

Indeed $H_0 \Phi_{h, \mathbf{t}} = -D_{\mathbf{t}} \Phi_{h, \mathbf{t}}$. The electric–magnetic terms therefore have the displayed positive derivative sign. Moving H_0 to the opposite packet by self-adjointness does not move it through the multiplication operator W_p . This accounts for both derivatives of B_p , including the commutator contribution that a commuting substitution would lose. Smooth packets belong to all operator domains, so this is also the H^2 kernel.

Apply the four-term reflection operation in (PK39) to $G, K, \mathcal{L}$ and then set all $t_e = s_e = t$. The first two moments of the actual spectral measure (PK44) are

$$\begin{aligned} \int_0^\infty \lambda d\nu(\lambda) &= K^-(h, h) - E_0\Gamma, \\ \int_0^\infty \lambda^2 d\nu(\lambda) &= \mathcal{L}^-(h, h) - 2E_0K^-(h, h) + E_0^2\Gamma. \end{aligned} \quad (\text{PK83})$$

The total derivatives commute with reflection's edge permutation. Equations (PK79)–(PK83) retain the whole magnetic square, both mixed terms and the actual vacuum shift. We next estimate these actual moments by locating their spectral weight.

9.10. A vacuum bound with the original couplings

Write a link as $U = x_0I + 2\sum_{a=1}^3 x_a T_a$, where $\sum_{a=0}^3 x_a^2 = 1$. For a trial parameter $\zeta > 0$ define

$$v_\zeta(U) = \exp\left(\zeta \sum_e \text{tr } U_e\right), \quad Z_\zeta = \frac{2}{\pi} \int_{-1}^1 e^{4\zeta x} \sqrt{1-x^2} dx, \quad m_\zeta = \frac{2}{\pi Z_\zeta} \int_{-1}^1 x e^{4\zeta x} \sqrt{1-x^2} dx. \quad (\text{PK84})$$

These formulas use the Haar probability measure on each original $SU(2)$. The marginal follows from the spherical area element of the unit S^3 : the slice at $x_0 = x$ has density proportional to $\sqrt{1-x^2}$, whose integral is $\pi/2$. The full squared norm is Z_ζ^N . Reflection permutes the links and inverts some of them; trace is unchanged, so v_ζ is even. Its gauge invariance is not assumed. The unique ground vector on the whole link space is gauge invariant by (PK37); therefore a trial on that whole space bounds the same physical ground energy.

For one link,

$$\sum_a (X_a \text{tr } U)^2 = 1 - x_0^2, \quad E(\text{tr } U) = \frac{3}{4} \text{tr } U.$$

Integration of the derivative of $(1-x^2)^{3/2}e^{4\zeta x}$, whose endpoints vanish, gives

$$4\zeta \mathbb{E}_\zeta(1-x_0^2) = 3m_\zeta, \quad \frac{\langle v_\zeta, H_0 v_\zeta \rangle}{Z_\zeta^N} = \frac{3N\zeta m_\zeta}{4}. \quad (\text{PK85})$$

The marginal is even before tilting; pairing x with $-x$ shows $0 < m_\zeta < 1$. For its other useful bound set $y = 1-x$. Its density is proportional to $y^{1/2}e^{-4\zeta y}\sqrt{2-y}\mathbf{1}_{[0,2]}(y)$. Start with the gamma probability density proportional to $y^{1/2}e^{-4\zeta y}$ on $[0, \infty)$. Its mean is $3/(8\zeta)$, as one integration by parts shows. The extra factor $g(y) = \sqrt{2-y}\mathbf{1}_{[0,2]}(y)$ is nonincreasing. For independent copies Y, Y' ,

$$2 \text{Cov}(Y, g(Y)) = \mathbb{E}[(Y - Y')(g(Y) - g(Y'))] \leq 0.$$

Dividing by $\mathbb{E}g(Y) > 0$ proves

$$0 < 1 - m_\zeta \leq \frac{3}{8\zeta}. \quad (\text{PK86})$$

The tilted link distribution is central, so $\mathbb{E}_\zeta U = \mathbb{E}_\zeta U^{-1} = m_\zeta I$. Four distinct links occur in every face. Independence in the squared trial density gives $\mathbb{E}_\zeta W_p = 2m_\zeta^4$, including every boundary face. Consequently

$$\begin{aligned} \frac{\langle v_\zeta, H v_\zeta \rangle}{Z_\zeta^N} &= \frac{3\kappa N\zeta m_\zeta}{4} + 2bM(1 - m_\zeta^4) \\ &\leq \frac{3\kappa N\zeta}{4} + \frac{3bM}{\zeta}. \end{aligned} \quad (\text{PK87})$$

The inequality uses $1-m^4 = (1-m)(1+m+m^2+m^3) \leq 4(1-m)$. At the specific value $\zeta = 2\sqrt{bM/(\kappa N)}$, both upper-bound terms are equal. Combining with the constant trial proves the unconditional bound

$$0 \leq E_0 \leq U_{\text{vac}} := \min\{2bM, 3\sqrt{\kappa bNM}\}. \quad (\text{PK88})$$

The same upper bound for L at least two appears in the retained spatial-continuum Section 12, equations (93)–(99). PK101 gives the exact trial-parameter and scalar-factor comparison. Its rederivation here also covers $L=1$; the packet-escape application below is unchanged.

This also tightens every earlier vacuum interval by intersecting its upper endpoint with U_{vac} ; no vacuum eigenvalue is replaced by this bound.

9.11. The small-potential region and a spectral projection

Select the original 12-faces based at

$$(-L + 2u, -L + 2v, -L + w), \quad 0 \leq u, v \leq L - 1, \quad 0 \leq w \leq 2L. \quad (\text{PK89})$$

They use pairwise disjoint links: distinct layers use different links, and at a fixed layer the length-one squares have base coordinates separated by two. Their number is $n_L = L^2(2L + 1) = M/12$. Under product Haar measure their holonomies are independent Haar elements, since each face product contains four independent Haar links. For one such face put $D = 2 - \text{tr } U$. For $s > 0$,

$$\begin{aligned} \mathbb{E}e^{-sD} &= \frac{2}{\pi} \int_0^2 e^{-2sy} \sqrt{y(2-y)} dy \\ &\leq \frac{2\sqrt{2}}{\pi} \int_0^\infty e^{-2sy} y^{1/2} dy = \frac{1}{2\sqrt{\pi} s^{3/2}}. \end{aligned} \quad (\text{PK90})$$

All $2 - W_p$ are nonnegative. For the complete potential $V = b \sum_p (2 - W_p)$, any $\delta > 0$ therefore satisfies

$$\begin{aligned} \text{Haar}\{V \leq bM\delta\} &\leq \Pr\left\{ \sum_{\text{selected } p} D_p \leq 12n_L\delta \right\} \\ &\leq \min\left\{ 1, \left[8\sqrt{\frac{2}{\pi}} e^{3/2} \delta^{3/2} \right]^{n_L} \right\}. \end{aligned} \quad (\text{PK91})$$

To obtain the second line multiply the indicator by $\exp(s(12n_L\delta - \sum D_p))$, use independence and (PK90), and take $s = 1/(8\delta)$. This estimate uses a subset only to bound the complete nonnegative potential; the Hamiltonian and all its faces are unchanged.

Let $P_E = \mathbf{1}_{[0,E]}(H)$, let $f \in L^2(Q_L)$, and let $q > 0$. Set $v = P_E f$ and $\chi = \mathbf{1}_{\{V \leq q\}}$. Since $H \geq V \geq 0$, $\|(1 - \chi)v\|^2 \leq E\|v\|^2/q$. Orthogonality of the spectral projection and Cauchy–Schwarz, with both regions retained, give

$$\begin{aligned} \|v\|^2 &= \langle f, v \rangle \\ &\leq (\|\chi f\| + \sqrt{E/q} \|f\|) \|v\|, \\ \|P_E f\|^2 &\leq (\|\mathbf{1}_{\{V \leq q\}} f\| + \sqrt{E/q} \|f\|)^2. \end{aligned} \quad (\text{PK92})$$

Absolute values in the middle line justify it also for complex f . If $v = 0$ the last inequality is immediate; otherwise divide by $\|v\|$. A bounded-energy spectral vector is in the form domain, so the quadratic-form comparison used here is justified without a pointwise eigenfunction assumption.

9.12. The fixed-heat packet actually escapes to high energies

Use the exact path (PK43), its threshold (PK43b), and the raw lower bound γ_j in (PK72). Write $N_j, M_j^{\text{face}}, b_j$ exactly as there, and set

$$\begin{aligned} n_j &= j^8(2j^4 + 1) = M_j^{\text{face}}/12, & Z_{t_*} &= p_{t_*}(I), & C_H &= 8\sqrt{2/\pi} e^{3/2}, \\ \varepsilon_j &= \kappa_* j, & q_j &= \frac{b_j M_j^{\text{face}}}{\sqrt{j}}, \\ A_j &= \frac{Z_{t_*}^{2N_j}}{\gamma_j} \min\{1, (C_H j^{-3/4})^{n_j}\}, \\ B_j &= \frac{3\kappa_*}{100\sqrt{j}} \sqrt{\frac{N_j}{M_j^{\text{face}}}} + \frac{\kappa_*^2}{10000 M_j^{\text{face}} \sqrt{j}}, & R_j &= \min\{1, (\sqrt{A_j} + \sqrt{B_j})^2\}. \end{aligned} \tag{PK93}$$

No spatial lattice spacing or covering degree is denoted by q_j or n_j ; these are the displayed potential threshold and selected face count.

For every U , character unitarity gives $|p_{t_*}(U)| \leq p_{t_*}(I) = Z_{t_*}$. The heat kernel is positive. Each averaged packet is therefore between zero and $Z_{t_*}^{N_j}$, and

$$|\eta_j(U)| \leq Z_{t_*}^{N_j}, \quad \|\mathbf{1}_{\{V_j \leq q_j\}} \eta_j\|^2 \leq Z_{t_*}^{2N_j} \min\{1, (C_H j^{-3/4})^{n_j}\}. \tag{PK94}$$

The difference of two numbers in that interval has absolute value at most its length; no extra factor two is required. Apply (PK92) with $E = E_{0,j} + \varepsilon_j$. Reflection oddness removes the ground vector exactly. Using (PK88), $\Gamma_j \geq \gamma_j$, and the original $\kappa_j = \kappa_*$, $b_j = 10000j^2/\kappa_*$ gives

$$0 \leq \nu_j((0, \kappa_* j]) \leq \Gamma_j R_j, \quad R_j \longrightarrow 0. \tag{PK95}$$

Indeed $(U_{\text{vac}} + \kappa_* j)/q_j \leq B_j$; direct substitution gives both terms of (PK93), with no change of coupling. Moreover $N_j/M_j^{\text{face}} = 1 + 1/(2j^4)$, so $B_j \rightarrow 0$. To prove $A_j \rightarrow 0$, retain the explicit positive constant c_γ from the second line of (PK49), so $\gamma_j \geq c_\gamma j^{-15}$. For all sufficiently large j ,

$$\frac{\log A_j}{n_j} \leq 2 \left(12 + \frac{6}{j^4} \right) \log Z_{t_*} + \log C_H - \frac{3}{4} \log j + \frac{15 \log j - \log c_\gamma}{n_j} \longrightarrow -\infty. \tag{PK96}$$

Here $N_j/n_j = 12 + 6/j^4$ exactly. This proves (PK95) with the full volume counts. Ratios in this comparison do not replace η_j or its actual raw norm by a different object.

The consequences for the moments (PK83) and the actual resolvent are now definite:

$$\begin{aligned} K_j^- - E_{0,j} \Gamma_j &\geq \kappa_* j \Gamma_j (1 - R_j), \\ \mathcal{L}_j^- - 2E_{0,j} K_j^- + E_{0,j}^2 \Gamma_j &\geq (\kappa_* j)^2 \Gamma_j (1 - R_j), \\ 0 \leq \frac{s_j F_{h_j, t_*}(s_j)}{\Gamma_j} &\leq R_j + \frac{1}{j^2 + 1}, \quad s_j = \kappa_* / j, \\ 0 \leq \frac{s_j f_j}{\Gamma_j} &\leq R_j + \frac{1}{j^2 + 1} + \frac{1}{j^2} \longrightarrow 0. \end{aligned} \tag{PK97}$$

The moment inequalities integrate over $(\kappa_* j, \infty)$. For the resolvent split at $\kappa_* j$; on the low interval $s_j/(\lambda + s_j) \leq 1$, and on its complement it is at most $1/(j^2 + 1)$. The last line uses (PK73). Thus the lower expression inside the maximum in (PK75) is negative for all sufficiently large j . Its inability to certify positive mass is explained here by the actual packet, not by cutoff error.

Equation (PK95) proves that the fraction of this packet's spectral mass in every fixed bounded excitation interval tends to zero; even the expanding interval $(0, \kappa_* j]$ loses that fraction. It does not assert that its un-divided raw mass tends to zero: Γ_j has not been replaced by its lower bound as an asymptotic equality. It also does not bound the spectral measures of other states or decide the intended continuum Yang–Mills problem. It rules out the fixed- t_* packet path as a source of a nonvanishing fraction of low-energy weight in the present construction.

9.13. An exact interacting evolution of the escaping packet

The next operation can be constructed on the same Hilbert space and classical labels. For each original finite lattice and every $u > 0$ put $A = H - E_0$ and

$$\xi_{h,t}(u) = e^{-uA} \eta_{h,t}, \quad Q_{h,t}(u) = \|\xi_{h,t}(u)\|^2 = \int_0^\infty e^{-2u\lambda} d\nu(\lambda). \quad (\text{PK98})$$

The operator commutes with the original gauge action and reflection, so this map stays physical and odd. It depends on the complete interacting Hamiltonian. Every spectral multiplier is positive, so $\xi(u) \neq 0$ whenever $\eta \neq 0$. It is smooth by the elliptic spectral domain characterization on the compact link manifold. Its Gram kernel, with both original labels retained, is $\langle \eta_h, e^{-2u(H-E_0)} \eta_k \rangle$. The product formula (PK45), with its original potential, evaluates this kernel; it is not the free heat operation (PK28).

Let $Q_n(u) = \int \lambda^n e^{-2u\lambda} d\nu(\lambda)$. Differentiation is justified by the exponential bound on $\lambda^n e^{-u\lambda}$, and gives the exact calculation

$$\begin{aligned} Q'_0(u) &= -2Q_1(u), & Q'_1(u) &= -2Q_2(u), \\ \frac{d}{du} \frac{Q_1(u)}{Q_0(u)} &= -2 \frac{Q_2(u)Q_0(u) - Q_1(u)^2}{Q_0(u)^2} \leq 0. \end{aligned} \quad (\text{PK99})$$

The numerator is one half of the double integral of $(\lambda - \mu)^2 e^{-2u(\lambda+\mu)}$ against $d\nu(\lambda)d\nu(\mu)$, so the sign is proved with every weight retained.

This operation has an exactly identifiable limit and defect. The compact-resolvent spectrum is discrete; η has zero vacuum component. Let $\alpha = \min\{\lambda > 0 : \mathbf{1}_{\{\lambda\}}(A)\eta \neq 0\}$ and $c_\alpha = \|\mathbf{1}_{\{\alpha\}}(A)\eta\|^2 > 0$. Then

$$e^{2u\alpha} Q_0(u) \longrightarrow c_\alpha, \quad \frac{Q_1(u)}{Q_0(u)} \longrightarrow \alpha > 0, \quad \mathbf{1}_I(A)\xi(u) = e^{-uA} \mathbf{1}_I(A)\eta. \quad (\text{PK100})$$

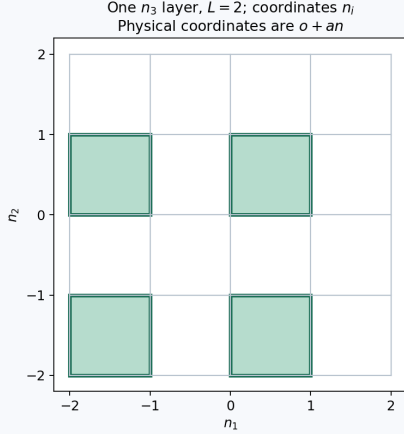
To prove the limits, split off the atom at α . For $u \geq 1$, each remaining summand of the first expression is bounded by its original squared coefficient and tends to zero. For the energy numerator use $\lambda e^{-2(\lambda-\alpha)} \leq C_\alpha$ on $[\alpha, \infty)$ and the same summable coefficients. Dominated convergence proves both limits. The last identity follows from the commuting spectral multipliers and shows exactly that no previously absent spectral support is created.

We have therefore constructed and calculated the interacting evolution, rather than assuming that it supplies the missing state. On the original path take, for example, $u_j = j/\kappa_*$; (PK98)–(PK100) give its actual raw norms, moments and support maps. The next unresolved calculation is the scale of the actual supported bottom α_j and the corresponding weights for this new map, using the full odd finite operator and the vacuum intervals. The fixed- t_* escape theorem requires this new analysis; it does not answer it or close the research programme.

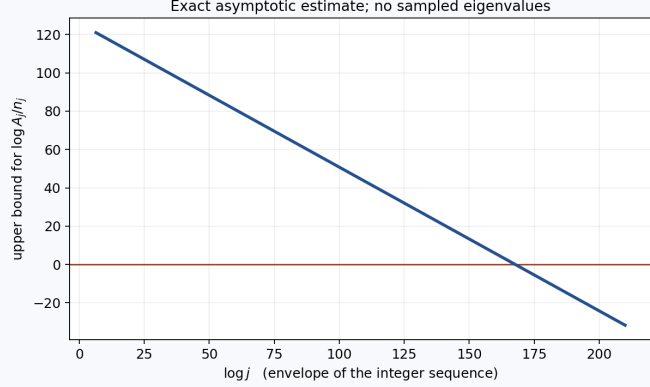
The coordinate diagram shows one exact lattice layer and its physical-coordinate map. The curve is the PK96 bound, using the specified PK49 constant and the PK70 upper bound for the heat value. It is an estimate, not sampled eigenvalues. Complete proofs are PK79–PK100. Reproducible figure source.

The fixed-heat packet leaves low-energy spectral intervals

$$H_j = \kappa \cdot H_{0,j} + 2b_j M_j^{\text{face}} - b_j \mathcal{W}_j \quad | \quad b_j = 10000j^2/\kappa \quad | \quad \text{original packet norm } \Gamma_j \text{ retained}$$



Green faces have no common links.
All layers give $n_L = L^2(2L + 1) = M_L/12$.
The full Hamiltonian keeps every face.



$t_* = 1$, $r = 1/10$, $R = 1$; $Z_i \leq (1 + e^{-1/4})/(1 - e^{-1/4})^3$
The complete expression (PK96) is plotted,
using the retained constant from (PK49).

$$\nu_j((0, \kappa * j]) \leq \Gamma_j R_j, \quad R_j \rightarrow 0 \quad (\text{PK93-PK96})$$

The actual packet loses its fraction of mass below an expanding energy threshold.
This does not assert decay of its raw mass or exclude different interacting states.

Proof: PERIOD_COUPLING_AND_PHYSICAL_KERNELS.md, PK79-PK100. Graph and estimates derived there.
Packet conventions: Bahr-Thiemann, arXiv:0709.4636v1, as compared in PK27-PK36.

Figure 4: Original edge-disjoint faces and the full packet escape estimate

9.14. Exact receiving maps from the retained weak-coupling calculation

The retained programme already proves the full fixed-open-box weak-coupling comparison. We use its exact objects, rather than deriving a second boundary model. The complete providers are:

- Finite-box proof, Sections 1–7: original Hamiltonian, maximal-tree/Haar map, full kinetic tensor, all open-boundary modes and actual eigenvalue convergence.
- Actual vacuum and spectral maps, Sections 1–4: compactness, strong vacuum convergence and norm convergence of finite spectral projections, with the density and dilation retained.
- Cubic symmetry, Sections 1–3: original signed graph automorphisms, the three lowest cochains, and their exact nonlinear-chart and limiting intertwiners.

All three providers have been read at these loci. Their mathematical proofs are included in this repository. Historical references to a literature audit do not mean that an external PDF was read again here. Their fixed L, a quantifiers remain in every use below.

There is also a provenance correction to (PK84)–(PK88). The same vacuum upper bound was proved in spatial-continuum Section 12, equations (93)–(99). For its parameter t_{WC} , one-link integral Z_t^{WC} and product trial $F_{t_{\text{WC}}}$, the exact comparison is

$$t_{\text{WC}} = 2\zeta, \quad Z_\zeta = Z_{2\zeta}^{\text{WC}}, \quad m_\zeta = q_{2\zeta}^{\text{WC}}, \quad v_\zeta = (Z_{2\zeta}^{\text{WC}})^{N/2} F_{2\zeta}, \quad 3\sqrt{\kappa b N M} = \frac{3\sqrt{NM}}{a}. \quad (\text{PK101})$$

The last equality follows from the original $\kappa b = 1/a^2$. Thus the bound was rederived, not newly obtained in this programme. The current proof also covers $L = 1$; the earlier stated domain was $L \geq 2$. Its use in the new packet-escape argument is unchanged.

Fix now $L \geq 2, a > 0$ and let $g \downarrow 0$, retaining

$$H_g = \frac{2g^2}{a} H_0 + \frac{1}{2g^2 a} \sum_p (2 - W_p), \quad A_g = H_g - E_{0,g}, \quad \psi_g > 0, \quad \|\psi_g\| = 1. \quad (\text{PK102})$$

For clarity, the complete original coordinate maps used below are specified here. Root the tree at $(-L, -L, -L)$, with parent obtained by decreasing the first coordinate in the order 1, 2, 3 that exceeds $-L$. If t_v is its ordered tree holonomy, set $Z_c = t_{s(c)} U_c t_{t(c)}^{-1}$ for each chord. The inverse retains the tree links and uses $U_c = t_s^{-1} Z_c t_t$. Independent left/right Haar translations on the chords preserve the entire product measure. Based gauge averaging removes the tree variables; the residual action is one simultaneous adjoint rotation on all chord variables.

There are $d_L = 2(2L)^3 + 3(2L)^2$ chords. On real cochains let $(Tx)_c = p_s(x) + x_c - p_t(x)$, where p_v is the signed additive tree-path integral, let j_C insert zero tree entries, and retain

$$G = TT^*, \quad C = d_1 j_C, \quad O^T G^{1/2} C^* C G^{1/2} O = \text{diag}(\sigma_1^2, \dots, \sigma_{d_L}^2), \quad z^\alpha = O^T G^{-1/2} x^\alpha. \quad (\text{PK103})$$

These are maps on the original counting inner products. The source proves $Tj_C = I$, $\ker T = \text{im } d_0 = \ker d_1$ and $d_1 T^* G^{-1} = C$; hence the positive matrices and all inverses in (PK103) exist.

In the unique logarithm chart $Z_c = \exp(y_c^\alpha T_\alpha)$, $|y_c| < 2\pi$, set $y = gx$. The exact Hilbert map, including its zero extension, is

$$\mathcal{J}(y) = (16\pi^2)^{-d_L} \prod_c \left[\frac{\sin(|y_c|/2)}{|y_c|/2} \right]^2, \quad \Omega_g = \{x : |x_c| < 2\pi/g \text{ for every } c\}, \quad (\text{PK104})$$

$$(\mathcal{B}_g F)(x) = \mathbf{1}_{\Omega_g}(x) g^{3d_L/2} \mathcal{J}(gx)^{1/2} F(\exp(gx)).$$

It is an isometry onto the subspace supported in Ω_g , not onto all Euclidean L^2 . Its adjoint has the reciprocal density and dilation on that subspace. The complete transformed form is

$$\frac{2}{a} \int A^{ij}(gx) \left(\partial_i f - \frac{g}{2} (\partial_i \log \mathcal{J})(gx) f \right)^* \left(\partial_j f - \frac{g}{2} (\partial_j \log \mathcal{J})(gx) f \right) dx + \int \frac{W(gx)}{2g^2 a} |f|^2 dx. \quad (\text{PK105})$$

Here $A(0) = G \otimes I_3$ and $W(y) = \frac{1}{4} \sum_\alpha \|C y^\alpha\|^2 + O_L(|y|^3)$. The full $A, \mathcal{J}, W$, including their remainders, remain in (PK105). The retained proof localizes this form and proves convergence of actual eigenvalues, vacua and finite spectral projections at fixed L, a .

9.15. The bottom of the actual reflection-odd sector

Let $m = 2L$, $\lambda_q = 4 \sin^2(\pi q / [2(m+1)])$. On the one-dimensional vertex and edge counting spaces use

$$\begin{aligned} v_0(l) &= (m+1)^{-1/2}, \\ v_q(l) &= \sqrt{2/(m+1)} \cos \frac{\pi q(l+1/2)}{m+1}, \quad 1 \leq q \leq m, \\ w_q(l) &= -\sqrt{2/(m+1)} \sin \frac{\pi q(l+1)}{m+1}, \quad 1 \leq q \leq m. \end{aligned} \quad (\text{PK106})$$

The index is $l = n_i + L$; original physical coordinates are still $o + an$ and edge midpoints $o + a(n + e_i/2)$. The difference map satisfies $Dv_q = \sqrt{\lambda_q} w_q$. At a frequency triple q , the transverse edge multiplicity is $\#\{i : q_i > 0\} - 1$ when at least two components are positive, with squared frequency $\sum_i \lambda_{q_i}$. This follows from the exact curl identity $\sum_{i < h} |s_i x_h - s_h x_i|^2 = |s|^2 |x|^2 - |s \cdot x|^2$ on the complement of the gradient $s_i = \sqrt{\lambda_{q_i}}$. These tensor bases exhaust all original cochains.

For $i < j$ let b_i^{ij} be the edge tensor with w_1 in direction i , v_1 in direction j , and v_0 in the other direction; define b_j^{ij} with i, j interchanged. The three lowest transverse cochains are

$$V_{ij} = (b_i^{ij} - b_j^{ij})/\sqrt{2}, \quad V_{ji} = -V_{ij}, \quad \sigma_* = \sqrt{8} \sin \frac{\pi}{4L+2}, \quad \delta_* = \frac{2\sigma_*}{a}. \quad (\text{PK107})$$

Choose the first three columns of O in (PK103) to be $G^{-1/2}TV_{12}, G^{-1/2}TV_{13}, G^{-1/2}TV_{23}$; complete them by any fixed orthonormal spectral basis. This specifies the three coordinates z_1, z_2, z_3 used below without a basis ambiguity.

A graph reflection in coordinate r maps a vertex index to $m-l$ and an edge index to $m-1-l$, reversing that directed edge. From (PK106), $v_1(m-l) = -v_1(l)$ and $w_1(m-1-l) = w_1(l)$. Including the directed-edge sign gives

$$V_{ij} \mapsto \eta_i \eta_j V_{ij} \quad \text{under } \text{diag}(\eta_1, \eta_2, \eta_3). \quad (\text{PK108})$$

Coordinate permutations send V_{ij} to $V_{\pi(i)\pi(j)}$ with its displayed antisymmetric sign. Thus reflection $\mathcal{R}$ in coordinate one acts on (z_1, z_2, z_3) by $(-z_1, -z_2, z_3)$. These are spatial mode signs; every z_μ still has three colour components.

The comparison oscillator retains all modes:

$$H_{\text{osc}} = \sum_{\nu=1}^{d_L} \left[-\frac{2}{a} \Delta_{z_\nu} + \frac{\sigma_\nu^2}{8a} |z_\nu|^2 \right], \quad \mu_0 = \frac{3}{2a} \sum_{\nu=1}^{d_L} \sigma_\nu. \quad (\text{PK109})$$

In the original x coordinates its vacuum is exactly

$$\Phi_0(x) = (\det G)^{-3/4} \prod_{\nu=1}^{d_L} \left(\frac{\sigma_\nu}{4\pi} \right)^{3/4} \exp \left[-\frac{1}{8} \sum_{\nu=1}^{d_L} \sigma_\nu |z_\nu|^2 \right]. \quad (\text{PK109a})$$

Indeed $x = G^{1/2}Oz$ in each colour component has absolute Jacobian $(\det G)^{3/2}$; its square-root factor is retained in this wavefunction. Its physical vectors are simultaneous-colour-rotation invariants. One quantum transforms as an adjoint vector and has no invariant component. The first physical cluster therefore has excitation δ_* , with six basis vectors

$$D_\mu = \frac{\sigma_*}{2\sqrt{6}} (|z_\mu|^2 - 6/\sigma_*) \Phi_0, \quad O_{\mu\nu} = \frac{\sigma_*}{2\sqrt{3}} (z_\mu \cdot z_\nu) \Phi_0, \quad \mu < \nu. \quad (\text{PK110})$$

Here Φ_0 is the exact product vacuum after the full constant Jacobian in (PK103); each component of the three selected modes has variance $2/\sigma_*$. These factors make the six vectors orthonormal. Under $\mathcal{R}$, the odd vectors are exactly O_{13}, O_{23} . The other four are even.

Let $E_-(g)$ be the lowest eigenvalue of the actual H_g on its physical reflection-odd subspace. Then

$$\lim_{g \downarrow 0} (E_-(g) - E_{0,g}) = \delta_*. \quad (\text{PK111})$$

Here is the passage from the comparison vectors to that exact sector. Take an interval around $\mu_0 + \delta_*$ with endpoints outside the oscillator spectrum, containing no other level. The retained spectral projection proof gives an actual six-dimensional cluster for sufficiently small positive g , and its chart images converge in norm to the span of (PK110). The original graph reflections commute with H_g . Their exact chart maps are $\mathcal{B}_g \mathcal{S}_R \mathcal{B}_g^*$, not a substituted linear action. The rooted-tree map after a reflection is the original finite word map β_R , with $g^{-1} \log \beta_R(\exp(gx)) = D\beta_R(I)x + O_L(g|x|^2)$. The density maps (PK104) and that expansion give the strong limiting action (PK108) on polynomial Gaussian vectors, as proved in the cubic source. Uniform boundedness of the unitaries and approximation by those vectors give convergence on the six-dimensional cluster.

For the group of all eight coordinate sign flips, each character projection is its exact finite average $8^{-1} \sum_R \chi(R) \mathcal{S}_R$. Its trace on the cluster is an integer and converges to the corresponding trace in (PK110); it therefore equals that limiting integer for all sufficiently small g . The trivial sign character has dimension three (the D_μ); the other three occurring characters each have dimension one (the $O_{\mu\nu}$). Signed coordinate permutations carry these three lines transitively to one another. Since H_g commutes with those permutations, its eigenvalue on these three one-dimensional lines is the same number, denoted $E_T(g)$.

The actual vacuum is even. Eigenvalue convergence excludes all levels between it and this first cluster. Within the cluster the trivial sign character is even under $\mathcal{R}$, while two of the three nontrivial character lines are odd. Consequently

$$E_-(g) = E_T(g), \quad \dim \ker(H_g - E_-(g))|_{\text{physical, odd}} = 2 \quad (\text{PK112})$$

for all sufficiently small $g > 0$ at fixed L, a . Possible equality with an even eigenvalue does not alter that dimension. The cluster limits prove (PK111). This identifies the actual odd bottom; it does not yet identify its overlap with a chosen state.

9.16. A bounded odd observable on the actual interacting vacuum

Construct that overlap explicitly. Let $\rho = \pi/4$, let $b_0(s) = e^{-1/s}$ for $s > 0$ and $b_0(s) = 0$ for $s \leq 0$, and set

$$\chi(y) = \frac{b_0(4\rho^2 - |y|^2)}{b_0(4\rho^2 - |y|^2) + b_0(|y|^2 - \rho^2)}. \quad (\text{PK113})$$

The denominator never vanishes: its two arguments cannot both be nonpositive. The function is smooth, equals one on $|y| \leq \rho$, and vanishes on $|y| \geq 2\rho$. Its support is strictly inside every single-chord logarithm boundary. It is invariant under simultaneous colour rotation.

Write $\tilde{z}^\alpha(y) = O^T G^{-1/2} y^\alpha$. On the logarithm chart define, with every scale shown,

$$f_g(U) = \sin\left(\frac{\sigma_*}{2g^2} \chi(y(U)) \tilde{z}_1(y(U)) \cdot \tilde{z}_3(y(U))\right), \quad (\text{PK114})$$

and set it equal to zero outside the chart. The support property makes this extension smooth, including near any chord $-I$. All tree and chord maps are the ones specified above. Residual simultaneous conjugation rotates both three-vectors by the same orthogonal matrix; their dot product is fixed. Thus f_g is physical. It is bounded in absolute value by one.

Let the reflection act on functions by the original graph unitary, and put

$$F_g = \frac{1}{2}(f_g - \mathcal{S}_{\mathcal{R}} f_g), \quad v_g = F_g \psi_g, \quad \Gamma_g^{\text{odd}} = \|v_g\|^2. \quad (\text{PK115})$$

This is the exact nonlinear antisymmetrization, including the reflected root and chord words. It proves $|F_g| \leq 1$, $\mathcal{S}_{\mathcal{R}} F_g = -F_g$ and $\langle \psi_g, v_g \rangle = 0$. Every vector is smooth and physical on the original compact link space. No trial vacuum is substituted for ψ_g .

At fixed $x, y = gx$, the exact scale in (PK114) tends to $(\sigma_*/2)z_1 \cdot z_3$. Under the exact reflected chart it tends to its negative, by (PK108). The strong actual-vacuum limit and boundedness therefore prove

$$\mathcal{B}_g v_g \longrightarrow v_0 := \sin\left(\frac{\sigma_*}{2} z_1 \cdot z_3\right) \Phi_0 \quad \text{in } L^2. \quad (\text{PK116})$$

For detail, subtract the limiting multiplier times the actual chart vacuum; its norm tends to zero by dominated convergence against $|\Phi_0|^2$ and strong convergence of that vacuum, since all multipliers are

bounded by one. The remaining term tends to zero by the same vacuum convergence. This argument also applies to the reflected multiplier, with the exact density map, and then to their half-difference.

Under $|\Phi_0|^2$, the vectors z_1, z_3 are independent three-dimensional Gaussians with component variance $2/\sigma_*$. Conditioning on z_1 , and then doing the three elementary Gaussian integrals, gives for real θ

$$\mathbb{E}e^{i\theta z_1 \cdot z_3} = (1 + 4\theta^2/\sigma_*^2)^{-3/2}. \quad (\text{PK117})$$

Taking $\theta = \sigma_*$ in the cosine in $\sin^2$ proves

$$\|v_0\|^2 = \frac{1}{2}(1 - 5^{-3/2}). \quad (\text{PK118})$$

Differentiating (PK117) at $\theta = \sigma_*/2$, with the exact coefficient in O_{13} , gives

$$\langle O_{13}, v_0 \rangle = \frac{\sigma_*}{2\sqrt{3}} \mathbb{E}[(z_1 \cdot z_3) \sin(\sigma_* z_1 \cdot z_3/2)] = \sqrt{3/32}, \quad \langle O_{23}, v_0 \rangle = 0. \quad (\text{PK119})$$

The second equality follows by integrating the independent centered vector z_2 . The other four first-cluster vectors have even parity. Combining (PK116), norm convergence of the actual cluster projection and (PK112) proves the actual raw limits

$$\Gamma_g^{\text{odd}} \longrightarrow \frac{1}{2}(1 - 5^{-3/2}), \quad \|\mathbf{1}_{\{E_-(g) - E_{0,g}\}}(A_g)v_g\|^2 \longrightarrow \frac{3}{32}. \quad (\text{PK120})$$

In particular the second quantity is positive for every sufficiently small $g > 0$ at this fixed box. The supported spectral bottom of v_g is exactly $E_-(g) - E_{0,g}$, rather than an unspecified larger odd eigenvalue.

9.17. Every raw spectral weight of the limiting odd observable

The first weight is part of a completely calculable measure. Define the exact orthogonal change on the two selected spatial modes by

$$q_+ = (z_1 + z_3)/\sqrt{2}, \quad q_- = (z_1 - z_3)/\sqrt{2}, \quad z_1 \cdot z_3 = (|q_+|^2 - |q_-|^2)/2. \quad (\text{PK121})$$

Its absolute Jacobian is one for each colour component; it preserves both equal-frequency kinetic and potential terms. The other modes remain in their original vacuum. For $r_{\pm} = \sigma_*|q_{\pm}|^2/4$, the two independent radial densities are $r^{k-1}e^{-r}/\Gamma(k)$ with $k = 3/2$. Thus the multiplier in (PK116) is exactly $\sin(r_+ - r_-)$.

Define $L_n^{k-1}(r)$ by the finite polynomial

$$L_n^{k-1}(r) = \sum_{\ell=0}^n \frac{(-1)^\ell (k)_n}{(k)_\ell (n-\ell)! \ell!} r^\ell, \quad (k)_n = k(k+1) \cdots (k+n-1), \quad (k)_0 = 1. \quad (\text{PK122})$$

Integration by parts in the identity $r^{k-1}e^{-r}L_n^{k-1}(r) = n!^{-1} \frac{d^n}{dr^n}(e^{-r}r^{n+k-1})$ proves orthogonality and squared norm $(k)_n/n!$. All boundary terms vanish first on the appropriate monomial integrals; the exponential controls infinity and $k > 0$ controls zero. The normalized radial polynomial is therefore $\ell_n = \sqrt{n!/(k)_n}L_n^{k-1}$. The finite coefficients satisfy $rL_n'' + (k-r)L_n' + nL_n = 0$. With $r = \sigma_*|q|^2/4$, direct differentiation of the polynomial times the original Gaussian gives

$$(H_{\text{osc}}^{(q)} - 3\sigma_*/(2a))(L_n(r)\Phi^{(q)}) = \frac{2\sigma_*}{a}[-rL_n'' - (3/2 - r)L_n']\Phi^{(q)} = \frac{2n\sigma_*}{a}L_n(r)\Phi^{(q)}.$$

For the specific vector here, exhaustion of its full norm by these orthogonal eigenvectors is proved below by equality of the sum of their squared coefficients with its directly computed norm.

Termwise integration of (PK122), or its displayed derivative identity, gives the exact coefficient

$$a_n(\beta) := \langle \ell_n, e^{i\beta r} \rangle = \sqrt{\frac{(k)_n}{n!}} \frac{(-i\beta)^n}{(1-i\beta)^{k+n}}. \quad (\text{PK123})$$

The power branch is the one continuous from $\beta = 0$. At $\beta = 1$, the coefficient of the two-mode radial state (m, n) in $\sin(r_+ - r_-)$ is

$$c_{mn} = \frac{a_m(1)a_n(-1) - a_m(-1)a_n(1)}{2i}, \quad |c_{mn}|^2 = 2^{-3-m-n} \frac{(3/2)_m(3/2)_n}{m!n!} \sin^2 \frac{(n-m)\pi}{4}. \quad (\text{PK124})$$

No products between the two radial factors are dropped. Reflection interchanges them, with a minus on both Cartesian vectors that does not affect their radii; the coefficients are accordingly antisymmetric.

For $l \geq 0$ put $w_l = \sum_{m+n=l} |c_{mn}|^2$. Vandermonde's coefficient identity and $\sin^2 x = (1 - \cos 2x)/2$ give

$$w_l = 2^{-l-4} \left[\frac{(l+1)(l+2)}{2} - \mathbf{1}_{\{l \text{ even}\}} (-1)^{l/2} \frac{(3/2)_{l/2}}{(l/2)!} \right],$$

$$\sum_{l=0}^{\infty} w_l s^l = \frac{1}{16} \left[(1-s/2)^{-3} - (1+s^2/4)^{-3/2} \right], \quad |s| < 2. \quad (\text{PK125})$$

To verify the cosine term, its generating function is $(1-is)^{-3/2}(1+is)^{-3/2} = (1+s^2)^{-3/2}$. The first term comes from $(1-s)^{-3}$. This proves the displayed coefficients, not merely their sum. The nonnegativity of every w_l also follows directly from (PK124); $w_0 = 0$, $w_1 = 3/32$, and $w_2 = 15/128$. At $s = 1$ the sum is exactly (PK118).

Consequently the complete raw comparison spectral measure and its Euclidean-time function are

$$\nu_0^{\text{odd}} = \sum_{l=1}^{\infty} w_l \delta_{2l\sigma_*/a}, \quad (\text{PK126})$$

$$\langle v_0, e^{-\tau(H_{\text{osc}} - \mu_0)} v_0 \rangle = \frac{1}{16} \left[\left(1 - \frac{1}{2}e^{-2\tau\sigma_*/a}\right)^{-3} - \left(1 + \frac{1}{4}e^{-4\tau\sigma_*/a}\right)^{-3/2} \right], \quad \tau \geq 0.$$

The actual raw measures of v_g converge weakly to this measure at fixed L, a . Indeed (PK116) gives convergence of their total masses, and each finite spectral projection converges in norm. Choosing an oscillator cutoff beyond all but a small tail of the explicit summable weights then proves tightness and convergence against every bounded continuous test function. This argument retains the raw total mass (PK118).

The interacting evolution $\tilde{v}_g(u) = e^{-uA_g} v_g$ has all the exact moment-flow identities (PK98)–(PK100), now with a supported bottom known to be $E_-(g) - E_{0,g}$ for sufficiently small g at fixed box. Its first atom is positive by (PK120). This constructs a vacuum-based state family that addresses the original packet's missing-overlap problem.

9.18. A receiving map from the full higher-carrier domain

The domain need not be abandoned when changing the quantum state map. Retain $Y = \text{Tot}(E_D)$, $D = F_4/\rho(\text{Sp}(1))$, and the exact ordered rank-24 decomposition and three selected quaternionic coordinates α, β, γ of the moment-map proof, Sections 4–7 and HC4–HC9. Their other twelve real fibre coordinates remain in the domain. Define

$$\mathfrak{a}(y) = |\alpha(y)|^2 + |\beta(y)|^2 + |\gamma(y)|^2, \quad d_{\text{vert}} \mathfrak{a}(\dot{y}) = 2 \text{Re}(\dot{\alpha}\bar{\alpha} + \dot{\beta}\bar{\beta} + \dot{\gamma}\bar{\gamma}). \quad (\text{PK127})$$

Although the displayed coordinates use a local bundle frame, each selected quaternion transforms by the prescribed unit-quaternion representation. Its norm is unchanged, so both the function and its differential glue globally. Its zero set is exactly the retained rank-12 zero subbundle of the original colour-profile map. All three selected inputs occur; no global colour frame is chosen.

In (PK114), replace the multiplier of the sine by $\mathbf{a}(y)$ times that entire multiplier, including $\sigma_*/(2g^2)$, and apply the same exact half-difference (PK115). Denote the resulting vector by

$$\mathfrak{V}_g(y) = \frac{1}{2} \left[\sin \left(\mathbf{a}(y) \frac{\sigma_*}{2g^2} \chi(y_{\text{ch}}) \tilde{z}_1(y_{\text{ch}}) \cdot \tilde{z}_3(y_{\text{ch}}) \right) - \mathcal{S}_{\mathcal{R}} \sin \left(\mathbf{a}(y) \frac{\sigma_*}{2g^2} \chi(y_{\text{ch}}) \tilde{z}_1(y_{\text{ch}}) \cdot \tilde{z}_3(y_{\text{ch}}) \right) \right] \psi_g. \quad (\text{PK128})$$

Here $y \in Y$, whereas y_{ch} is the chord-logarithm argument of the physical configuration U ; their domains are explicitly different. Reflection acts only on U , not on the parameter y . The formula extends by the same zero cutoff. It defines a global smooth map from Y to the smooth physical odd vectors at each fixed regulator. For example its first parameter derivative is the same half-difference with each sine replaced by $d\mathbf{a}$ times the entire chord multiplier times its cosine. Higher derivatives follow from that formula, with bounded configuration coefficients on each compact parameter set at fixed $g > 0$. The unit-quaternion transition law leaves $\mathbf{a}$ invariant, so no parameter derivative creates a frame inconsistency.

This map factors through the explicitly proved scalar map $\mathbf{a}$; it is not asserted to distinguish configurations with the same scalar. The original rank-24 moment map and its angular data remain available as separate proved maps. At fixed parameter y , the preceding Gaussian calculation, with $A = \mathbf{a}(y)$, gives

$$\begin{aligned} \|\mathfrak{V}_g(y)\|^2 &\longrightarrow \frac{1}{2} [1 - (1 + 4A^2)^{-3/2}], \\ \|\mathbf{1}_{\{E_-(g) - E_{0,g}\}}(A_g) \mathfrak{V}_g(y)\|^2 &\longrightarrow \frac{3A^2}{(1 + A^2)^5}. \end{aligned} \quad (\text{PK129})$$

These follow from (PK117) at $\theta = A\sigma_*/2$ and its derivative; the mode O_{23} still has zero coefficient. For every fixed $A > 0$ the second limit is strictly positive. At $A = 0$ the vector is exactly zero for every g . The selected input with $A = 1$ gives (PK118)–(PK126). For the original equal-colour input $|\alpha| = |\beta| = |\gamma| = 1$, the first-band weight instead tends to $27/100000$, with raw total mass $\frac{1}{2}(1 - 37^{-3/2})$. No division by the parameter amplitude has been introduced.

9.19. The original joint path and the remaining estimate

On the original geometric sequence $L_j = j^4$, $a_j = 1/(100j)$, the exactly computed comparison threshold is

$$\delta_{*,j} = 400\sqrt{2}j \sin \frac{\pi}{4j^4 + 2} < 100\sqrt{2}\pi j^{-3}, \quad \delta_{*,j} \sim 100\sqrt{2}\pi j^{-3}. \quad (\text{PK130})$$

The inequality is $\sin x < x$; the asymptotic follows from $\sin x/x \rightarrow 1$, with its original denominator retained. Equations (PK111) and (PK120) take $g \rightarrow 0$ after fixing L, a . They therefore do not yet give these limits along $g_j^2 = \kappa_*/(200j)$. That path keeps $\kappa_j = \kappa_*$, $b_j = 10000j^2/\kappa_*$ and the previous classical dilation $\lambda_j = j^2$; none has been replaced by a smaller coupling selected from a fixed-box existence statement.

The exact new quantities to estimate on that path are now specified:

$$\begin{aligned} \mathfrak{e}_j &= E_-(g_j; L_j, a_j) - E_{0,j} - \delta_{*,j}, \\ \mathfrak{w}_j &= \|\mathbf{1}_{\{E_-(g_j) - E_{0,j}\}}(A_j) v_{g_j}\|^2, \\ \Gamma_j^{\text{odd}} &= \|v_{g_j}\|^2, \end{aligned} \quad (\text{PK131})$$

where the observable, exact tree/Haar maps, reflection and actual vacuum are all given above. The fixed-box limits of these quantities are $0, 3/32, \frac{1}{2}(1-5^{-3/2})$, respectively. The L, a -dependent localization constants are still present in (PK105); they have not been assumed uniform. This leaves a concrete original-operator calculation, while replacing the escaping fixed-heat packet by an explicit bounded observable on the actual vacuum. The new family is distinct from $\eta_{h,t}$, and (PK115) specifies its exact receiving map. A continuum state, reconstruction and the intended theory identification remain to be constructed.

Subsequent joint-path calculation: PK137–PK142 below prove that this chart-supported family's raw total mass and first weight tend to zero on the prescribed path. PK143–PK157 construct and calculate a different global observable. The preceding fixed-box theorems retain their exact quantifiers.

A bounded odd observable on the interacting vacuum

Fixed $L \geq 2$, $a > 0$; actual coupling limit $g \rightarrow 0$.

$$v_g = \frac{1}{2}(f_g - \mathcal{S}_R f_g)\psi_g \rightarrow \sin[(\sigma_*/2)z_1 \cdot z_3]\Phi_0$$

$$\sigma_* = \sqrt{8} \sin[\pi/(4L+2)], \quad \delta_* = 2\sigma_*/a$$

Full tree, Haar-density, cutoff and reflection maps: PK102–PK116.

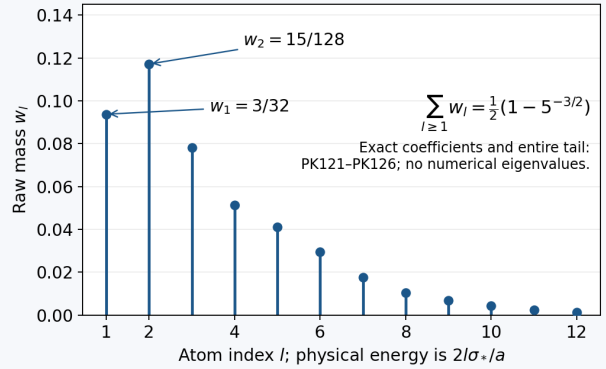
Exact first-cluster reflection

Mode label	Sign	Physical singlets	Sign
$z_1 (V_{12})$	–	D_1, D_2, D_3	+
$z_2 (V_{13})$	–	O_{12}	+
$z_3 (V_{23})$	+	O_{13}, O_{23}	–

Two odd vectors form the actual lowest odd cluster for sufficiently small g at each fixed box.

Raw first weight $\rightarrow 3/32$; supported bottom $\rightarrow \delta_*$.
Proof: PK108–PK112 and PK117–PK120.

Full comparison measure: first 12 atoms



Global map from the retained higher carrier

$$Y = \text{Tot}(E_D), \quad D = F_4/\rho(\text{Sp}(1)) \rightarrow A = |\alpha|^2 + |\beta|^2 + |\gamma|^2 \rightarrow \mathfrak{B}_g(Y)$$

All 24 fibre coordinates retained; $A^{-1}(0)$ is the original rank-12 zero subbundle.

This particular receiving map factors through A. Full formulas and proof: PK127–PK129.

Joint path $L_j = j^4$, $a_j = 1/(100j)$, $g_j^2 = \kappa_*/(200j)$:

The actual odd energy and raw weight still require estimates uniform along this path (PK130–PK131).

Figure 5: Exact reflection signs, raw spectral weights and the higher-carrier receiving map

The mode labels identify the cochains specifying each coordinate; they do not equate cochains with coordinates. The plot shows the exact first twelve atoms from PK125, with the full tail given by PK125–PK126. The actual fixed-box limits and the separate original joint path have their stated quantifiers. The parameter map keeps the full domain and explicitly factors through PK127. Reproducible figure source.

9.20. An actual-vacuum density estimate with all dimensions retained

The order of limits in PK111–PK129 is essential. We now calculate the raw mass on the original simultaneous path, rather than selecting a smaller coupling from a fixed-box existence statement. This calculation uses the original operator, its positive vacuum, the Haar map PK104 and the earlier vacuum upper bound PK88/PK101.

Put $m = 2L$, and retain every boundary contribution:

$$N = 3m(m+1)^2, \quad M = 3m^2(m+1), \quad d = 2m^3 + 3m^2, \quad H = \kappa H_0 + V, \quad V = b \sum_p (2 - W_p), \quad \kappa = \frac{2g^2}{a}, \quad b = \frac{1}{2g^2 a}. \quad (\text{PK132})$$

Here N is the number of links, M the number of faces, and d the number of chords. The actual positive unit vacuum is ψ , with energy E_0 ; write $d\mu = \psi^2 dU$. In particular

$$0 \leq E_0 \leq \min\{2bM, B\}, \quad B = \frac{3\sqrt{NM}}{a}.$$

The full $2bM$ scalar and all face terms remain in H .

For the one-link electric operator $E = -\sum_{\alpha=1}^3 X_{\alpha}^2$, with $T_{\alpha} = -i\sigma_{\alpha}/2$, its heat kernel relative to Haar probability has identity value

$$K_s(I) = \sum_{n=1}^{\infty} n^2 e^{-s(n^2-1)/4} \leq e^{s/4} \left(\frac{2\sqrt{\pi}}{s^{3/2}} + \frac{4}{s} + \frac{\sqrt{\pi}}{s^{1/2}} \right). \quad (\text{PK133})$$

The equality follows from the complete matrix coefficients in PK65: the spin $(n-1)/2$ block has dimension n^2 and Casimir $(n^2-1)/4$. For the bound, on $x \in [n-1, n]$, $n^2 e^{-sn^2/4} \leq (x+1)^2 e^{-sx^2/4}$. Sum the integrals over these intervals. The three terms of the resulting integral are, respectively, the integrals of $x^2, 2x, 1$ times $e^{-sx^2/4}$; their displayed values follow by substitution and integration by parts. The factor $e^{s/4}$ from the original Casimir has not been discarded. For $0 < s \leq 1$ this also gives

$$K_s(I) \leq C_0 s^{-3/2}, \quad C_0 = e^{1/4}(3\sqrt{\pi} + 4). \quad (\text{PK134})$$

Here is the passage from this free kernel to the actual interacting vacuum. The free product semigroup $P_t = e^{-t\kappa H_0}$ preserves nonnegative functions. For a smooth nonnegative initial function this follows from the heat equation: apply the minimum argument to the solution plus $\varepsilon(1+t)$; a first nonpositive spatial minimum would have nonpositive time derivative and nonnegative Laplacian, contradicting the positive ε term. Let ε decrease to zero and then use approximation for nonnegative L^2 functions. Its kernel is the product of the one-link kernels, whose series converge with all derivatives for positive time.

Differentiate $P_{t-r} e^{-rE_0} \psi$ in r , using $(\kappa H_0 - E_0)\psi = -V\psi$, and integrate. Every function is smooth on this compact finite-dimensional manifold. The exact identity is

$$e^{-tE_0} \psi = P_t \psi - \int_0^t P_{t-r} e^{-rE_0} V \psi \, dr \leq P_t \psi.$$

The integral is nonnegative because $V \geq 0$ and $\psi > 0$. Cauchy–Schwarz in the original Haar measure, and the convolution identity for the square of the product heat kernel, therefore give

$$\|\psi\|_{\infty}^2 \leq e^{2tE_0} K_{2\kappa t}(I)^N \leq e^{2tB} K_{2\kappa t}(I)^N. \quad (\text{PK135})$$

There is no product-vacuum substitution in this inequality. Set $t = N/(2B)$ and $s = \kappa N/B = (\kappa a/3)\sqrt{N/M}$. For every finite box with $s \leq 1$, the resulting explicit bound is

$$\|\psi\|_{\infty}^2 \leq [eC_0 s^{-3/2}]^N, \quad s = \frac{\kappa a}{3} \sqrt{\frac{N}{M}}. \quad (\text{PK136})$$

Using the full expression in PK133 instead of PK134 gives a bound for every positive s , with no small- s restriction.

9.21. The chart cutoff loses its entire raw mass on the original path

For $R > 0$, define the exact configuration set

$$S_R = \left\{ U : Z_c = \exp(y_c^\alpha T_\alpha), |y_c| < 2\pi \text{ for all } c, \quad \sum_c |y_c|^2 \leq R^2 \right\}.$$

Its definition is independent of based gauge variables and is invariant under residual simultaneous conjugation. The exact tree map preserves product Haar measure; integrating its tree variables gives one. Since every squared sinc in PK104 is at most one, integration on the intersection of the logarithm domain with the Euclidean $3d$ -ball gives

$$\begin{aligned} \lambda(S_R) &\leq (16\pi^2)^{-d} \frac{\pi^{3d/2} R^{3d}}{\Gamma(3d/2 + 1)} \\ &= \frac{R^{3d}}{16^d \pi^{d/2} \Gamma(3d/2 + 1)}, \\ \mu(S_R) &\leq \min \left\{ 1, [eC_0 s^{-3/2}]^N \frac{R^{3d}}{16^d \pi^{d/2} \Gamma(3d/2 + 1)} \right\} =: \mathcal{E}_{L,a,g}(R) \quad (s \leq 1). \end{aligned} \tag{PK137}$$

The first inequality holds also when $R > 2\pi$: replacing the chart intersection by the whole Euclidean ball only increases the integral. The Euclidean volume factor follows by evaluating the Cartesian Gaussian integral in polar coordinates; neither the Haar scalar nor the dimension has been replaced.

For PK113–PK115, f_g is supported in $S_{\pi/2}$ and $|f_g| \leq 1$. Let $P_- = (I - \mathcal{S}_{\mathcal{R}})/2$. The actual vacuum is reflection invariant, so $v_g = P_-(f_g\psi)$. Orthogonal projection is contractive. Consequently, with the original raw norm and spectral measure,

$$\Gamma_g^{\text{odd}} = \|v_g\|^2 \leq \|f_g\psi\|^2 \leq \mu(S_{\pi/2}) \leq \mathcal{E}_{L,a,g}(\pi/2), \quad 0 \leq \nu_g^{\text{odd}}(J) \leq \Gamma_g^{\text{odd}} \tag{PK138}$$

for every Borel energy set J . This avoids any extra support-volume factor from a union with the reflected chart. The identical inequality holds for PK128 at every parameter y , even if y varies along the sequence: each sine stays bounded by one.

Now substitute the original path, including all its coefficients:

$$\begin{aligned} L_j &= j^4, \quad m_j = 2j^4, \quad a_j = \frac{1}{100j}, \quad g_j^2 = \frac{\kappa_*}{200j}, \quad \kappa_j = \kappa_*, \quad b_j = \frac{10000j^2}{\kappa_*}, \\ N_j &= 3(2j^4)(2j^4 + 1)^2, \quad M_j = 3(2j^4)^2(2j^4 + 1), \quad d_j = 2(2j^4)^3 + 3(2j^4)^2, \\ s_j &= \frac{\kappa_*}{300j} \sqrt{1 + \frac{1}{2j^4}}. \end{aligned} \tag{PK139}$$

For any fixed $\kappa_* > 0$, $s_j \leq 1$ eventually. The classical dilation is still $\lambda_j = j^2$; it has no effect on this exact bound for configuration support in the given operator.

The logarithm of the second entry of the minimum in PK137 is exactly

$$\begin{aligned} D_j(R) &= N_j[1 + \log C_0 - \tfrac{3}{2} \log s_j] + 3d_j \log R - d_j \log 16 - \tfrac{d_j}{2} \log \pi \\ &\quad - \log \Gamma(3d_j/2 + 1). \end{aligned} \tag{PK140}$$

Here $v_j = 3d_j/2$ is an integer. Since $\log x$ is increasing,

$$\log(v_j!) = \sum_{k=1}^{v_j} \log k \geq \int_1^{v_j} \log x \, dx = v_j \log v_j - v_j + 1.$$

Also

$$\frac{N_j}{d_j} = \frac{3(m_j + 1)^2}{m_j(2m_j + 3)} \longrightarrow \frac{3}{2}, \quad \frac{\log d_j}{\log j} \longrightarrow 12, \quad \frac{-\log s_j}{\log j} \longrightarrow 1.$$

For fixed $R > 0$, these formulas prove

$$\limsup_{j \rightarrow \infty} \frac{D_j(R)}{d_j \log j} \leq \frac{9}{4} - 18 = -\frac{63}{4}. \quad (\text{PK141})$$

In particular $\mathcal{E}_{L_j, a_j, g_j}(R) \rightarrow 0$. More generally, if $R_j \leq Cj^p$, where $C > 0$ and $p < 21/4$, the right side of PK141 becomes $-63/4 + 3p < 0$. Thus every uniformly bounded family supported in these sets, followed by the same physical odd projection, has raw norm tending to zero. This is a quantitative statement about an entire space of supported observables. More precisely, let $\mathcal{C}_j(R)$ consist of physical L^∞ functions vanishing almost everywhere outside S_R , and define

$$\mathcal{T}_j : \mathcal{C}_j(R) \longrightarrow \mathcal{H}_{\text{phys}, -}, \quad f \longmapsto P_-(f\psi_j), \quad \|\mathcal{T}_j f\|^2 \leq \|f\|_\infty^2 \mathcal{E}_j(R). \quad (\text{PK141a})$$

Multiplication followed by the exact physical odd projection proves linearity and the displayed norm bound. Thus a common bound $\|f_j\|_\infty \leq C_f$ retains the full prefactor $C_f^2 \mathcal{E}_j(R_j)$.

For the particular original cutoff,

$$\Gamma_j^{\text{odd}} \longrightarrow 0, \quad \mathfrak{w}_j \longrightarrow 0, \quad \|\nu_j^{\text{odd}}\|_{\text{TV}} = \Gamma_j^{\text{odd}} \longrightarrow 0. \quad (\text{PK142})$$

Every parameter family PK128 has the same bound. Any later contraction $e^{-u_j(H_j - E_{0,j})}$, $u_j \geq 0$, has norm no larger. The positive fixed-box limit in PK120 and PK129 remains correct: it takes $g \rightarrow 0$ at fixed L, a , whereas PK142 uses PK139. The vanishing raw mass is a proved defect of this chart-supported map. No statement about all physical states or all interacting continuum limits follows.

9.22. A global group-coordinate map and its exact nonzero state

The next construction removes the support restriction just identified. Retain the original rooted chord words $Z_c = t_s U_c t_t^{-1}$. Define their global real coordinates by

$$q_c^\alpha(U) = -2 \operatorname{tr}(T_\alpha Z_c(U)), \quad \mathbf{M} = O^T G^{-1/2}, \quad u_\mu^\alpha(U) = \sum_c \mathbf{M}_{\mu c} q_c^\alpha(U). \quad (\text{PK143})$$

For the first three rows there is an exact original-cochain formula:

$$\mathbf{M}_{\mu c} = V_\mu(c), \quad (V_1, V_2, V_3) = (V_{12}, V_{13}, V_{23}), \quad c \in \mathcal{C}. \quad (\text{PK143a})$$

Indeed $T^*G^{-1}T$ is the orthogonal projection onto $(\ker T)^\perp$: it is self-adjoint, its square equals itself by $TT^* = G$, and its kernel is $\ker T$. Each original transverse cochain V_μ lies in this range. The specified column $O_\mu = G^{-1/2}TV_\mu$ therefore gives

$$O_\mu^T G^{-1/2} = V_\mu^T T^* G^{-1} = V_\mu^T T^* G^{-1} T j_{\mathcal{C}} = V_\mu^T j_{\mathcal{C}}.$$

This identifies every coefficient with the value of the original oriented transverse cochain on that chord. It does not discard tree holonomies from $q_c(U)$. For $Z = \exp(y^\alpha T_\alpha)$, their complete formula is

$$q^\alpha(Z) = \frac{\sin(|y|/2)}{|y|/2} y^\alpha, \quad |q(Z)|^2 = 4 \left[1 - \left(\frac{\text{tr } Z}{2} \right)^2 \right]. \quad (\text{PK144})$$

At $y = 0$ the sinc has its analytic value one. These equalities follow by inserting the exponential with its original sign:

$$Z = \cos(|y|/2)I + \frac{2 \sin(|y|/2)}{|y|} \sum_\alpha y^\alpha T_\alpha.$$

This last displayed expression is the exponential for $T_\alpha = -i\sigma_\alpha/2$, and $-2\text{tr}(T_\alpha T_\beta) = \delta_{\alpha\beta}$ then gives PK144. The defining trace formula PK143 is globally smooth even at $Z = -I$; it does not require a logarithm chart. Under simultaneous conjugation the three real coordinates transform by the same adjoint orthogonal matrix.

For $A \geq 0$ define, on the entire original compact link space,

$$\begin{aligned} \Theta_g^A(U) &= \frac{A\sigma_*}{2g^2} u_1(U) \cdot u_3(U), \\ \widehat{F}_g^A(U) &= \frac{1}{2} [\sin \Theta_g^A(U) - \sin \Theta_g^A(\mathcal{R}U)], \quad \widehat{v}_g^A = \widehat{F}_g^A \psi_g. \end{aligned} \quad (\text{PK145})$$

The tree/chord inverse, G, O, σ_* , graph reflection, physical spacing, coupling and actual vacuum are precisely those of PK102–PK115. In particular reflection acts on the whole original graph configuration, not by imposing a linear sign map on the global q coordinates. Gauge invariance follows from the common adjoint action at the root. The exact half-difference proves oddness, vacuum orthogonality and $|\widehat{F}_g^A| \leq 1$. Smoothness holds on the entire configuration space for every positive g, a .

This vector is nonzero for every $A > 0$ and every positive g, a at each fixed box $L \geq 2$, without a small-coupling assumption. To prove this, use a sufficiently small chord chart only to evaluate the Taylor coefficient of the globally defined function. From PK144, $q_c(y) = y_c + O(|y_c|^3)$. The reflected chord word has linear part from PK108 and a remainder $O_L(|y|^2)$. Thus

$$\widehat{F}_g^A(\exp y) = \frac{A\sigma_*}{2g^2} (\mathbb{M}y)_1 \cdot (\mathbb{M}y)_3 + O_{L,g,A}(|y|^3). \quad (\text{PK146})$$

Take $y^1 = tG^{1/2}O(e_1 + e_3)$ and $y^2 = y^3 = 0$; for sufficiently small t every chord stays in its chart. Then

$$\lim_{t \rightarrow 0} t^{-2} \widehat{F}_g^A(\exp y(t)) = \frac{A\sigma_*}{2g^2} > 0. \quad (\text{PK147})$$

Consequently the function is nonzero on an open set. Haar measure is positive on that set, and $\psi_g > 0$, so its raw norm is positive. The same Taylor calculation applies to the old cutoff, because $\chi = 1$ in a neighborhood of the identity. This strengthens the earlier nonzero-norm assertion for both finite-regulator maps: the raw norm is positive for every positive coupling, although PK142 proves that the old map loses that norm along PK139.

For fixed L, a, A , substitute $y = gx$ in PK144: $q_c(\exp(gx))/g = \text{sinc}(g|x_c|/2)x_c \rightarrow x_c$. Boundedness and the actual-vacuum convergence PK116 therefore give

$$\mathcal{B}_g \widehat{v}_g^A \longrightarrow \sin[(A\sigma_*/2)z_1 \cdot z_3] \Phi_0 \quad \text{strongly in } L^2. \quad (\text{PK148})$$

The proof is exactly dominated convergence against $|\Phi_0|^2$ plus strong convergence of $\mathcal{B}_g \psi_g$; the reflected multiplier uses the original word map and its established derivative. Hence PK129 gives the same

full raw mass and first-cluster weight for this global map. At $A = 1$, every limiting weight and the correlation PK125–PK126 are unchanged. The actual supported bottom equals the actual odd bottom for sufficiently small g at fixed box, by the same norm-projection and positive-overlap argument.

Finally use the full higher-carrier parameter $A = \mathfrak{a}(y) = |\alpha|^2 + |\beta|^2 + |\gamma|^2$ of PK127:

$$\widehat{\mathfrak{Y}}_g : Y \longrightarrow C^\infty(\mathcal{Q}_L)_{\text{physical, odd}}, \quad y \longmapsto \widehat{F}_g^{\mathfrak{a}(y)} \psi_g. \quad (\text{PK149})$$

The invariant scalar glues, and all derivatives of the trigonometric formula are bounded on each compact parameter set at each fixed regulator; this proves smoothness as a map into the space with all smooth-function seminorms. PK147 proves its exact zero set, for every positive coupling:

$$\widehat{\mathfrak{Y}}_g^{-1}(0) = \{\alpha = \beta = \gamma = 0\} = \mathcal{V}_{r,D} \oplus 0_{\mathcal{L}_{q,1}} \oplus 0_{\mathcal{L}_{q,2}} \oplus 0_{\mathcal{L}_{p,1}} \oplus \mathcal{L}_{p,2}. \quad (\text{PK150})$$

This is the original rank-12 subbundle in the full rank-24 domain. The remaining coordinates are present; this receiving map still factors through the explicitly displayed scalar. It does not supply an identification of every angular parameter with a physical observable.

9.23. Complete derivatives and raw moments of the global replacement

We next calculate its actual finite-regulator energy, retaining the unknown interacting vacuum rather than substituting the Gaussian one. This yields exact quantities for the next path estimate.

Write each rooted chord as its full ordered word $Z_c = A_1 \cdots A_{\ell_c}$, where each $A_r = U_{e_r}^{\varepsilon_r}$, $\varepsilon_r \in \{1, -1\}$. For $X = X_{e,\alpha}$, define

	XA_r	X^2A_r
$e_r \neq e$	0	0
$e_r = e, \varepsilon_r = 1$	$T_\alpha U_e$	$T_\alpha^2 U_e$
$e_r = e, \varepsilon_r = -1$	$-U_e^{-1} T_\alpha$	$U_e^{-1} T_\alpha^2$

(PK151)

The derivative convention is the original left field $X_{e,\alpha} F = \frac{d}{dt} F(\dots, e^{tT_\alpha} U_e, \dots)|_0$. Direct differentiation gives

$$\begin{aligned} XZ_c &= \sum_r A_1 \cdots (XA_r) \cdots A_{\ell_c}, \\ X^2Z_c &= \sum_r A_1 \cdots (X^2A_r) \cdots A_{\ell_c} \\ &\quad + 2 \sum_{r < s} A_1 \cdots (XA_r) \cdots (XA_s) \cdots A_{\ell_c}, \\ Xu_\mu^\beta &= -2 \sum_c M_{\mu c} \text{tr}(T_\beta XZ_c), \\ X^2u_\mu^\beta &= -2 \sum_c M_{\mu c} \text{tr}(T_\beta X^2Z_c). \end{aligned} \quad (\text{PK152})$$

Repeated occurrences of the same physical link, their inverse traversals and the order of every matrix product are retained. These formulas also apply to the reflected chord words in PK145.

With $\Theta = \Theta_g^A$, $\Theta_R = \Theta_g^A \circ \mathcal{R}$, and $F = \widehat{F}_g^A$, the complete first and second derivatives are

$$\begin{aligned} X\Theta &= \frac{A\sigma_*}{2g^2} [(Xu_1) \cdot u_3 + u_1 \cdot Xu_3], \\ X^2\Theta &= \frac{A\sigma_*}{2g^2} [(X^2u_1) \cdot u_3 + 2(Xu_1) \cdot Xu_3 + u_1 \cdot X^2u_3], \\ XF &= \frac{1}{2} [\cos \Theta X\Theta - \cos \Theta_R X\Theta_R], \\ X^2F &= \frac{1}{2} [\cos \Theta X^2\Theta - \sin \Theta (X\Theta)^2 - \cos \Theta_R X^2\Theta_R + \sin \Theta_R (X\Theta_R)^2]. \end{aligned} \quad (\text{PK153})$$

Thus no derivative of a tree word, inverse, sine or reflected configuration is omitted.

Let $\mathcal{A} = H - E_0$. The complete product rule and the actual vacuum equation give

$$\mathcal{A}(F\psi) = \kappa\psi \left[H_0 F - 2 \sum_{e,\alpha} (X_{e,\alpha} \log \psi) X_{e,\alpha} F \right]. \quad (\text{PK154})$$

The potential and its constant contribution cancel here only after using $(\kappa H_0 + V)\psi = E_0\psi$ with their original values. They still determine the full ψ and E_0 . Integration by parts on each Haar factor proves

$$\begin{aligned} \widehat{\Gamma}_g^A &= \int F^2 \psi^2 dU, \\ \widehat{m}_{1,g}^A &= \langle F\psi, \mathcal{A}F\psi \rangle = \kappa \sum_{e,\alpha} \int (X_{e,\alpha} F)^2 \psi^2 dU, \\ \widehat{m}_{2,g}^A &= \|\mathcal{A}F\psi\|^2 \\ &= \kappa^2 \int \left[H_0 F - 2 \sum_{e,\alpha} (X_{e,\alpha} \log \psi) X_{e,\alpha} F \right]^2 \psi^2 dU. \end{aligned} \quad (\text{PK155})$$

For the middle equality, expand $-\int F\psi^2 X^2 F = \int \psi^2 (XF)^2 + 2 \int F\psi(X\psi)XF$ for each field. The second term cancels the displayed drift term in PK154. This is the exact ground-state transform already used in the retained local-energy proof, equation (1.3), here applied to the complete new global observable with all its derivatives. It is not a new general transform theorem.

For every $A > 0$, PK147 shows $\widehat{\Gamma}_g^A > 0$. Also $\widehat{m}_{1,g}^A > 0$: otherwise each derivative of F vanishes everywhere, since the nonnegative integrands are continuous and $\psi > 0$. Connectedness would make F constant, while its oddness and nonzero Taylor coefficient exclude that. The complete spectral measure $\widehat{\nu}_g^A$ has these raw zeroth, first and second moments, and no vacuum atom. For every $\varepsilon > 0$, spectral integration gives the unconditional bounds

$$\max \left\{ 0, \widehat{\Gamma}_g^A - \frac{\widehat{m}_{1,g}^A}{\varepsilon} \right\} \leq \widehat{\nu}_g^A((0, \varepsilon]) \leq \widehat{\Gamma}_g^A, \quad \widehat{\nu}_g^A((\varepsilon, \infty)) \leq \frac{\widehat{m}_{2,g}^A}{\varepsilon^2}. \quad (\text{PK156})$$

For example, the first inequality is the identity for the total mass minus the high-energy mass, with $\int_{(\varepsilon, \infty)} d\widehat{\nu} \leq \varepsilon^{-1} \int \lambda d\widehat{\nu}$. No positivity of its displayed lower bound along PK139 is assumed.

The joint-path question is now the original-operator calculation

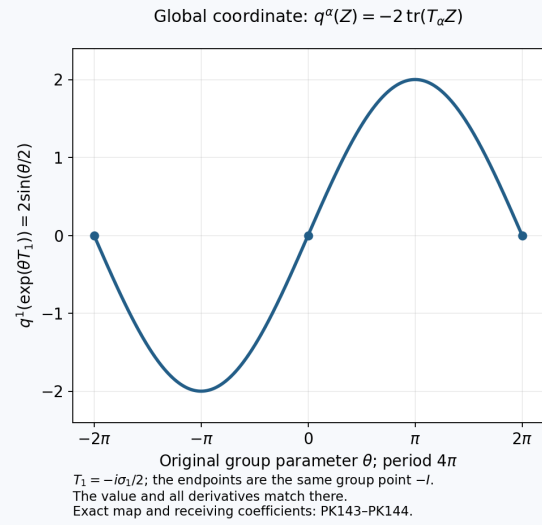
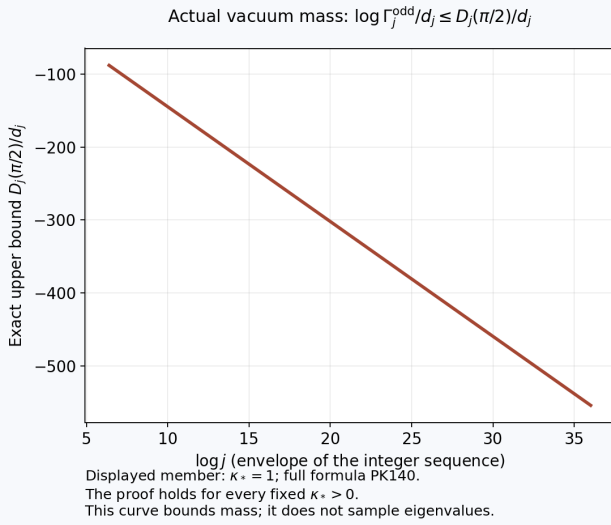
$$\left(\widehat{\Gamma}_{g_j}^1, \widehat{m}_{1,g_j}^1, \widehat{m}_{2,g_j}^1, \widehat{\nu}_{g_j}^1 \right) \quad \text{for exactly the } (L_j, a_j, g_j) \text{ in PK139.} \quad (\text{PK157})$$

PK152–PK155 specify every derivative and vacuum-dependent term in these quantities. The chart-support collapse has been removed from the construction, and nonzero norm at each finite regulator is proved. A nonvanishing limiting norm, a closing actual spectral threshold, continuum reconstruction and theory identification are still unfinished. The former cutoff cannot be reused as a source of nonzero raw limiting mass: PK142 has settled that calculation.

The left curve evaluates the full PK140 bound for the displayed member $\kappa_{**}=1$; PK141 proves the result for every fixed positive κ_{**} . The right curve is the exact PK144 coordinate on the original 4-pi-periodic group circle. This is a bound and an exact map, not sampled eigenvalues. The complete proofs and remaining global-state estimates are PK132–PK157. Reproducible figure source.

The original joint path: cutoff mass and a global replacement

$$L_j = j^4, \quad a_j = 1/(100j), \quad g_j^2 = \kappa_*/(200j), \quad b_j = 10000j^2/\kappa_*$$



$\limsup D_j(R)/(d_j \log j) \leq -63/4$ for fixed R : the cutoff's raw measure tends to zero.

$$\hat{v}_g^A = \frac{1}{2} \{ \sin[(A\sigma_*/2g^2) u_1 \cdot u_3] - \mathcal{S}_R \sin[(A\sigma_*/2g^2) u_1 \cdot u_3] \} \psi_g$$

The global replacement is smooth and nonzero for every $A > 0$ at each finite regulator.
Its joint-path mass and energy still require estimates. Complete proofs: PK132-PK157.

All heat, Haar, boundary-count, trace and derivative factors are derived in the accompanying proof.
The actual ground-state transform retains its earlier programme source: local_energy_current_true_vacuum.md, equation (1.3).

Figure 6: Actual joint-path support-mass bound and the global group-coordinate map

9.24. Exact chord coefficients and their full Gram matrix

We apply the original-tree formula PK143a before estimating the global observable. Set $m = 2L$, $n = m + 1$, and

$$h = \frac{\pi}{2n}, \quad c = \cos h, \quad C = \cot h, \quad v_0(l) = n^{-1/2}, \quad v(l) = \sqrt{2/n} \cos \frac{\pi(l+1/2)}{n}, \quad w(l) = -\sqrt{2/n} \sin \frac{\pi(l+1)}{n}. \quad (\text{PK158})$$

Vertex indices are $l_i = n_i + L \in \{0, \dots, m\}$; an edge in direction i has $l_i < m$. These indices do not replace the original physical coordinates $o + a(n_1, n_2, n_3)$. The root and parent rule in PK103 imply that the tree contains every direction-one edge, the direction-two edges with $l_1 = 0$, and the direction-three edges with $l_1 = l_2 = 0$. Indeed each such edge is exactly the edge to the prescribed parent of its positive endpoint; each other edge fails that parent rule. Hence the complete chord sets are

$$\begin{aligned} \mathcal{C}_2 &= \{(l, 2) : 1 \leq l_1 \leq m, 0 \leq l_2 < m, 0 \leq l_3 \leq m\}, \\ \mathcal{C}_3 &= \{(l, 3) : 0 \leq l_1, l_2 \leq m, (l_1, l_2) \neq (0, 0), 0 \leq l_3 < m\}. \end{aligned} \quad (\text{PK159})$$

Their sizes $m^2(m+1)$ and $m[(m+1)^2 - 1]$ add to the original $d = 2m^3 + 3m^2$. No omitted tree edge is omitted from a chord word.

Write $a_c = M_{1c}$, $b_c = M_{3c}$, and retain the second row M_{2c} as well. PK106–PK108 and PK143a give

	M_{1c}	M_{2c}	M_{3c}	
$c = (l, 2) \in \mathcal{C}_2$	$-v(l_1)w(l_2)v_0(l_3)/\sqrt{2}$	0	$v_0(l_1)w(l_2)v(l_3)/\sqrt{2}$	(PK160)
$c = (l, 3) \in \mathcal{C}_3$	0	$-v(l_1)v_0(l_2)w(l_3)/\sqrt{2}$	$-v_0(l_1)v(l_2)w(l_3)/\sqrt{2}$	

All signs and endpoint exclusions are part of this formula. Let $R_{\mu\nu} = \sum_c M_{\mu c} M_{\nu c}$. The full first-three-row Gram matrix and the contraction needed below are

$$R = \begin{pmatrix} \frac{1}{2} - \frac{c^2}{n} & 0 & 0 \\ 0 & \frac{1}{2} - \frac{c^2}{n^2} & -\frac{c^2}{n^2} \\ 0 & -\frac{c^2}{n^2} & 1 - \frac{1}{2n} - \frac{c^2}{n^2} \end{pmatrix}, \quad D_{13} := \sum_c a_c^2 b_c^2 = \frac{3}{8n^3} \left(1 - \frac{2c^2}{n}\right). \quad (\text{PK161})$$

Here is the complete summation. The orthonormal one-dimensional vectors satisfy $\sum v^2 = \sum w^2 = 1$, $\sum v_0^2 = 1$, and $\sum vv_0 = 0$. Also $v(0)^2 = 2c^2/n$ and $v_0(0)^2 = 1/n$. The first row has only its direction-two component, so its squared norm is $\frac{1}{2}(1 - v(0)^2)$. The second row loses only the corner term from its direction-three component, giving $\frac{1}{2}(1 - v(0)^2/n)$. The third row's direction-two component has squared norm $\frac{1}{2}(1 - 1/n)$, and its direction-three component has squared norm $\frac{1}{2}(1 - v(0)^2/n)$. The first/second cross product vanishes because their directions differ. The first/third product contains the full factor $\sum_{l_3} v_0 v = 0$. The second/third product on the full direction-three rectangle is zero; subtracting its excluded corner leaves $-v(0)^2/(2n)$. These are precisely the six independent entries in PK161.

For the fourth-order contraction only direction-two chords contribute. Its three separate factors are

$$\frac{1}{4} \left[\frac{1 - v(0)^2}{n} \right] \left[\sum_{l=0}^{m-1} w(l)^4 \right] \left[\frac{1}{n} \right].$$

The finite geometric sums of $e^{2\pi i r/n}$ and $e^{4\pi i r/n}$, $r = 0, \dots, n-1$, are zero because $n = 2L + 1 \geq 5$. Inserting $\sin^4 x = (3 - 4 \cos 2x + \cos 4x)/8$ therefore gives $\sum_{r=1}^{n-1} \sin^4(\pi r/n) = 3n/8$ and $\sum w^4 = 3/(2n)$. Substitution proves D_{13} .

The absolute row sums $S_\mu = \sum_c |\mathbf{M}_{\mu c}|$ also have complete closed expressions:

$$\begin{aligned} S_1 &= \sqrt{\frac{2}{n}} C(C - c), \\ S_2 &= \sqrt{\frac{2}{n}} C(C - c/n), \\ S_3 &= \sqrt{\frac{2}{n}} [(2 - 1/n)C^2 - (c/n)C]. \end{aligned} \tag{PK162}$$

To verify them, sum the positive sine values on the edges to obtain $\sum |w| = \sqrt{2/n} C$. The vertex cosine has one central zero; its two equal absolute half-sums give $\sum |v| = \sqrt{2/n} C$. For example, the first half-sum is $\sum_{l=0}^{L-1} \cos[\pi(l + 1/2)/n] = \sin(L\pi/n)/(2 \sin h) = c/(2 \sin h)$. The first row excludes the vertex $l_1 = 0$ and thus subtracts $\sqrt{2/n} c$ in that factor. The second row subtracts its single corner. The third row adds the direction-two contribution with $n - 1$ values of l_1 and the direction-three contribution with its single excluded corner. Multiplying the displayed factors gives all three formulas in PK162.

9.25. The complete fourth moment and the actual receiving inequality

Let

$$\mathcal{S}(U) = u_1(U) \cdot u_3(U) = \sum_{c,d} a_c b_d q_c(U) \cdot q_d(U). \tag{PK163}$$

The original tree/Haar isomorphism makes the chord variables independent under Haar measure. This statement is only about Haar measure. The actual vacuum measure is still $d\mu = \psi^2 dU$. For one Haar chord, write $Z = x_0 I + 2 \sum_\alpha x_\alpha T_\alpha$, $x_0^2 + x_1^2 + x_2^2 + x_3^2 = 1$. Its measure is the invariant probability measure on that unit sphere, and $q^\alpha = 2x_\alpha$. Rotational invariance and sign symmetry give

$$\begin{aligned} \mathbb{E}_\lambda q^\alpha &= 0, \quad \mathbb{E}_\lambda q^\alpha q^\beta = \delta_{\alpha\beta}, \\ \mathbb{E}_\lambda q^\alpha q^\beta q^\gamma q^\delta &= \frac{2}{3} (\delta_{\alpha\beta} \delta_{\gamma\delta} + \delta_{\alpha\gamma} \delta_{\beta\delta} + \delta_{\alpha\delta} \delta_{\beta\gamma}). \end{aligned} \tag{PK164}$$

For the coefficient in the last line, let $A_4 = \mathbb{E} x_1^4$ and $B_4 = \mathbb{E} x_1^2 x_2^2$. Rotation of x_1 to $(x_1 + x_2)/\sqrt{2}$, and sign symmetry, give $A_4 = 3B_4$. Squaring $\sum_{i=0}^3 x_i^2 = 1$ gives $4A_4 + 12B_4 = 1$. Thus $B_4 = 1/24$, $A_4 = 1/8$; the factor 2^4 in q yields PK164.

For arbitrary real chord coefficients a_c, b_c , put $A_2 = \sum a_c^2$, $B_2 = \sum b_c^2$, $C_{12} = \sum a_c b_c$, $D_{12} = \sum a_c^2 b_c^2$. Independence between distinct chords and PK164 give

$$\mathbb{E}_\lambda \mathcal{S} = 3C_{12}, \quad \mathbb{E}_\lambda \mathcal{S}^2 = 3A_2 B_2 + 12C_{12}^2 - 5D_{12}. \tag{PK165}$$

To check every contraction, first use covariance pairing with coefficient one for each of the three Kronecker pairings. Summing the two colour indices gives $3A_2 B_2 + (9 + 3)C_{12}^2$. When all four chord indices agree, their actual fourth moment has coefficient $2/3$ instead of one. The necessary correction is $-(1/3)(9 + 3 + 3) \sum a_c^2 b_c^2 = -5D_{12}$. Other coincidences with an unpaired index vanish by sign symmetry. This proves the formula without dropping coincident-chord terms.

For the exact original rows PK160, PK161 gives $C_{12} = 0$. Their full Haar moment is

$$\mathcal{H}_m := \mathbb{E}_\lambda \mathcal{S}^2 = 3 \left(\frac{1}{2} - \frac{c^2}{n} \right) \left(1 - \frac{1}{2n} - \frac{c^2}{n^2} \right) - \frac{15}{8n^3} \left(1 - \frac{2c^2}{n} \right) \rightarrow \frac{3}{2}. \tag{PK166}$$

Every scalar $c = \cos[\pi/(2n)]$, endpoint and mixed term is retained.

Now use the actual vacuum and the global observable PK145 with parameter $A \geq 0$. Define

$$\beta = \frac{A\sigma_*}{2g^2} = \frac{A\delta_*}{2\kappa}, \quad \mathcal{Q}_{4,L,a,g} = \int \mathcal{S}^2 \psi^2 dU. \quad (\text{PK167})$$

The equality for β uses the original $\delta_* = 2\sigma_*/a$, $\kappa = 2g^2/a$. Reflection invariance of the actual vacuum and contraction of P_- give the unconditional actual estimate

$$0 < \hat{\Gamma}_g^A = \|P_-(\psi \sin(\beta\mathcal{S}))\|^2 \leq \int \sin^2(\beta\mathcal{S}) d\mu \leq \beta^2 \mathcal{Q}_{4,L,a,g} \quad (A > 0). \quad (\text{PK168})$$

The strict inequality is PK147; $|\sin x| \leq |x|$ proves the last one. Applying the same projection argument to Haar instead gives $\|P_- \sin(\beta\mathcal{S})\|_{L^2(\lambda)}^2 \leq \beta^2 \mathcal{H}_m$. No Haar expectation is substituted into PK168.

On the unchanged path PK139 with $A = 1$,

$$\begin{aligned} \beta_j &= \frac{200\sqrt{2}j}{\kappa_*} \sin \frac{\pi}{4j^4 + 2}, \\ j^3 \beta_j &\longrightarrow \frac{50\sqrt{2}\pi}{\kappa_*}, \quad j^6 \beta_j^2 \mathcal{H}_{2j^4} \longrightarrow \frac{7500\pi^2}{\kappa_*^2}, \\ \mathcal{Q}_{4,j} &\geq \frac{4\kappa_*^2}{\delta_{*,j}^2} \hat{\Gamma}_{g_j}^1. \end{aligned} \quad (\text{PK169})$$

These follow from $\sin x/x \rightarrow 1$, PK166 and PK168. The final inequality is for the actual interacting vacuum at every finite regulator. In particular the factor multiplying its raw norm grows asymptotically as $\kappa_*^2 j^6 / (5000\pi^2)$. This identifies the actual fourth-moment quantity that must be calculated; the bounded Haar moment alone cannot supply it.

The exact physical loop content of this fourth moment is also explicit:

$$B_{cd} := q_c \cdot q_d = 2 \operatorname{tr}(Z_c Z_d^{-1}) - \operatorname{tr} Z_c \operatorname{tr} Z_d, \quad \mathcal{Q}_4 = \sum_{c,d,e,f} a_c b_d a_e b_f \mu(B_{cd} B_{ef}). \quad (\text{PK170})$$

Multiplying $Z = x_0 I + 2x_\alpha T_\alpha$ and $Z'^{-1} = x'_0 I - 2x'_\alpha T_\alpha$ proves the first identity, including $c = d$. For full expansion put $T_{cd} = \operatorname{tr}(Z_c Z_d^{-1})$, $t_c = \operatorname{tr} Z_c$; then

$$\mu(B_{cd} B_{ef}) = 4\mu(T_{cd} T_{ef}) - 2\mu(T_{cd} t_e t_f) - 2\mu(t_c t_d T_{ef}) + \mu(t_c t_d t_e t_f). \quad (\text{PK171})$$

All four original word expectations, all ordered index tuples and their coincident cases remain. Simultaneous conjugation at the root fixes each trace, so these are actual physical vacuum observables.

9.26. Original-word bounds for both actual energy moments

For an original link e , let n_{ce} count its occurrences, with either orientation, in the full rooted word $Z_c = t_s U_c t_t^{-1}$. Define the following complete nonnegative finite sums for $\mu = 1, 3$:

$$\begin{aligned} B_\mu &= 2S_\mu, \quad D_{\mu e} = \sum_c |\mathbf{M}_{\mu c}| n_{ce}, \quad J_{\mu e} = \frac{1}{2} \sum_c |\mathbf{M}_{\mu c}| n_{ce}^2, \\ K_e &= B_3 D_{1e} + B_1 D_{3e}, \\ Q_e &= B_3 J_{1e} + 2D_{1e} D_{3e} + B_1 J_{3e}. \end{aligned} \quad (\text{PK172})$$

The words contain each tree link at most twice, once in each simple root path, and their own chord once. Thus $n_{ce} \leq 2$; this does not cancel a tree prefix or remove an occurrence from the sums.

For each original generator $X = X_{e,\alpha}$, PK151–PK152 give

$$|u_\mu| \leq B_\mu, \quad |Xu_\mu| \leq D_{\mu e}, \quad |X^2 u_\mu| \leq J_{\mu e}, \quad |XS| \leq K_e, \quad |X^2 S| \leq Q_e. \quad (\text{PK173})$$

Here every norm of a colour vector is Euclidean. To prove the derivative constants, note that $2T_\alpha = -i\sigma_\alpha$ is itself in $SU(2)$. A single differentiated occurrence in a word is therefore $\pm 1/2$ times an $SU(2)$ matrix, after retaining all surrounding unitary factors. Its q -vector has norm at most one. A twice differentiated occurrence or a product of two first derivatives is $\pm 1/4$ times an $SU(2)$ matrix and has q -vector norm at most $1/2$. The full second derivative has $n_{ce} + 2\binom{n_{ce}}{2} = n_{ce}^2$ such contributions. Triangle inequalities prove the first three bounds; the complete product rule for $u_1 \cdot u_3$ proves the last two, with the displayed factor two retained.

Use the previously proved all-positive-coupling score estimate

$$\|\nabla_e \log \psi\|_\infty \leq 2r_e \xi, \quad r_e = \#\{p : e \in \partial p\}, \quad \xi = \frac{b}{\kappa} = \frac{1}{4g^4}. \quad (\text{PK174})$$

Its complete proof is the retained spatial-continuum Section 10.1, equations (77)–(78): on each radius-two $SU(2)$ factor, the partial Bochner identity has curvature term $+|\nabla_e \log \psi|^2$. At its maximum the nonnegative mixed-Hessian sum and $|\nabla_e V| \leq br_e$ give $w_e - 2r_e \xi \sqrt{w_e} \leq 0$, which is exactly PK174. The original radius, metric, all plaquettes and exterior variables remain those of PK132.

Set

$$\mathcal{C} = 3 \sum_e K_e^2, \quad \mathcal{D} = 3 \sum_e Q_e + 4\sqrt{3} \xi \sum_e r_e K_e. \quad (\text{PK175})$$

The three colour derivatives in PK173 give $|\nabla_e S| \leq \sqrt{3}K_e$ and $|H_0 S| \leq 3 \sum_e Q_e$. Under multiplication by the actual ψ , the excitation operator is $\mathcal{L} = \kappa[H_0 - 2\nabla \log \psi \cdot \nabla]$. Consequently

$$|\mathcal{L}S| \leq \kappa \mathcal{D}, \quad \sum_{e,\alpha} |X_{e,\alpha} S|^2 \leq \mathcal{C}.$$

The original reflection commutes with $\mathcal{L}$: this follows from its commutation with H , its preservation of ψ , and the exact multiplication intertwiner. Thus its odd projection contracts both the energy form and the norm of $\mathcal{L}f$. One can see the form assertion directly from the decomposition into even and odd functions: reflection invariance makes their form cross term equal to its own negative.

The complete chain rule is

$$\mathcal{L} \sin(\beta S) = \beta \cos(\beta S) \mathcal{L}S + \kappa \beta^2 \sin(\beta S) \sum_{e,\alpha} |X_{e,\alpha} S|^2.$$

Combining it with PK155 gives actual, finite, all-coupling bounds

$$\begin{aligned} 0 &< \widehat{m}_{1,g}^A \leq \kappa \beta^2 \mathcal{C}, \\ 0 &< \widehat{m}_{2,g}^A \leq \kappa^2 (\beta \mathcal{D} + \beta^2 \mathcal{C})^2 \quad (A > 0). \end{aligned} \quad (\text{PK176})$$

The positivity follows from the nonzero odd vector and the nonnegative compact excitation operator with its single even zero eigenspace. These are bounds in the actual interacting vacuum; they use no approximation of its density by Haar or by a Gaussian.

For a fully evaluated coarse bound, $n_{ce} \leq 2$ gives $D_{\mu e} \leq 2S_\mu$, $J_{\mu e} \leq 2S_\mu$, $K_e \leq 8S_1 S_3$, $Q_e \leq 16S_1 S_3$. Each original face has exactly four links, including at the boundary, so $\sum_e r_e = 4M$. Therefore

$$\mathcal{C} \leq 192N S_1^2 S_3^2, \quad \mathcal{D} \leq S_1 S_3 (48N + 128\sqrt{3} \xi M). \quad (\text{PK177})$$

PK162 makes these fully explicit trigonometric functions of the original box size. They also record why the local score bound cannot be read as a uniform bound along the prescribed weak-coupling path: $\xi_j = 10000j^2/\kappa_*^2$ still appears, and both N_j, M_j grow with all their boundary terms. The sharper sums PK172–PK175 remain available; PK177 does not replace them.

The next receiving calculation is the actual four-loop combination PK170–PK171, together with the corresponding gradient-weighted expectations in PK155. PK169 has determined its necessary scale relative to the unaltered raw norm. Its growth in the true vacuum has not been established. The current estimates do not imply a closing actual odd gap or nonvanishing limiting spectral weight. The full interacting continuum state and the lesson series remain the same programme objective.

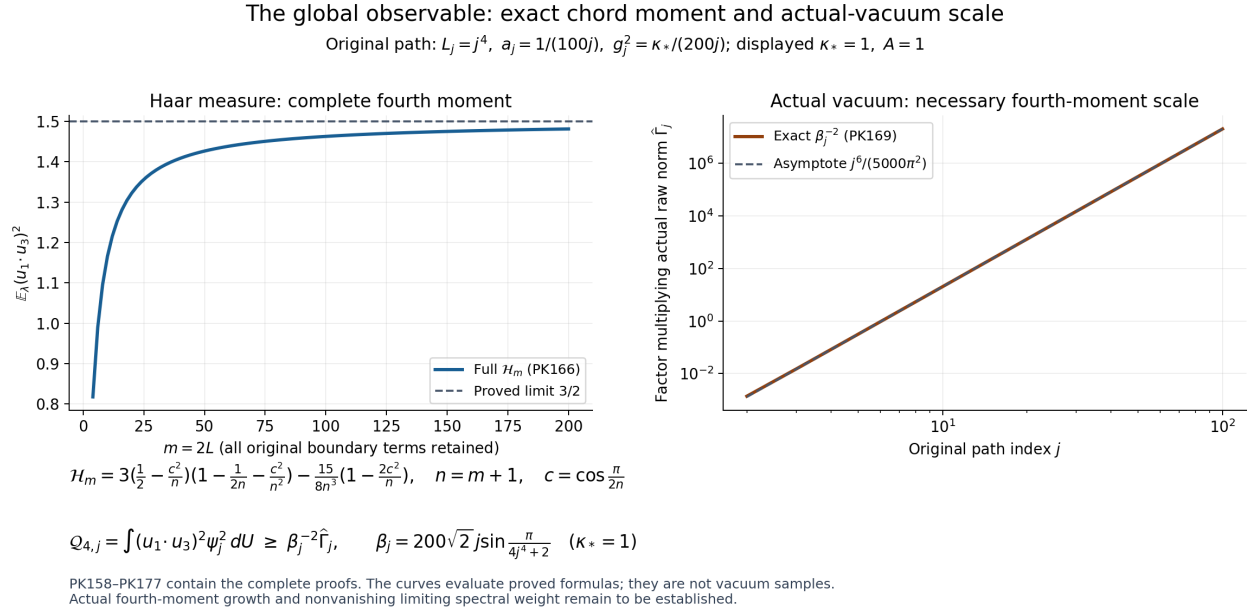


Figure 7: Exact chord fourth moment and original-path actual-vacuum receiving scale

The left curve evaluates the complete Haar moment PK166. The right curve evaluates the exact coefficient in the actual-vacuum inequality PK169 for the displayed $\kappa_* = 1$; PK169 retains every positive κ_* . Neither curve is an actual-vacuum sample or an energy eigenvalue. The complete proofs are PK158–PK177. Reproducible figure source.

9.26a. Full face fillings and an evaluated actual fourth-moment bound

We first apply the retained actual plaquette estimates to the original rooted chords. Put $D_p = 2 - W_p \geq 0$, and retain m, n, v, w, c, C from PK158. For a direction-two chord $c_2 = (i, k, l; 2)$, its rooted loop bounds the strip of direction-(1, 2) faces

$$\mathcal{F}(c_2) = \{(a, k, l; 12) : 0 \leq a < i\}.$$

For a direction-three chord $c_3 = (i, k, l; 3)$, use the bent strip

$$\mathcal{F}(c_3) = \{(a, k, l; 13) : 0 \leq a < i\} \cup \{(0, b, l; 23) : 0 \leq b < k\}. \quad (\text{PK177a})$$

The cardinalities are respectively i and $i + k$. These follow from the original root paths, which increase coordinates in order 3, 2, 1. Their common initial direction-three segment is retained as a conjugating

path. For the second strip first attach the direction-(2, 3) faces in increasing b , then the direction-(1, 3) faces in increasing a . For the first strip attach direction-(1, 2) faces in increasing a . Each attachment replaces a shared boundary segment q by r . The exact new/old boundary ratio is $A(rq^{-1})A^{-1}$. Induction therefore writes the rooted chord word as an ordered product of conjugated, oriented face words, with exactly the faces in PK177a. Reversing a boundary reverses the entire order and inverts every factor. This is the original non-Abelian attachment map, with no commutation of its factors.

For $\mu = 1, 3$, define every weight before evaluating its sum:

$$W_{\mu p} = \sum_{c: p \in \mathcal{F}(c)} |\mathbf{M}_{\mu c}|, \quad T_\mu = \sum_p W_{\mu p} = \sum_c |\mathbf{M}_{\mu c}| |\mathcal{F}(c)|, \quad w_\mu = \max_p W_{\mu p}. \quad (\text{PK177b})$$

Write $V_a = \sum_{i=a+1}^m |v(i)|$. The complete nonzero weights, with all other face weights zero, are

$$\begin{aligned} W_{1,(a,k,l;12)} &= |w(k)|V_a/\sqrt{2n}, & W_{3,(a,k,l;12)} &= (m-a)|w(k)v(l)|/\sqrt{2n}, \\ W_{3,(a,k,l;13)} &= (m-a)|v(k)w(l)|/\sqrt{2n}, & W_{3,(0,b,l;23)} &= \sqrt{n/2}|w(l)|V_b. \end{aligned} \quad (\text{PK177c})$$

Indeed, a face in a direction-one strip occurs for exactly the chords with $i > a$. A direction-(2, 3) face occurs for exactly the direction-three chords with $k > b$, and then all n values of i are allowed: its excluded tree corner $(i, k) = (0, 0)$ is absent already because $k > b \geq 0$. Inserting every coefficient from PK160 gives PK177c.

The exact sums and maxima are

$$T_1 = \frac{mC^2}{\sqrt{2n}}, \quad T_3 = 3T_1, \quad w_1 = \frac{\sqrt{2}c(C-c)}{n^{3/2}}, \quad w_3 = nw_1. \quad (\text{PK177d})$$

To prove the sums, PK162's one-dimensional calculation gives $\sum |v| = \sum |w| = \sqrt{2/n}C$. The reflection $|v(m-i)| = |v(i)|$ gives $\sum i|v(i)| = (m/2)\sum |v(i)|$. For T_1 , sum the tails V_a , then all k and all n values of l in PK177c. The result is its displayed value. Each of the three direction contributions to T_3 is exactly T_1 , using $\sum_{a=0}^{m-1} (m-a) = mn/2$ for the first two contributions. For the maxima, $\max |v| = \max |w| = \sqrt{2/nc}$, and $\max_a V_a = V_0 = \sqrt{2/n}(C-c)$. The largest direction-(1, 2) or (1, 3) third-row weight is $\sqrt{2}mc^2/n^{3/2}$; the largest direction-(2, 3) weight is $\sqrt{2/nc}(C-c)$. The latter is larger because $n(C-c)/(mc) = n(\csc(\pi/(2n)) - 1)/m > 1$ for $n \geq 5$. This proves both maxima with their original face supports.

The identity $I - XY = (I - X) + X(I - Y)$, unitary invariance, and $|q(Z)|^2 \leq 2\|I - Z\|_{\text{HS}}^2$ give

$$|u_\mu| \leq 2 \sum_p W_{\mu p} \sqrt{D_p}, \quad |u_\mu|^2 \leq 4T_\mu \sum_p W_{\mu p} D_p.$$

Here $\|I - U_p\|_{\text{HS}}^2 = 2D_p$, and all conjugations in the proved ordered filling are retained before applying the norm. The actual potential is $V = b \sum_p D_p$. Its retained exact vacuum bound PK132 and $0 \leq \sum_p D_p \leq 4M$ therefore prove

$$\begin{aligned} \mathcal{Q}_4 &\leq \min\{\mathcal{U}_{\text{point}}, \mathcal{U}_{\text{face}}\}, \\ \mathcal{U}_{\text{point}} &= 16S_1^2 S_3^2, \\ \mathcal{U}_{\text{face}} &= 16T_1 T_3 w_1 w_3 4M \min\{2M, 6g^2 \sqrt{NM}\}. \end{aligned} \quad (\text{PK177e})$$

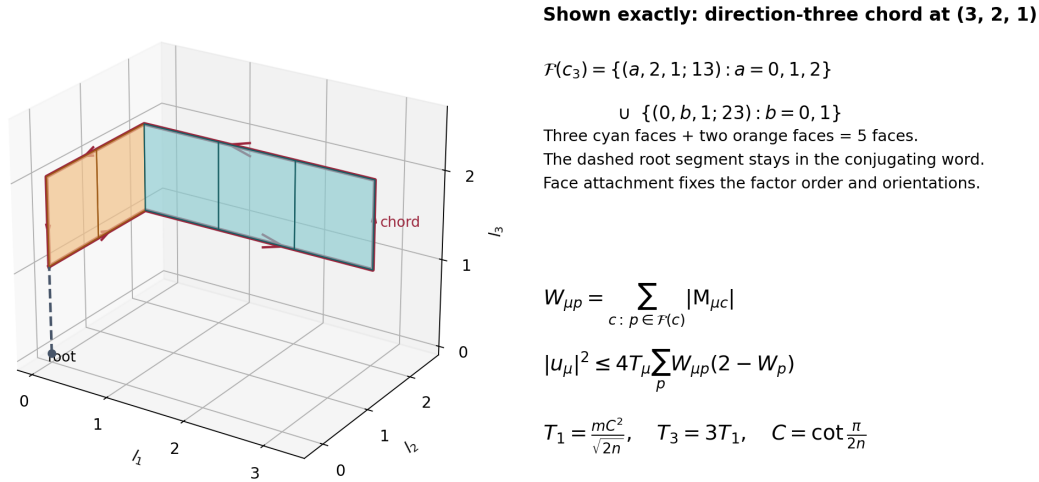
In detail, the product of the two weighted face sums is at most $w_1 w_3 (\sum_p D_p)^2$. Its actual expectation is at most $4M\mu(\sum_p D_p) \leq 4ME_{0,g}/b$. The bound $E_{0,g}/b \leq \min\{2M, 6g^2 \sqrt{NM}\}$ retains the original vacuum, not a Wilson-density replacement. The first bound uses $|u_\mu| \leq 2S_\mu$.

These bounds are now fully evaluated on PK139. Elementary $\sin x/x \rightarrow 1$ and $\cos x \rightarrow 1$, applied to the exact formulas above with $m = 2j^4, n = 2j^4 + 1$, give

$$\frac{\mathcal{U}_{\text{point}}}{j^{24}} \rightarrow \frac{2^{22}}{\pi^8}, \quad \frac{\mathcal{U}_{\text{face}}}{j^{19}\mathcal{U}_{\text{point}}} \rightarrow \frac{81\pi^2\kappa_*}{50}, \quad \frac{\beta_{0,j}^2\mathcal{U}_{\text{point}}}{j^{18}} \rightarrow \frac{5000\,2^{22}}{\kappa_*^2\pi^6}. \quad (\text{PK177f})$$

For example $C/n \rightarrow 2/\pi$, $S_1/n^{3/2} \rightarrow 4\sqrt{2}/\pi^2$, $S_3/n^{3/2} \rightarrow 8\sqrt{2}/\pi^2$, and $T_1/n^{5/2} \rightarrow 2\sqrt{2}/\pi^2$. Also $w_1\sqrt{n} \rightarrow 2\sqrt{2}/\pi$, $w_3/\sqrt{n} \rightarrow 2\sqrt{2}/\pi$; $N = 3m(m+1)^2, M = 3m^2(m+1)$ retain their full boundary terms throughout. The smaller potential branch is eventually $6g_j^2\sqrt{N_jM_j}$, since its ratio to $2M_j$ is $3g_j^2\sqrt{N_j/M_j} \rightarrow 0$. Substitution gives all three constants in PK177f. Thus the rigorous estimate $\Gamma_j(1) \leq \min\{1, \beta_{0,j}^2 \min(\mathcal{U}_{\text{point}}, \mathcal{U}_{\text{face}})\}$ eventually gives only its already known upper bound one. This proves the precise limitation of this attempted use of the full actual plaquette moments. It proves neither growth nor decay of the actual $\mathcal{Q}_{4,j}$.

Every face in the original rooted chord



$m = 4, L = 2, n = 5$ in the drawing. Original physical coordinates: $o + a(l_1 - L, l_2 - L, l_3 - L)$.

PK177a-PK177f prove the ordered non-Abelian filling, all face weights and the actual-vacuum bound. The bound retains every face and boundary term; it does not settle the original-path fourth-moment growth.

Figure 8: Exact ordered face filling of an original direction-three chord

The drawing is the exact five-face bent strip for $m=4$ and chord $(3,2,1;3)$, with the original root segment and physical coordinate map retained. PK177a-PK177f prove every face weight and the actual-vacuum bound. Reproducible figure source.

9.27. The original reflected phase is not its negative globally

Fix one of the original finite regulators $L \geq 2, a > 0, g > 0$. Throughout this section the measure is the actual vacuum measure $d\mu = \psi_g^2 dU$, and the excitation operator is $\mathcal{A} = H_g - E_{0,g}$. Set

$$\beta_0 = \frac{\sigma_*}{2g^2}, \quad p(U) = \beta_0 \mathcal{S}(U), \quad r(U) = p(\mathcal{R}U), \quad F_A(U) = \frac{\sin(Ap(U)) - \sin(Ar(U))}{2}. \quad (\text{PK178})$$

Thus $F_A \psi_g$ is exactly PK145, with its original amplitude A , not a vector obtained by dividing out its raw norm. Every tree word, coupling and reflection remains unchanged. Both p and r are real analytic:

they are finite sums of traces of products and inverses of the original group variables. They are gauge invariant. The reflection interchanges them.

It is necessary to determine the global relation between them before taking a large-amplitude limit. Their quadratic Taylor terms have opposite signs by PK146. Globally, however, $p + r$ is not zero. Here is an exact configuration proving this for every $L \geq 2$. Use $m = 2L, n = m + 1$, the original root and tree, and the functions v_0, v, w of PK158. Write

$$d_0 = \frac{v_0(0)w(0)v(0)}{\sqrt{2}}, \quad d_0^2 = \frac{2 \cos^2(\pi/(2n)) \sin^2(\pi/n)}{n^3} > 0. \quad (\text{PK179})$$

The letter d_0 here denotes this displayed coefficient, not the number $d = 2m^3 + 3m^2$ of chords. Set all physical links to I , except the two direction-two chords based at $(m, 0, 0)$ and $(L, 0, 0)$, whose values are respectively

$$h = \exp(\pi T_1) = 2T_1, \quad k = \exp(2\pi T_1) = -I.$$

Coordinates in this display are the vertex indices l ; the physical vertices remain $o + a(l_1 - L, l_2 - L, l_3 - L)$. All tree links are initially I . The first chord has $q(h) = 2(1, 0, 0)$, and the second has $q(k) = 0$. Since $v(m) = -v(0)$ and $v(L) = 0$, PK160 gives

$$u_1(U) = 2d_0(1, 0, 0), \quad u_3(U) = 2d_0(1, 0, 0).$$

For completeness, track the reflected words, including their tree segments. Reflection moves the link h to the direction-two tree edge based at $(0, 0, 0)$, and fixes the location of the middle chord k . First omit k . With only one physical link different from I , each rooted word is I, h , or h^{-1} : each simple root path uses this edge at most once, and if both use it their identical contribution cancels in this particular word. Thus each of its q -vectors is exactly $0, q(h)$, or $-q(h)$. The coefficient of this one $q(h)$ is the derivative at the identity. PK143a and the original cochain reflection rule PK108 therefore give the exact single-link values $u_1(\mathcal{R}U) = -2d_0(1, 0, 0)$ and $u_3(\mathcal{R}U) = 2d_0(1, 0, 0)$ before restoring k . This argument uses the full linearized tree map only to determine the coefficients of an expression already proved exactly linear in this single $q(h)$.

Restore the middle chord k . A chord is absent from every tree path, so exactly its own rooted chord word changes. Its starting root path is I ; its ending root path is h . Its previous word h^{-1} becomes $kh^{-1} = h$. Its q -vector consequently increases by $4(1, 0, 0)$. The first-row coefficient of this chord is zero and the third-row coefficient is d_0 . Hence the full values are

$$\begin{aligned} u_1(\mathcal{R}U) &= -2d_0(1, 0, 0), & u_3(\mathcal{R}U) &= 6d_0(1, 0, 0), \\ p(U) &= 4\beta_0 d_0^2, & r(U) &= -12\beta_0 d_0^2, \\ (p + r)(U) &= -8\beta_0 d_0^2, & (p - r)(U) &= 16\beta_0 d_0^2. \end{aligned} \quad (\text{PK180})$$

At the identity configuration all four functions $p, r, p + r, p - r$ vanish. PK180 proves that each is nonconstant. It also proves the precise defect of extending the quadratic reflection sign to the global coordinates.

9.28. Actual amplitude limits, all raw moments, and escaping weight

We first supply the oscillatory integral argument used below. On a connected compact real analytic manifold with smooth volume, a nonconstant real analytic function s has a critical set of volume zero. To see the required zero-set fact, work in a coordinate box and expand a nonzero real analytic function in the last variable around an interior slice. At least one coefficient is a nonzero analytic function of

the other variables. By induction on dimension its zero set has measure zero. Outside that exceptional set the one-variable analytic function is not identically zero and has isolated, hence countably many, zeros. Fubini proves zero measure in the box. Smaller boxes on which the series converges give a countable cover. In dimension one this is precisely the isolated-zero property from the first nonzero Taylor coefficient. Analytic continuation across overlapping coordinate boxes shows that a function zero on an open set of a connected manifold is zero everywhere. Accordingly, for a nonconstant s , in each connected coordinate box some first derivative is not identically zero. The critical set in that box is contained in its zero set. Smooth positive volume densities preserve these null sets.

For any integrable weight w , this proves

$$\int w(U) e^{iAs(U)} dU \longrightarrow 0 \quad (A \longrightarrow +\infty). \quad (\text{PK181})$$

Here are the integration details, so that no stationary-phase estimate is assumed. First take w smooth. Choose a smooth cutoff χ_δ , equal to zero where $|\nabla s|^2 \leq \delta^2$ and one where $|\nabla s|^2 \geq 2\delta^2$. The vector field $\chi_\delta w \nabla s / |\nabla s|^2$, extended by zero near the critical set, is smooth. Integration by parts on the compact manifold gives

$$\left| \int \chi_\delta w e^{iAs} dU \right| \leq \frac{1}{A} \int \left| \operatorname{div} \frac{\chi_\delta w \nabla s}{|\nabla s|^2} \right| dU.$$

For fixed $\delta > 0$ the last integral is finite. The omitted integral is bounded by $\int_{\{|\nabla s|^2 < 2\delta^2\}} |w| dU$, which tends to zero as $\delta \downarrow 0$ by the null critical set and dominated convergence. Take $A \rightarrow \infty$ first and then $\delta \downarrow 0$. General L^1 weights follow by smooth approximation in finitely many coordinate charts and the bound $|e^{iAs}| = 1$. This proves PK181 for the original compact product $SU(2)^N$. In particular it applies to weights containing the actual ψ_g .

Let

$$\Gamma(A) = \|F_A \psi_g\|^2, \quad \nu_A(B) = \langle F_A \psi_g, \mathbf{1}_B(\mathcal{A}) F_A \psi_g \rangle.$$

The exact reflection invariance of μ gives

$$\Gamma(A) = \frac{1}{4} [1 - \mu(\cos(2Ap)) - \mu(\cos(A(p-r))) + \mu(\cos(A(p+r)))] \longrightarrow \frac{1}{4}. \quad (\text{PK182})$$

The equality follows by expanding the entire squared half-difference, using $2 \sin x \sin y = \cos(x-y) - \cos(x+y)$, and retaining all three cosine terms. Each nonconstant phase was proved in PK180; PK181 gives its actual-vacuum limit. This holds at every positive coupling and every original finite box.

It has an exact order-of-limits consequence. PK129/PK148 prove, for each fixed A ,

$$\lim_{g \downarrow 0} \Gamma(A) = \frac{1}{2} \left[1 - (1 + 4A^2)^{-3/2} \right].$$

At fixed L, a , the two iterated limits are therefore

$$\lim_{g \downarrow 0} \lim_{A \rightarrow \infty} \Gamma(A) = \frac{1}{4}, \quad \lim_{A \rightarrow \infty} \lim_{g \downarrow 0} \Gamma(A) = \frac{1}{2}. \quad (\text{PK183})$$

The nonlinear reflected words, already evaluated in PK180, cannot be omitted in the first limit. No uniformity in g is inferred.

The complete first two energy moments can also be evaluated in this amplitude limit. Write

$$G_{ij} = \nabla p_i \cdot \nabla p_j, \quad (p_1, p_2) = (p, r), \quad \mathcal{L} = \kappa[H_0 - 2\nabla \log \psi_g \cdot \nabla], \quad d_i = \mathcal{L} p_i. \quad (\text{PK184})$$

Every gradient sums all original links and all three original colour generators. Thus $G_{11} = \beta_0^2 \sum_{e,\alpha} (X_{e,\alpha} \mathcal{S})^2$; the full word derivatives are PK151–PK153. Reflection sends G_{11} to G_{22} and fixes the integral of G_{12} . The exact formulas before any limit are

$$\begin{aligned} m_1(A) &= \frac{\kappa A^2}{4} \mu [\cos^2(Ap)G_{11} + \cos^2(Ar)G_{22} - 2 \cos(Ap) \cos(Ar)G_{12}] , \\ m_2(A) &= \frac{1}{4} \mu (A[\cos(Ap)d_1 - \cos(Ar)d_2] + \kappa A^2[\sin(Ap)G_{11} - \sin(Ar)G_{22}])^2 . \end{aligned} \quad (\text{PK185})$$

These follow from PK155 and its full ground-state product rule. In particular the sine term in the second line has its positive sign from $H_0 = -\sum X^2$. No score, mixed term or reflected word has been removed.

All G_{ij}, d_i are bounded smooth functions at this fixed regulator. Apply PK181 to the phases $2p, p - r, p + r$, with weights $G_{ij}\psi_g^2$ or $G_{ij}G_{kh}\psi_g^2$. In the second moment, the terms with factors A^2 and A^3 vanish after division by A^4 by boundedness. The two squared sine terms each tend to half their weighted integrals, and the cross term tends to zero. Consequently

$$\begin{aligned} \frac{m_1(A)}{A^2} &\longrightarrow \frac{\kappa}{4} \mu(G_{11}) > 0, \\ \frac{m_2(A)}{A^4} &\longrightarrow \frac{\kappa^2}{4} \mu(G_{11}^2) > 0, \\ \frac{m_1(A)}{A^2 \Gamma(A)} &\longrightarrow \kappa \mu(G_{11}), \\ \frac{m_2(A)}{A^4 \Gamma(A)} &\longrightarrow \kappa^2 \mu(G_{11}^2). \end{aligned} \quad (\text{PK186})$$

Nonconstancy of p and strict positivity of the actual vacuum make both displayed integrals strictly positive. The raw moments and raw norm remain explicit, including the factors $1/4$. These equalities do not replace the raw vector by a unit vector.

There is a stronger spectral statement than growth of moments. For any $h \in L^2(dU)$, $\bar{h}\psi_g \in L^1(dU)$; PK181 applied separately to p and r shows $F_A \psi_g \rightarrow 0$. The spectral projection of the original elliptic compact Hamiltonian onto a bounded energy interval has finite rank, also after restriction to physical odd vectors. Weak convergence and an orthonormal basis of that finite-dimensional range give

$$\nu_A([0, \Lambda]) \longrightarrow 0 \quad \text{for every finite } \Lambda \geq 0, \quad \nu_A([0, \infty)) \longrightarrow \frac{1}{4}. \quad (\text{PK187})$$

This concerns the full interacting spectral measure.

The same result produces an actual sequence on the original path PK139, using the amplitude already present in the full higher-carrier map. For each integer $j \geq 2$, define the integer

$$A_j = \min \left\{ n \in \mathbb{N} : n \geq j, |\Gamma_j(n) - 1/4| < \frac{1}{8j}, \nu_{j,n}([0, \kappa_* j]) < \frac{1}{j} \right\}. \quad (\text{PK188})$$

The set is nonempty by PK182 and PK187 at that particular finite regulator, so this is an unconditional, well-defined choice from the actual operator. It uses neither a Gaussian vacuum nor an unknown limiting gap. It gives the exact conclusions

$$\begin{aligned} \Gamma_j(A_j) &\longrightarrow \frac{1}{4}, \quad \nu_{j,A_j}([0, \Lambda]) \longrightarrow 0 \quad (\Lambda < \infty), \\ m_{1,j}(A_j) &\geq \kappa_* \left(\frac{j}{4} - \frac{9}{8} \right), \quad m_{2,j}(A_j) \geq \kappa_*^2 \left(\frac{j^2}{4} - \frac{9j}{8} \right). \end{aligned} \quad (\text{PK189})$$

Indeed, the raw mass above $\kappa_* j$ is at least $1/4 - 1/(8j) - 1/j$; integrate E or E^2 there. For a fixed finite Λ , eventually $\Lambda \leq \kappa_* j$. One obtains an actual point of the original rank-24 domain over any chosen base point by taking one selected quaternion of length $\sqrt{A_j}$, the other two selected quaternions zero, and specified zero values for the other coordinates. Its invariant scalar $\mathfrak{a}$ is exactly A_j . The full domain map and its other fibres remain PK149–PK150. Thus amplitude can retain raw mass, but this particular proved choice sends that mass to high energies. It does not alter the unfinished fixed- $A = 1$ calculation.

9.29. The exact two-phase space and its actual energy form

We continue from PK189 by constructing a space in which the phase profile can be varied while its full raw norm and energy are kept. Define

$$\Theta : \mathcal{Q}_L \longrightarrow \mathbb{R}^2, \quad \Theta(U) = (p(U), r(U)), \quad K = \Theta(\mathcal{Q}_L), \quad \rho = \Theta_* \mu. \quad (\text{PK190})$$

The compact image is contained in $[-T, T]^2$, where $T = 4\beta_0 S_1 S_3$, using the exact sums PK162. Reflection sends $\Theta(U)$ to $(r(U), p(U))$, so both K and ρ are invariant under interchange of the two coordinates. No rank, Lebesgue density or nonsingularity of this pushforward is assumed.

Let $L_-^2(\rho)$ consist of the functions satisfying $f(y, x) = -f(x, y)$ as equivalence classes. The map

$$J : L_-^2(\rho) \longrightarrow L^2(\mathcal{Q}_L, dU)_{\text{physical, odd}}, \quad (Jf)(U) = \psi_g(U)f(\Theta(U)) \quad (\text{PK191})$$

is an isometry, since $\|Jf\|^2 = \int |f|^2 d\rho$ by the definition of pushforward. It therefore has closed range. Gauge invariance follows from that of p, r, ψ_g ; oddness follows from the coordinate interchange. Its vectors are exactly vacuum orthogonal because $\int f d\rho = 0$. In particular $f_A(x, y) = [\sin(Ax) - \sin(Ay)]/2$ maps to the original $F_A \psi_g$. This proves the exact inclusion of the previous amplitude family, with every original scalar retained.

For a physical odd vector h , its adjoint is

$$(J^* h)(x, y) = \mathbb{E}_\mu[h/\psi_g \mid \Theta = (x, y)]. \quad (\text{PK192})$$

Conditional expectation is defined here as the orthogonal projection in $L^2(\mu)$ onto functions measurable with respect to Θ . The quotient is legitimate because $\psi_g > 0$ and $\int |h/\psi_g|^2 d\mu = \int |h|^2 dU$. Testing against bounded functions of Θ proves the adjoint identity; density extends it to L^2 . Reflection invariance proves its antisymmetry. Thus JJ^* is the exact orthogonal projection onto the constructed range; it is not asserted to commute with $\mathcal{A}$.

Construct finite signed measures on K by integrating the functions G_{ij} and d_i of PK184 over inverse images of Borel sets. They are absolutely continuous with respect to ρ : their densities are bounded by the corresponding suprema on $\mathcal{Q}_L$. Denote their Radon–Nikodym densities by

$$a_{ij} = \mathbb{E}_\mu[G_{ij} \mid \Theta], \quad b_i = \mathbb{E}_\mu[d_i \mid \Theta] \quad (1 \leq i, j \leq 2). \quad (\text{PK193})$$

The matrix a is real symmetric and positive semidefinite ρ -almost everywhere. To verify the last assertion, for every rational vector $z \in \mathbb{Q}^2$, the defining measure of $z^T a z$ integrates $|z_1 \nabla p + z_2 \nabla r|^2 \geq 0$. Its density is nonnegative. Remove the union of the resulting countably many null sets and pass to real vectors by continuity.

For restrictions to K of smooth antisymmetric functions on $\mathbb{R}^2$, the complete actual form is

$$\begin{aligned} \mathfrak{q}[f, h] &:= \langle Jf, \mathcal{A}Jh \rangle \\ &= \kappa \int_K [a_{11} \overline{f_1} h_1 + a_{12} \overline{f_1} h_2 + a_{21} \overline{f_2} h_1 + a_{22} \overline{f_2} h_2] d\rho. \end{aligned} \quad (\text{PK194})$$

Here f_i and h_i are the ordinary coordinate derivatives before restriction. The ground-state identity PK154 gives the first line as $\kappa \int \nabla(f \circ \Theta)^* \nabla(h \circ \Theta) d\mu$; the full chain rule and PK193 give the second line. This also proves independence of the chosen smooth extension: a smooth function zero ρ -almost everywhere pulls back to a smooth function zero μ -almost everywhere, hence everywhere since $\psi_g > 0$ and Haar has full support. Its derivatives on $\mathcal{Q}_L$ are zero. Individual $f_i|_K$ need not be determined by $f|_K$, but the displayed contracted form is.

The form is closable in $L_-^2(\rho)$. In fact, a form-Cauchy sequence f_n converging to zero in $L^2(\rho)$ has Jf_n Cauchy in the closed excitation form and converging to zero in the original Hilbert norm. Closedness there forces $\mathfrak{q}[f_n, f_n] \rightarrow 0$. Take this closure as the form domain $\mathcal{D}_{\text{ph}}$. It is dense in $L_-^2(\rho)$: continuous functions on the compact support are dense in $L^2(\rho)$, their extensions to a containing rectangle are uniformly approximated by polynomials, and averaging $[f(x, y) - f(y, x)]/2$ preserves approximation to an antisymmetric function. Smooth functions agreeing with these polynomials near the compact rectangle give the stated core.

At this fixed regulator, the form norm of $\mathcal{A}$, plus the original L^2 norm, is equivalent to the H^1 norm on the compact link manifold: the potential and $E_{0,g}$ are bounded and $\kappa > 0$. Its embedding into L^2 is compact. For example, expansion in the complete free product-Casimir eigenbasis bounds the squared norm of the tail above electric energy R by the bounded electric form divided by R ; each remaining eigenspace is finite dimensional. Finite-dimensional subsequence extraction then proves compactness. Because J is an isometry, the closed phase form also has compact embedding and defines a nonnegative self-adjoint operator with compact resolvent, denoted B_{ph} . Explicitly, on the Hilbert space with inner product $\mathfrak{q}[f, h] + \langle f, h \rangle_\rho$, the Riesz representation theorem assigns to each $u \in L_-^2(\rho)$ a unique Ru satisfying $\mathfrak{q}[h, Ru] + \langle h, Ru \rangle_\rho = \langle h, u \rangle_\rho$ for every form-domain h . This operator is bounded, positive, self-adjoint and injective on $L_-^2(\rho)$; injectivity follows from density of the form domain. Its range is dense because its orthogonal complement is its kernel. The just-proved compact embedding makes R compact. Its orthogonal spectral decomposition therefore defines $B_{\text{ph}} = R^{-1} - I$, with domain consisting of the spectral coefficient sequences for which this expression is square summable. The variational identity gives precisely the closed form above, proves self-adjointness and nonnegativity, and identifies $(B_{\text{ph}} + I)^{-1} = R$. Its lowest eigenvalue is strictly positive. Otherwise a sequence of phase functions with norm one and energies tending to zero would have, after applying J , a strongly convergent subsequence. Its limit would have norm one, be odd, and lie in the zero eigenspace of $\mathcal{A}$, which consists only of the even vacuum. This is impossible.

9.30. Full second moment, residual interaction, and finite receiving matrices

On the smooth phase core the full chain rule, before conditional expectation, is

$$\mathcal{L}(f \circ \Theta) = \sum_{i=1}^2 d_i f_i(\Theta) - \kappa \sum_{i,j=1}^2 G_{ij} f_{ij}(\Theta). \quad (\text{PK195})$$

The sign of the Hessian term follows from $H_0 = -\sum X^2$. Define the measurable phase function and the complete residual on the original configuration manifold by

$$\begin{aligned} Bf &= b_1 f_1 + b_2 f_2 - \kappa(a_{11} f_{11} + a_{12} f_{12} + a_{21} f_{21} + a_{22} f_{22}), \\ R_f(U) &= \sum_{i=1}^2 [d_i(U) - b_i(\Theta(U))] f_i(\Theta(U)) \\ &\quad - \kappa \sum_{i,j=1}^2 [G_{ij}(U) - a_{ij}(\Theta(U))] f_{ij}(\Theta(U)). \end{aligned} \quad (\text{PK196})$$

Every term is square integrable at a fixed finite regulator. Their exact conditional expectation identities give $\mathbb{E}_\mu[R_f \mid \Theta] = 0$. Integration against another smooth phase function, followed by PK194, proves $B_{\text{ph}}f = Bf$: the functional $\mathbf{q}[h, f]$ is represented by this bounded $L^2(\rho)$ function for every core h , and extends to the closed form domain. This also proves that Bf is independent of the extension of f . The original excitation vector decomposes exactly as

$$\mathcal{A}Jf = J(B_{\text{ph}}f) + \psi_g R_f, \quad J^*(\psi_g R_f) = 0. \quad (\text{PK197})$$

Thus the complete raw moments for these new profiles are

$$\begin{aligned} m_0[f] &= \int_K |f|^2 d\rho, \\ m_1[f] &= \kappa \int_K \sum_{i,j=1}^2 a_{ij} \overline{f_i} f_j d\rho, \\ m_2[f] &= \int_K |B_{\text{ph}}f|^2 d\rho + \int_{\mathcal{Q}_L} |R_f|^2 d\mu. \end{aligned} \quad (\text{PK198})$$

The second term in m_2 is the exact coupling to the complement of the phase range. It cannot be omitted by calling the phase operator the whole Hamiltonian. PK197 proves their connecting map. For f_A , these formulas recover every term of PK185.

There is a finite, convergent variational construction within this space. For integers $0 \leq q < p$, put

$$\begin{aligned} P_{pq}(x, y) &= x^p y^q - x^q y^p, \quad d_\ell(t) = \begin{cases} 0, & \ell = 0, \\ \ell t^{\ell-1}, & \ell \geq 1, \end{cases} \\ D_{1,pq} &= d_p(x) y^q - d_q(x) y^p, \\ D_{2,pq} &= x^p d_q(y) - x^q d_p(y). \end{aligned} \quad (\text{PK199})$$

The separate zero case retains the constant monomials without introducing undefined negative powers at $x = 0$ or $y = 0$. For all p, q, s, t in this indexing set define

$$\begin{aligned} \mathcal{G}_{pq, st} &= \int_K [x^{p+s} y^{q+t} - x^{p+t} y^{q+s} - x^{q+s} y^{p+t} + x^{q+t} y^{p+s}] d\rho, \\ \mathcal{C}_{pq, st} &= \kappa \int_K [a_{11} D_{1,pq} D_{1, st} + a_{12} D_{1,pq} D_{2, st} \\ &\quad + a_{21} D_{2,pq} D_{1, st} + a_{22} D_{2,pq} D_{2, st}] d\rho. \end{aligned} \quad (\text{PK200})$$

These are exact Gram and energy matrices in the actual interacting vacuum; all four mixed terms in each expression remain. For $f = \sum c_{pq} P_{pq}$, the original raw norm and energy are exactly $c^* \mathcal{G} c$ and $c^* \mathcal{C} c$. A vector in the kernel of $\mathcal{G}$ gives a smooth pullback identically zero on $\mathcal{Q}_L$, by the full-support argument after PK194. It is also in the kernel of $\mathcal{C}$. Hence quotienting only this exact common zero vector space makes the following finite minimum well-defined:

$$\begin{aligned} \lambda_R^{\text{ph}} &= \min_{\substack{c^* \mathcal{G}_{RC} > 0 \\ 0 \leq q < p \leq R}} \frac{c^* \mathcal{C}_{RC} c}{c^* \mathcal{G}_{RC} c}, \\ 0 < \Delta_{\text{odd}} &\leq \lambda_{\text{ph}} \leq \lambda_{R+1}^{\text{ph}} \leq \lambda_R^{\text{ph}}, \\ \lambda_R^{\text{ph}} &\downarrow \lambda_{\text{ph}}. \end{aligned} \quad (\text{PK201})$$

Here Δ_{odd} is the actual lowest odd excitation of the original Hamiltonian and λ_{ph} is the lowest eigenvalue of the proved phase-form operator. The space for $R = 1$ is nonzero because $p - r$ is nonconstant and has

nonzero square integral. Every such trial vector belongs to the actual physical odd form domain, proving the first variational inequality. Increasing R increases the finite trial space, proving monotonicity.

To prove the final limit with its required form density, approximate a smooth core function on a rectangle containing K , together with its first derivatives, by two-variable polynomials. One direct construction uses tensor Bernstein polynomials after the affine coordinate map from $[0, 1]^2$ onto $[-\tau, \tau]^2$, where $\tau = T + 1$:

$$(s, t) \mapsto (2\tau s - \tau, 2\tau t - \tau), \quad \tilde{f}(s, t) = f(2\tau s - \tau, 2\tau t - \tau).$$

Write $b_{R,i}(s) = \binom{R}{i} s^i (1-s)^{R-i}$. The approximating polynomial is $\sum_{i,j=0}^R \tilde{f}(i/R, j/R) b_{R,i}(s) b_{R,j}(t)$. Its first s -derivative is exactly

$$R \sum_{i=0}^{R-1} \sum_{j=0}^R [\tilde{f}((i+1)/R, j/R) - \tilde{f}(i/R, j/R)] b_{R-1,i}(s) b_{R,j}(t);$$

the t -derivative has the analogous displayed-coordinate interchange. The fundamental theorem of calculus writes each bracket times R as the integral average of that first derivative over the interval of length $1/R$. Uniform continuity of that derivative, and the binomial variance bound $t(1-t)/R \leq 1/(4R)$, prove uniform derivative convergence; the use of degree $R-1$ shifts the corresponding mean by at most $1/R$, which also tends to zero. The same binomial argument proves function convergence. Return to the original coordinates by $s = (x+\tau)/(2\tau)$, $t = (y+\tau)/(2\tau)$; each first derivative has the factor $1/(2\tau)$. Antisymmetrize the polynomials. Their coefficients then pair exactly into the P_{pq} of PK199. Since ρ is finite and a is bounded, uniform function and derivative convergence implies convergence in $m_0 + \mathfrak{q}$. The core defines the closed form, so these polynomial spaces are dense in that full form domain. Approximate a lowest phase eigenfunction in this norm; its Rayleigh quotients converge to λ_{ph} . Together with the preceding lower bound this proves PK201.

The first receiving matrix is particularly explicit in the original objects. Let

$$\mathcal{S}_{\mathcal{R}} = \mathcal{S} \circ \mathcal{R}, \quad \mathcal{Q}_{4,\mathcal{R}} = \mu(\mathcal{S}\mathcal{S}_{\mathcal{R}}), \quad \mathcal{K}_1 = \mu(|\nabla \mathcal{S}|^2), \quad \mathcal{K}_{\mathcal{R}} = \mu(\nabla \mathcal{S} \cdot \nabla \mathcal{S}_{\mathcal{R}}).$$

For $P_{10} = x - y$ one has exactly

$$\begin{aligned} \mathcal{G}_{10,10} &= 2\beta_0^2(\mathcal{Q}_4 - \mathcal{Q}_{4,\mathcal{R}}) > 0, \\ \mathcal{C}_{10,10} &= 2\kappa\beta_0^2(\mathcal{K}_1 - \mathcal{K}_{\mathcal{R}}) > 0, \\ \lambda_1^{\text{ph}} &= \frac{2\kappa\beta_0^2(\mathcal{K}_1 - \mathcal{K}_{\mathcal{R}})}{2\beta_0^2(\mathcal{Q}_4 - \mathcal{Q}_{4,\mathcal{R}})}. \end{aligned} \tag{PK202}$$

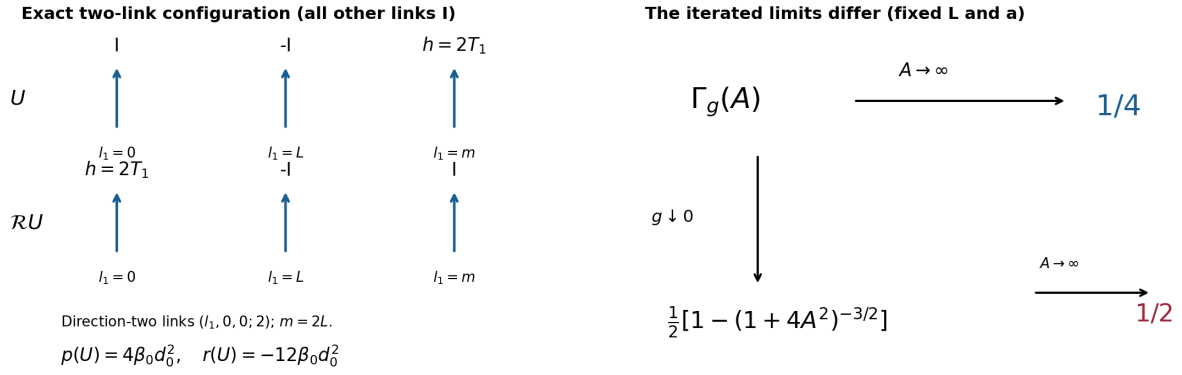
Reflection invariance supplies the two equal diagonal terms in each line; both cross terms remain. Strict positivity follows from the nonzero odd smooth function $p-r$, the full-support vacuum measure and the unique even zero eigenspace. PK170–PK171 give the full loop expansion for $\mathcal{Q}_4$; replace its second pair of chord words by their reflected words to obtain $\mathcal{Q}_{4,\mathcal{R}}$, with the same four trace terms and every ordered index tuple. PK151–PK153 give every derivative in the two $\mathcal{K}$ terms. Thus this finite entry involves the actual four-loop and gradient calculations already identified, and retains the complete $2, \kappa, \beta_0^2$ factors in its norm and energy.

The two-link configuration is evaluated exactly in PK179–PK180. The limits are proved in PK181–PK189; no plotted vacuum sample is used. PK190–PK202 prove the phase-space isometry, full Hamiltonian action, residual second moment and finite variational matrices. Reproducible figure source.

9.31. Full phase rank, actual density, and all boundary flux terms

The preceding construction did not require a rank hypothesis. For the present original objects its rank can in fact be proved. Keep the two physical links used in PK180, but allow their values to vary through

Full reflection, amplitude limits and the next variational space



Actual phase space and the full Hamiltonian action

$$(\mathcal{Q}_L, \mu = \psi_g^2 dU) \xrightarrow{\Theta = (p, r)} (K, \rho = \Theta_* \mu)$$

$$Jf = \psi_g f(p, r) \quad \mathcal{A}f = J(B_{\text{ph}} f) + \psi_g R_f$$

$$m_0 = \int |\hat{f}|^2 d\rho, \quad m_1 = \kappa \int \sum_{i,j=1}^2 a_{ij} \bar{f}_i f_j d\rho$$

$$m_2 = \int |B_{\text{ph}} \hat{f}|^2 d\rho + \int |R_f|^2 d\mu, \quad \mathbb{E}_\mu[R_f | \Theta] = 0$$

PK178-PK206 contain the complete proofs. High-amplitude raw mass survives but escapes every bounded energy interval. The phase form and finite polynomial matrices retain the actual vacuum and the entire residual interaction.

Figure 9: Full reflection, actual amplitude limits and complete phase variational map

$h(t) = \exp(tT_1)$ and $k(s) = \exp(sT_1)$. Put

$$u(t) = 2 \sin(t/2), \quad v(s) = 2 \sin(s/2), \quad w(t, s) = 2 \sin((s - t)/2).$$

These are the full group coordinates, including their 4π periods. The first-row coefficient of the middle chord is zero. Before reflection the two relevant mode vectors are $d_0 u(1, 0, 0)$ and $d_0(u + v)(1, 0, 0)$. After reflection their first components are $-d_0 u$ and $d_0(2u + w)$: the middle chord's word changes from h^{-1} to kh^{-1} , whose coordinate is w . Consequently the complete restricted phase map is

$$p(t, s) = \beta_0 d_0^2 u(t)[u(t) + v(s)], \quad r(t, s) = -\beta_0 d_0^2 u(t)[2u(t) + w(t, s)]. \quad (\text{PK203})$$

At $(t, s) = (\pi, 2\pi)$ it gives exactly PK180. At the different point $(t, s) = (\pi/2, \pi)$, it gives

$$(p, r) = ((2 + 2\sqrt{2})\beta_0 d_0^2, -6\beta_0 d_0^2),$$

$$\frac{\partial(p, r)}{\partial(t, s)} = \beta_0 d_0^2 \begin{pmatrix} 2 + \sqrt{2} & 0 \\ -4 & -1 \end{pmatrix}, \quad \det \frac{\partial(p, r)}{\partial(t, s)} = -(2 + \sqrt{2})\beta_0^2 d_0^4 \neq 0. \quad (\text{PK204})$$

For example $u = \sqrt{2}, v = 2, w = \sqrt{2}$ at this point, $u_t = \sqrt{2}/2, v_s = 0, w_t = -\sqrt{2}/2, w_s = \sqrt{2}/2$; inserting these four derivatives into both full products in PK203 proves all four matrix entries and its determinant. The inverse function theorem therefore puts an open neighbourhood of the displayed phase value inside K .

On the full link manifold the nonnegative analytic function $\det(G_{ij})$ is not identically zero, because PK204 gives two independent differentials at one configuration. The analytic zero-set argument preceding PK181 proves that its zero set has Haar measure zero, hence actual-vacuum measure zero. Thus the full phase map has rank two μ -almost everywhere. These facts give the strengthenings

$$d\rho(x, y) = \varpi(x, y) dx dy, \quad \varpi \geq 0, \quad \int_{\mathbb{R}^2} \varpi dx dy = 1, \quad \rho(\partial K) = 0,$$

$$a(x, y) > 0 \quad \rho\text{-almost everywhere.} \quad (\text{PK205})$$

Here $a > 0$ means positive definite; no uniform ellipticity constant on all of K is asserted.

To prove absolute continuity, cover the rank-two set by countably many coordinate neighbourhoods in which $(p, r, z_3, \dots, z_{3N})$ is a smooth coordinate system. Such neighbourhoods exist by the inverse function theorem, retaining the other local coordinates for an invertible minor. The smooth positive actual-vacuum density acquires the full coordinate Jacobian in each neighbourhood. A partition into countably many measurable chart pieces, followed by Fubini in these exact coordinates, shows that the inverse image of every planar Lebesgue-null set has μ -measure zero there. The complementary critical set has already been proved null. This proves the stated Radon–Nikodym density on the whole plane, extended by zero outside K ; its integral is one by the pushforward definition, with no mass rescaling. If a point of ∂K had a regular preimage, the submersion neighbourhood at that preimage would put an open neighbourhood of the point inside K , a contradiction. Thus the boundary inverse image is contained in the null critical set, proving $\rho(\partial K) = 0$.

Finally let $\ell(U)$ be the smaller eigenvalue of the matrix $G(U)$. It is bounded, measurable and strictly positive almost everywhere. Its conditional expectation $\bar{\ell} = \mathbb{E}_\mu[\ell \mid \Theta]$ is strictly positive ρ -almost everywhere: if it vanished on a set of positive ρ -measure, its defining integral would force the integral of the strictly positive ℓ over an inverse image of positive μ -measure to vanish. Conditional expectation of $G \geq \ell I$ gives $a \geq \bar{\ell} I$, proving the last assertion of PK205.

The support of ρ is exactly K . Indeed every neighbourhood of an image point has a nonempty open inverse image, which has positive μ -measure. Since K contains an open set by PK204, a nonzero

polynomial cannot vanish ρ -almost everywhere: continuity would make it zero on K , hence on an open set, and the analytic zero-set property would make it the zero polynomial. The polynomials P_{pq} are linearly independent because their ordered leading monomials $x^p y^q$, $p > q$, have distinct exponent pairs and the reversed terms have $p < q$. Therefore every finite $\mathcal{G}_R$ in PK200 is positive definite, and so is $\mathcal{C}_R$ by the actual odd spectral lower bound. For this original phase space the possible common kernel discussed before PK201 is exactly zero. This strengthening propagates PK201 to ordinary positive-definite generalized eigenvalue matrices.

The full boundary information has an exact distributional form. Regard ϖa_{ij} , ϖb_i and ϖBf as integrable functions on $\mathbb{R}^2$, extended by zero off the actual support. Then

$$\varpi b_i = -\kappa \sum_{j=1}^2 \partial_j(\varpi a_{ij}), \quad \varpi B_{\text{ph}} f = -\kappa \sum_{i,j=1}^2 \partial_j(\varpi a_{ij} f_i) \quad \text{in } \mathcal{D}'(\mathbb{R}^2). \quad (\text{PK206})$$

For the first identity, take an arbitrary smooth compactly supported test function $\eta(x, y)$. The actual ground-state identity on the compact link manifold gives

$$\int \eta \varpi b_i dx dy = \mu((\eta \circ \Theta) \mathcal{L} p_i) = \kappa \sum_j \mu(\eta_j(\Theta) G_{ij}) = \kappa \sum_j \int \eta_j \varpi a_{ij} dx dy.$$

No antisymmetry is needed for this identity; it is the same full chain and ground-state calculation as PK194. It is exactly the distributional derivative in PK206. Multiplying by the smooth f_i , applying the product rule to the distributions, and retaining all four $a_{ij} f_{ij}$ terms gives its second identity and the operator PK196. The tests range over the entire plane, so derivatives of the zero-extended coefficients retain all contributions at the actual phase boundary. No imposed Dirichlet condition, classical boundary trace, or deleted boundary flux enters this construction.

The amplitude construction has now supplied actual noncollapsing vectors and proved their high-energy escape, not the requested low-energy state. PK190–PK206 construct the next exact receiving space, its full interaction with the original Hamiltonian, and convergent finite variational matrices. The next calculation is the actual entries of these matrices and the residual term PK198 on the unchanged regulator path, starting with PK202 and the actual coefficient identities PK205–PK206. No estimate of their path limit or of continuum reconstruction is assumed. The globally smooth interacting gapless continuum state and the integrated lesson series remain the programme goal.

The grid is the full PK203 map at the displayed original regulator $L=2, a=1, g=1$. Its phase coordinates are not rescaled. PK204 proves the nonzero differential; PK205–PK206 prove actual absolute continuity, positive finite matrices and the operator identity with all boundary contributions. The plotted grid does not represent a computed vacuum density. Reproducible figure source.

9.32. Actual vacuum mass above an extensive electric cutoff

Keep all original objects of PK132: $m = 2L$, $N = 3m(m+1)^2$, $M = 3m^2(m+1)$, $a, g > 0$, $\kappa = 2g^2/a$, $b = 1/(2g^2a)$, $H = \kappa H_0 + b(2M - \mathcal{W})$, $\mathcal{W} = \sum_p W_p$, the physical unit vacuum $\psi > 0$, and $d\mu = \psi^2 dU$. In particular its full energy satisfies $E_0 \leq \min\{2bM, 3\sqrt{NM}/a\}$. No shift of H is made in the following electric estimates.

We first calculate a Haar tail for the full face sum, without treating all faces as independent. Make a graph whose vertices are the original faces and whose edges join faces sharing a link. Each face has four links; each link belongs to at most four faces; distinct faces share at most one link. Hence this graph has degree at most twelve. Greedily assigning one of thirteen colours gives classes $\mathcal{F}_1, \dots, \mathcal{F}_{13}$ in which all

An exact rank-two section of the actual phase image

Displayed original regulator: $L = 2$, $a = 1$, $g = 1$, $n = 5$; no phase coordinate rescaling.

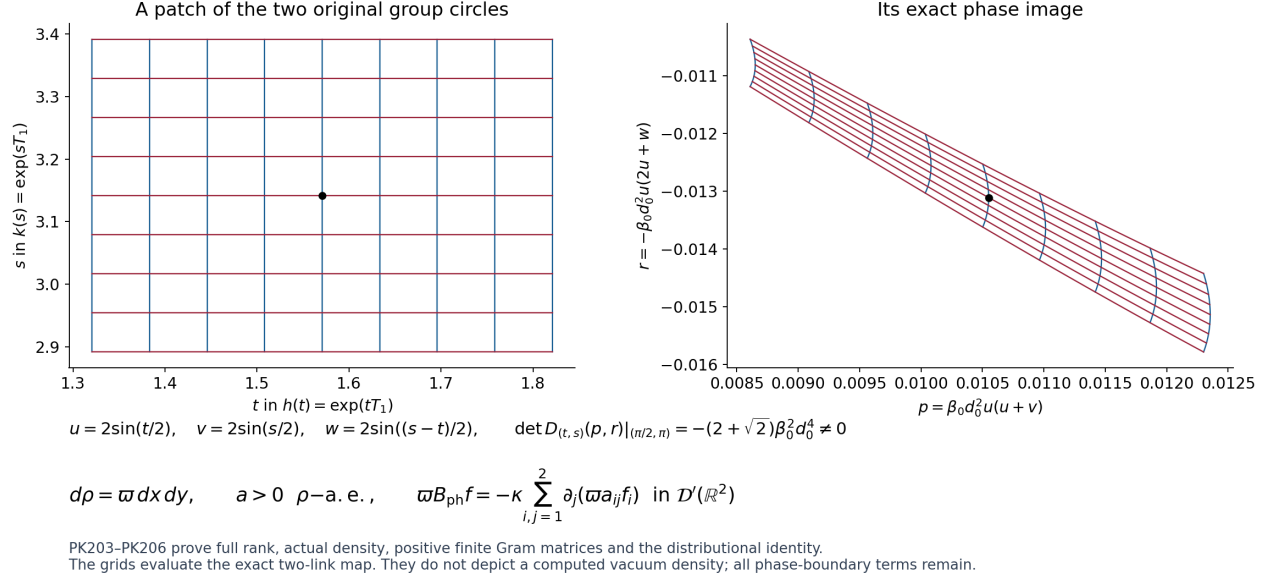


Figure 10: Exact rank-two phase section and the full distributional operator

link supports are disjoint. Empty classes are retained. In each class the traces are independent under product Haar, each has mean zero, and each lies in $[-2, 2]$.

For a mean-zero random variable $Y \in [-2, 2]$, convexity gives

$$\mathbb{E} e^{tY} \leq \mathbb{E} \left[\frac{2+Y}{4} e^{2t} + \frac{2-Y}{4} e^{-2t} \right] = \cosh(2t) \leq e^{2t^2}.$$

The last inequality follows by integrating $(\log \cosh(2t))' = 2 \tanh(2t) \leq 4t$ for $t \geq 0$; evenness covers negative t . Hölder's inequality, with all thirteen exponents equal to thirteen, now proves

$$\begin{aligned} \int e^{tW} dU &\leq \prod_{c=1}^{13} \left(\int \exp \left[13t \sum_{p \in \mathcal{F}_c} W_p \right] dU \right)^{1/13} \\ &\leq \exp(26Mt^2), \\ \int_{\{W \geq M\}} dU &\leq \exp(-M/104). \end{aligned} \tag{PK207}$$

For the last line take $t = 1/52$ in the exponential Markov inequality: the full exponent is $-M/52 + 26M/52^2 = -M/104$. This calculation retains the actual dependent face collection.

Let $\mathbf{P}_\Lambda = \mathbf{1}_{[0,\Lambda]}(H_0)$ on the physical space, now for any $\Lambda \geq 0$. Let $\mathbf{P}_\Lambda^{\text{all}}$ be the full, ungauged product-space projection and $D_\Lambda = \dim \text{Ran } \mathbf{P}_\Lambda^{\text{all}}$. In this subsection D_Λ is this dimension; the cutoff denominator of PK68 will below be written explicitly as $\kappa\Lambda - 2bM$. Every unit vector $v \in \text{Ran } \mathbf{P}_\Lambda$ satisfies

$$|v(U)|^2 \leq D_\Lambda, \quad D_\Lambda \leq e^{16\Lambda} K_{16}(I)^N \leq \exp(16\Lambda + 8Ne^{-12}). \tag{PK208}$$

Indeed the full electric basis of PK65 has constant reproducing diagonal D_Λ : on a single spin- j block, summing the squared absolute values of the entries $\sqrt{2j+1} D_{rs}^j(U)$ gives $(2j+1)^2$, by unitarity. Products

and then the complete cutoff sum give the constant diagonal. Cauchy–Schwarz in the coefficients of v proves the first inequality; the physical subspace is contained in this full space. For every retained eigenvalue $\lambda \leq \Lambda$, $1 \leq e^{16(\Lambda-\lambda)}$. Summing and retaining the whole product heat trace proves the second inequality. For its final bound put $q = e^{-12}$. In the exact one-link series, with $n = 2j \geq 1$,

$$\begin{aligned} K_{16}(I) - 1 &= \sum_{n \geq 1} (n+1)^2 e^{-4n(n+2)} \leq \sum_{n \geq 1} (n+1)^2 q^n \\ &= \frac{1+q}{(1-q)^3} - 1 \leq 8q, \quad 0 \leq q \leq 1/16. \end{aligned}$$

The last rational inequality follows after multiplication by $(1-q)^3$: the difference $8q(1-q)^3 - (4q - 3q^2 + q^3)$ is $q(4 - 21q + 23q^2 - 8q^3) \geq 0$ on this interval, since $4 - 21/16 - 8/16^3 > 0$. Finally $1 + 8q \leq e^{8q}$.

Define the actual, unshifted electric threshold

$$\Lambda_{\text{el}} = \frac{M}{4096}, \quad \alpha = \min \left\{ 1, 6g^2 \sqrt{\frac{N}{M}} \right\}, \quad B_{\text{el}} = \min \left\{ 1, \left(\sqrt{\alpha} + e^{-M/416} \right)^2 \right\}. \quad (\text{PK209})$$

These give the actual vacuum estimates

$$\begin{aligned} \|\mathbf{P}_{\Lambda_{\text{el}}} \psi\|^2 &\leq B_{\text{el}}, \\ \langle \psi, H_0 \psi \rangle &\geq \frac{M}{4096} (1 - B_{\text{el}}), \\ E_0 &\geq \frac{\kappa M}{4096} (1 - B_{\text{el}}). \end{aligned} \quad (\text{PK210})$$

To prove them let $A = \{\mathcal{W} \geq M\}$. On A^c the original potential is greater than bM , so $\mu(A^c) \leq E_0/(bM) \leq 6g^2 \sqrt{N/M}$; the probability bound supplies the additional cap one. For every unit v in the physical cutoff, PK207–PK208 give

$$\|\mathbf{1}_A v\|^2 \leq \exp(16\Lambda_{\text{el}} + 8Ne^{-12} - M/104) \leq e^{-M/208}.$$

Here $N/M = (m+1)/m \leq 3/2$, including the original boundary links. The remaining numerical inequality is

$$\frac{1}{256} + 12e^{-12} \leq \frac{1}{208}.$$

It follows from $e > 5/2$ and $12(2/5)^{12} \leq 3/3328 = 1/208 - 1/256$; the latter is the integer inequality $12 \cdot 2^{12} \cdot 3328 \leq 3 \cdot 5^{12}$. Splitting the original inner product over A and A^c and applying Cauchy–Schwarz gives

$$|\langle v, \psi \rangle| \leq \|\mathbf{1}_A v\| \|\mathbf{1}_A \psi\| + \|\mathbf{1}_{A^c} v\| \|\mathbf{1}_{A^c} \psi\| \leq e^{-M/416} + \sqrt{\alpha}.$$

Taking the supremum over this unit sphere proves the first line of PK210. Spectral calculus for H_0 on its full complement gives the second. The original $V \geq 0$ and $H = \kappa H_0 + V$ give the third.

On the unchanged path $L_j = j^4$, $a_j = 1/(100j)$, $g_j^2 = \kappa_*/(200j)$, retain

$$\begin{aligned} N_j &= 3(2j^4)(2j^4 + 1)^2, \quad M_j = 3(2j^4)^2(2j^4 + 1), \quad \kappa_j = \kappa_*, \\ \alpha_j &= \min \left\{ 1, \frac{3\kappa_*}{100j} \sqrt{\frac{2j^4 + 1}{2j^4}} \right\}, \\ B_{\text{el},j} &= \min \left\{ 1, \left(\sqrt{\alpha_j} + e^{-M_j/416} \right)^2 \right\} \longrightarrow 0, \\ \limsup_{j \rightarrow \infty} j \|\mathbf{P}_{M_j/4096} \psi_j\|^2 &\leq \frac{3\kappa_*}{100}, \quad \liminf_{j \rightarrow \infty} \frac{E_{0,j}}{j^{12}} \geq \frac{3\kappa_*}{512}. \end{aligned} \quad (\text{PK211})$$

The limits follow directly from $M_j = 24j^{12} + 12j^8$, with both terms retained in the finite estimates. The exponential term and its cross term vanish after multiplication by j . In particular every fixed electric cutoff eventually lies below $M_j/4096$ and has vanishing actual vacuum mass. The same is true of every varying cutoff bounded above by this displayed threshold. These statements concern H_0 and the full E_0 . The actual excitation operator remains $H - E_0$; PK211 does not replace its spectral measure by an electric one.

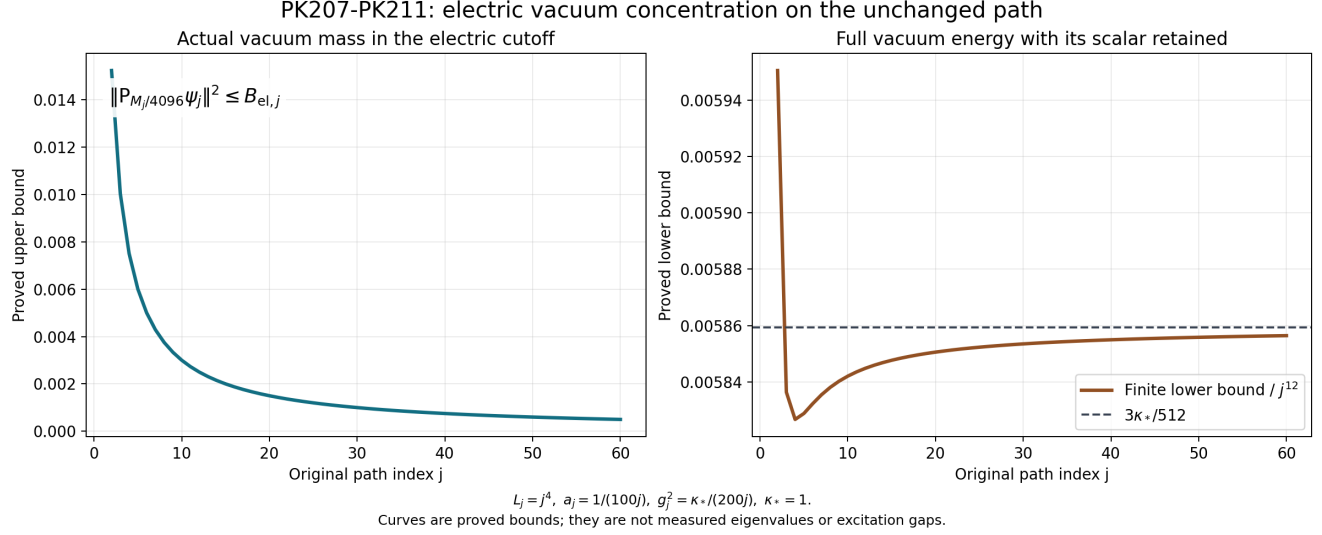


Figure 11: Actual vacuum electric mass and full energy bounds

These curves evaluate the proved PK209–PK211 bounds with the displayed original path and $\kappa_* = 1$. They are not sampled vacuum probabilities or excitation energies. All face and boundary terms remain in the plotted formulas. Reproducible source.

9.33. A fully explicit vacuum gap and the complete cutoff comparison

We continue by making the actual vacuum approximation quantitative at every finite regulator, with no small-coupling assumption. The retained score estimate PK174 is in the metric for which the generators $X_{e,\alpha}$ are orthonormal. Each group factor is the radius-two three-sphere and has diameter 2π . Join two configurations by changing one original link at a time along a shortest group geodesic. Integrating PK174 on each segment gives

$$\begin{aligned} \text{osc log } \psi &\leq 2\pi \sum_e 2r_e \xi = 16\pi \xi M, & \xi &= \frac{b}{\kappa} = \frac{1}{4g^4}, \\ \Omega &= 32\pi \xi M, & \frac{\max \psi^2}{\min \psi^2} &\leq e^\Omega, & e^{-\Omega} &\leq \psi^2(U) \leq e^\Omega. \end{aligned} \tag{PK212}$$

Every face contributes four to $\sum_e r_e = 4M$, including boundary faces. The final individual bounds use $\int \psi^2 dU = 1$; they do not change that original convention.

For every physical h of Haar mean zero, $\int |\nabla h|^2 dU \geq 3 \int |h|^2 dU$. Here the constant three is the original physical electric gap: a nontrivial spin support admitting vertex invariants cannot have a degree-one vertex, since a single nontrivial irreducible representation has no invariant vector. Its finite support therefore contains a cycle. The original cubic graph has no cycle shorter than four links, each nontrivial link costs at least $3/4$, and a fundamental plaquette has energy exactly three. Applying this argument to the complete Peter–Weyl decomposition of PK65 proves the assertion for the form domain.

Write $m_\psi = \min \psi^2$ and $M_\psi = \max \psi^2$. For physical f in the weighted form domain, minimizing over its constant and then using the Haar mean gives

$$\begin{aligned} \int |f - \mu(f)|^2 d\mu &\leq M_\psi \int |f - \int f dU|^2 dU \\ &\leq \frac{M_\psi}{3} \int |\nabla f|^2 dU \leq \frac{M_\psi}{3m_\psi} \int |\nabla f|^2 d\mu. \end{aligned}$$

Multiplication by the positive smooth ψ is a bijection of the fixed-regulator physical form domains, and the exact ground-state identity is PK154. Consequently the entire physical excitation gap, and hence also its odd restriction, obey

$$\delta_v = 3\kappa e^{-\Omega} > 0, \quad H - E_0 \geq \delta_v (I - |\psi\rangle\langle\psi|) \quad \text{on the physical space.} \quad (\text{PK213})$$

This bound uses the density ratio in PK212 once, not the product of its two less precise individual bounds.

Return to the original physical cutoff PK65, taking $\kappa\Lambda > 2bM$. Keep its full matrix H_Λ , both parities, $u_\Lambda, e_\Lambda, w_\Lambda$ from PK68–PK69, and choose a unit eigenvector ϕ_Λ at u_Λ . Let

$$\bar{E} = 2bM, \quad \ell = 2bM, \quad \eta_\Lambda = \frac{w_\Lambda}{\delta_v} \leq \frac{\ell^2}{(\kappa\Lambda - 2bM)\delta_v}. \quad (\text{PK214})$$

Whenever the final expression is less than one, ϕ_Λ has a nonzero vacuum component. The lowest cutoff eigenspace is then one-dimensional: two orthonormal vectors in it would provide a two-dimensional subspace of energies less than $E_0 + \delta_v$, contrary to PK213. Its symmetries are those of the vacuum, since its nonzero overlap with that vacuum excludes the other characters. In particular it can be chosen real and even. Choose its overall phase so that $\langle\psi, \phi_\Lambda\rangle > 0$ for the estimates below; computed quadratic expressions are independent of this phase.

Let $h_\Lambda = \phi_\Lambda - \psi$. The complete bounds are

$$\begin{aligned} 1 - |\langle\psi, \phi_\Lambda\rangle|^2 &\leq \eta_\Lambda, \\ \|h_\Lambda\| &\leq \sqrt{2\eta_\Lambda}, \\ \|\nabla h_\Lambda\|^2 &\leq \frac{w_\Lambda + 2\bar{E}\eta_\Lambda}{\kappa}, \\ |\mu(F) - \langle\phi_\Lambda, F\phi_\Lambda\rangle| &\leq 2\|F\|_\infty \sqrt{\eta_\Lambda} \quad (F \text{ bounded}). \end{aligned} \quad (\text{PK215})$$

Indeed PK213 and $\langle\phi_\Lambda, (H - E_0)\phi_\Lambda\rangle = u_\Lambda - E_0 \leq w_\Lambda$ prove the first. Writing the positive overlap as c , the second follows from $2(1 - c) \leq 2(1 - c^2)$. The excitation form of h_Λ is exactly that of ϕ_Λ , since it annihilates ψ . Thus $\langle h_\Lambda, Hh_\Lambda \rangle \leq w_\Lambda + E_0\|h_\Lambda\|^2$. Since $H \geq \kappa H_0$ and $E_0 \leq \bar{E}$, this proves the third line. For the last line the difference of the two rank-one unit-vector projections has its two nonzero eigenvalues equal to $\pm\sqrt{1 - |\langle\psi, \phi_\Lambda\rangle|^2}$. This follows by writing their two-by-two matrices in the basis consisting of ψ and the orthogonal component of ϕ_Λ . Taking its trace against multiplication by F bounds the absolute value by twice that eigenvalue times $\|F\|_\infty$. Approximation includes all bounded measurable F .

There is an explicit choice achieving any $0 < \varepsilon < 1$:

$$\Lambda(\varepsilon) = 1 + \left\lceil \max \left\{ 3, \frac{2bM + \ell^2/(\delta_v \varepsilon^2)}{\kappa} \right\} \right\rceil, \quad \eta_{\Lambda(\varepsilon)} \leq \varepsilon^2. \quad (\text{PK216})$$

This is a finite number for every original finite regulator. Its size is not claimed practical. All vacuum estimates are proved for the original H , without identifying its vacuum with Haar, a Gaussian, or the cutoff eigenvector.

9.34. Actual phase matrices with computed finite error bounds

Use the complete polynomials P_{pq} of PK199 and, for $0 \leq q < p \leq R$, define $F_{pq} = P_{pq} \circ \Theta$. Let $n = p + q$, $T = 4\beta_0 S_1 S_3 > 0$, $\mathcal{C} = 3 \sum_e K_e^2$, and $\mathcal{D}_0 = 3 \sum_e Q_e$, retaining every original word occurrence in PK172. Define

$$\begin{aligned} h_n(T) &= \begin{cases} 0, & n = 1, \\ n(n-1)T^{n-2}, & n \geq 2, \end{cases} \\ U_{pq} &= 2T^n, \quad V_{pq} = 2nT^{n-1}\beta_0\sqrt{\mathcal{C}}, \\ W_{pq} &= 2nT^{n-1}\beta_0\mathcal{D}_0 + 4h_n(T)\beta_0^2\mathcal{C}, \\ |F_{pq}| &\leq U_{pq}, \quad |\nabla F_{pq}| \leq V_{pq}, \quad |H_0 F_{pq}| \leq W_{pq}. \end{aligned} \tag{PK217}$$

The separate $n = 1$ case retains its zero Hessian term. For each of the two first derivatives of P_{pq} , the sum of absolute coefficients is $p + q = n$; hence its bound is nT^{n-1} . For each of its four second derivatives that sum is at most $n(n-1)$. The original phase gradients are bounded by $\beta_0\sqrt{\mathcal{C}}$, their two electric Laplacians by $\beta_0\mathcal{D}_0$, and each $|G_{ij}| \leq \beta_0^2\mathcal{C}$. The full first and second chain rules, including all four Hessian terms and the original minus sign of H_0 , give PK217. Neither $\mathcal{C}$ nor $\mathcal{D}_0$ is replaced by a different word or a removed root segment.

Write a, b for the complete pairs $(p, q), (s, t)$ in this finite index set. The following are entirely finite expressions from H_Λ, ϕ_Λ and the original polynomials:

$$\begin{aligned} \mathcal{G}_{ab}^\Lambda &= \langle \phi_\Lambda, F_a F_b \phi_\Lambda \rangle, \\ \mathcal{C}_{ab}^\Lambda &= \kappa \langle \phi_\Lambda, (\nabla F_a \cdot \nabla F_b) \phi_\Lambda \rangle, \\ \mathcal{Z}_a^\Lambda &= [H, F_a] \phi_\Lambda = \kappa [(H_0 F_a) \phi_\Lambda - 2 \nabla F_a \cdot \nabla \phi_\Lambda], \\ \mathcal{Q}_{ab}^\Lambda &= \langle \mathcal{Z}_a^\Lambda, \mathcal{Z}_b^\Lambda \rangle. \end{aligned} \tag{PK218}$$

Every spin matrix coefficient in the finite cutoff is a polynomial in the original link entries and their conjugates; the rooted words, q , and the two phases are finite such polynomials as well. Left differentiation preserves these finite representation products. Thus every displayed Haar integral reduces to the exact finite entry projectors PK35. All links and reflected words remain in that calculation. The potential commutes with F_a , which proves the stated commutator without removing it from H_Λ or from its eigenvector.

The actual matrices are

$$\mathcal{G}_{ab} = \langle F_a \psi, F_b \psi \rangle, \quad \mathcal{C}_{ab} = \langle F_a \psi, (H - E_0) F_b \psi \rangle, \quad \mathcal{Q}_{ab} = \langle (H - E_0) F_a \psi, (H - E_0) F_b \psi \rangle. \tag{PK219}$$

Here $\mathcal{G}, \mathcal{C}$ are exactly PK200. For $\mathcal{Q}$, $(H - E_0) F_a \psi = [H, F_a] \psi$, including the full actual vacuum energy. Put

$$\begin{aligned} e_a &= \kappa W_a \sqrt{2\eta_\Lambda} + 2\sqrt{\kappa} V_a \sqrt{w_\Lambda + 2\bar{E}\eta_\Lambda}, \\ E_R^2 &= \sum_a e_a^2, \quad A_R^2 = \sum_a \left(\kappa W_a + 2\sqrt{\kappa \bar{E}} V_a \right)^2, \\ \epsilon_G &= 2\sqrt{\eta_\Lambda} \sum_a U_a^2, \quad \epsilon_C = 2\kappa\sqrt{\eta_\Lambda} \sum_a V_a^2, \quad \epsilon_Q = 2A_R E_R + E_R^2. \end{aligned} \tag{PK220}$$

Then in the original coefficient Euclidean norm,

$$\|\mathcal{G} - \mathcal{G}^\Lambda\| \leq \epsilon_G, \quad \|\mathcal{C} - \mathcal{C}^\Lambda\| \leq \epsilon_C, \quad \|\mathcal{Q} - \mathcal{Q}^\Lambda\| \leq \epsilon_Q. \tag{PK221}$$

For a coefficient vector c , the full polynomial $F_c = \sum_a c_a F_a$ satisfies $\|F_c\|_\infty^2 \leq (\sum_a U_a^2) \|c\|^2$ and $\|\nabla F_c\|_\infty^2 \leq (\sum_a V_a^2) \|c\|^2$. Apply PK215 to these two real bounded multiplication functions. The supremum over coefficient vectors proves the first two Hermitian matrix bounds. For the third, PK215 and the exact commutator give $\|[H, F_a](\phi_\Lambda - \psi)\| \leq e_a$. Also $H \geq \kappa H_0$ and $\langle \phi_\Lambda, H \phi_\Lambda \rangle = u_\Lambda \leq \bar{E}$ give $\|Z_a^\Lambda\| \leq \kappa W_a + 2\sqrt{\kappa \bar{E}} V_a$. For arbitrary c , Cauchy–Schwarz over the coefficient indices bounds the difference of the two resulting squared norms by $(2A_R E_R + E_R^2) \|c\|^2$. This proves the last matrix bound and includes the entire complementary Hamiltonian action in PK198.

For completeness these errors can be driven to zero on the original simultaneous path by an explicit finite cutoff, even when the desired polynomial degree $R_j \geq 1$ varies. At each j use every original value in PK217 and set

$$\begin{aligned} \Omega_j &= \frac{320000\pi j^2 M_j}{\kappa_*^2}, & \delta_{v,j} &= 3\kappa_* e^{-\Omega_j}, & \bar{E}_j = \ell_j &= \frac{20000j^2 M_j}{\kappa_*}, \\ T_{R_j}^2 &= \sum_a \left(\sqrt{2} \kappa_* W_a + 2\sqrt{\kappa_*} V_a \sqrt{\delta_{v,j} + 2\bar{E}_j} \right)^2, \\ Z_{R_j} &= \max \left\{ 1, 2 \sum_a U_a^2, 2\kappa_* \sum_a V_a^2, 2A_{R_j} T_{R_j} + T_{R_j}^2 \right\}, & \varepsilon_j &= \frac{1}{2j Z_{R_j}}, \\ \Lambda_j^{\text{mat}} &= 1 + \left\lceil \max \left\{ 3, \frac{\bar{E}_j + \ell_j^2 / (\delta_{v,j} \varepsilon_j^2)}{\kappa_*} \right\} \right\rceil. \end{aligned} \tag{PK222}$$

The summation range is exactly $0 \leq q < p \leq R_j$; A_{R_j} is PK220 with the displayed path values. The scalar T_{R_j} in this equation is an error constant, not the phase-coordinate bound T in PK217. PK216 gives $\eta \leq \varepsilon_j^2$ and $w \leq \delta_{v,j} \varepsilon_j^2$, so $E_{R_j} \leq \varepsilon_j T_{R_j}$. Since $\varepsilon_j \leq 1$, all three errors in PK221 are at most $1/(2j)$. These are proved absolute operator-norm errors for the actual varying matrices. The cutoff may grow exponentially in Ω_j ; no practical evaluation of these large matrices is claimed.

9.35. Full residual matrices and certified low-energy receiving quantities

The complete Gram matrix is positive definite by PK205. Let $J_R c = \sum_a c_a F_a \psi$, so $J_R^* J_R = \mathcal{G}_R$. Its exact physical range projection is

$$\Pi_R = J_R \mathcal{G}_R^{-1} J_R^*, \quad \Sigma_R = \mathcal{Q}_R - \mathcal{C}_R \mathcal{G}_R^{-1} \mathcal{C}_R = J_R^* (H - E_0) (I - \Pi_R) (H - E_0) J_R \geq 0. \tag{PK223}$$

Idempotence and self-adjointness of Π_R follow by inserting $J_R^* J_R = \mathcal{G}_R$; its range is exactly that of J_R . Every $F_a \psi$ is smooth and in the domain of all powers needed in this identity. Expanding the projection proves the entire subtracted matrix. This is the residual of the full original Hamiltonian, with no invariant-subspace assumption.

It too has a finite error enclosure. Once

$$g_\Lambda = \lambda_{\min}(\mathcal{G}_R^\Lambda), \quad a_\Lambda = g_\Lambda - \epsilon_G > 0,$$

define

$$\begin{aligned} \Sigma_R^\Lambda &= \mathcal{Q}_R^\Lambda - \mathcal{C}_R^\Lambda (\mathcal{G}_R^\Lambda)^{-1} \mathcal{C}_R^\Lambda, \\ \epsilon_\Sigma &= \epsilon_Q + \frac{\epsilon_C (2\|\mathcal{C}_R^\Lambda\| + \epsilon_C)}{a_\Lambda} + \frac{\|\mathcal{C}_R^\Lambda\|^2 \epsilon_G}{a_\Lambda g_\Lambda}, \\ \|\Sigma_R - \Sigma_R^\Lambda\| &\leq \epsilon_\Sigma. \end{aligned} \tag{PK224}$$

At fixed regulator and R , success of this positivity test is guaranteed for all sufficiently large cutoffs: PK216–PK221 give norm convergence and PK205 gives $\lambda_{\min}(\mathcal{G}_R) > 0$. The inverse identity gives $\|\mathcal{G}_R^{-1} - (\mathcal{G}_R^\Lambda)^{-1}\| \leq \epsilon_G/(a_\Lambda g_\Lambda)$. Write $\mathcal{C}_R = \mathcal{C}_R^\Lambda + D$ and expand both products. Bounding the two linear D terms and the quadratic D term with $\|\mathcal{G}_R^{-1}\| \leq 1/a_\Lambda$, followed by the displayed inverse error, proves PK224. The finite Σ_R^Λ is an approximating matrix; it need not be positive. Its exact target is the positive matrix PK223.

For a fixed polynomial f contained in some degree R_0 , the complete residual also satisfies

$$\|(I - \Pi_R)(H - E_0)Jf\|^2 \downarrow \|(I - JJ^*)(H - E_0)Jf\|^2 = \int |R_f|^2 d\mu, \quad R \geq R_0, \quad R \rightarrow \infty. \quad (\text{PK225})$$

The polynomial spaces are nested and dense in the phase Hilbert space by PK199–PK201. Their projections therefore converge strongly to JJ^* : for a vector in that range approximate in norm by a finite polynomial vector, use the projection contraction, and then pass to the limit; on its orthogonal complement every projection is zero. Squared residual norms decrease by the Pythagorean identity. Apply this to the actual vector $(H - E_0)Jf$ and use PK197. Thus the finite matrices retain both the higher-phase component and the complete coupling outside the phase range. No rate for the degree limit is asserted.

The same computed matrices enclose the actual finite variational energy. At fixed regulator increase the electric cutoff until both $\mathcal{G}^\Lambda - \epsilon_G I > 0$ and $\mathcal{C}^\Lambda - \epsilon_C I > 0$. Termination follows from their proved positive definite limits. Then

$$\begin{aligned} \lambda_R^- &= \min_{c \neq 0} \frac{c^*(\mathcal{C}^\Lambda - \epsilon_C I)c}{c^*(\mathcal{G}^\Lambda + \epsilon_G I)c} \leq \lambda_R^{\text{ph}} \\ &\leq \min_{c \neq 0} \frac{c^*(\mathcal{C}^\Lambda + \epsilon_C I)c}{c^*(\mathcal{G}^\Lambda - \epsilon_G I)c} = \lambda_R^+, \\ \Delta_{\text{odd}} &\leq \lambda_{\text{ph}} \leq \lambda_R^{\text{ph}}. \end{aligned} \quad (\text{PK226})$$

For each c , PK221 bounds the numerator and denominator of its original Rayleigh quotient. Their positivity gives both displayed comparisons, and taking the minima proves the first two lines. PK201 gives the last line. In particular λ_R^- is a lower bound on this finite phase variational value; it is not a lower bound on the actual odd spectral bottom.

Finally the full raw spectral measure of J_{RC} is $\nu_c(B) = \langle J_{RC}c, \mathbf{1}_B(H - E_0)J_{RC} \rangle$. Its original mass and first moment are $c^*\mathcal{G}_{RC}$ and $c^*\mathcal{C}_{RC}$, with no rescaling of c or of the vector. Oddness removes the vacuum atom. For every energy $s > 0$,

$$\nu_c((0, s]) \geq \max \left\{ 0, c^*(\mathcal{G}_R^\Lambda - \epsilon_G I)c - \frac{c^*(\mathcal{C}_R^\Lambda + \epsilon_C I)c}{s} \right\}. \quad (\text{PK227})$$

Indeed the mass above s is at most the first moment divided by s . Subtract it from the full raw mass, then apply PK221 with the displayed signs. For the first polynomial $P_{10} = x - y$, the target entries are exactly the full four-loop and gradient differences of PK202. PK218 supplies finite original-word calculations for both, PK221 supplies their actual-vacuum errors, and PK223–PK225 retain the second-moment residual. The resulting lower endpoint has not been proved positive on a shrinking energy window along the original path. The next calculation is that endpoint with these certified vacuum and matrix comparisons, preserving the extensive electric-vacuum estimate PK211 and the full $E_{0,j}$ subtraction. The interacting continuum construction, reconstruction and intended theory identification remain the active endpoint.

PK212–PK227 prove all maps and error bounds in this diagram, including the full complementary return and the original raw spectral-weight expression. The shrinking-window lower endpoint remains to evaluate. Reproducible source.

Original actual state	$H = \kappa H_0 + b(2M - \mathcal{W}), \quad H\psi = E_0\psi, \quad \int \psi^2 dU = 1$
Complete finite provider	$H_\Lambda = P_\Lambda H P_\Lambda, \quad H_\Lambda \phi_\Lambda = u_\Lambda \phi_\Lambda$
Proved vacuum error	$\delta_v = 3\kappa e^{-32\pi(b/\kappa)M}, \quad \eta_\Lambda = (u_\Lambda - e_\Lambda)/\delta_v$
All three actual matrices	$\ G - G^\Lambda\ \leq \varepsilon_G, \quad \ C - C^\Lambda\ \leq \varepsilon_C, \quad \ Q - Q^\Lambda\ \leq \varepsilon_Q$
Full complementary return	$\Sigma_R = Q_R - C_R G_R^{-1} C_R \geq 0$
Original raw spectral weight	$\nu_c((0, s)) \geq \max\{0, c^*(G^\Lambda - \varepsilon_G I)C - s^{-1}c^*(C^\Lambda + \varepsilon_C I)C\}$

Every cutoff contains all permitted electric sectors and every original magnetic face.
The shrinking-window lower endpoint still has to be evaluated; a positive limit is not assumed.

Figure 12: Exact original vacuum, cutoff matrices and full residual

10. Reproducibility and scope

The original symbolic checker verifies the full angular period law, both matrix inverses, every inverse-metric block, cover conjugacy, the complete dual quadratic form, representation matrices and Haar projectors in the smallest relevant tensor sectors, the original plaquette trace and all path factors. Finite checks support these algebraic identities. The bundle, analytical convergence, injectivity, positivity, vacuum and nonzero-vector arguments are the written proofs above.

The heat-packet source conventions were read in the original author TeX; no PDF reading or whole-paper recertification is asserted. No novelty claim is made for heat-kernel coherent states, group averaging, or the finite-lattice ground-state facts. Their application here receives the actual corrected moment-map field, full cusp covering, and original interacting lattice Hamiltonian with the indicated exact factors.

The complementary-moment checker adds 43 exact checks for PK55–PK62: direct Haar polynomial integration, oriented cube matrix-entry contraction, exhaustive six-face cycles in the smallest original box, every sixth-moment entry on a cube plus another face, boundary incidence and reflection in three boxes, and the full projection and resolvent identities. General proofs, including the exact cubical filling map, are given in Sections 9.2–9.3. A finite check is not a continuum estimate.

The full-packet checker adds 39 exact checks for PK63–PK78, including complex packet cross terms, both returns with the original non-unit Gram, vacuum intervals, cutoff constants, and the original 37-dimensional physical cutoff in the smallest box. Infinite-space estimates are the complete written proofs in Sections 9.4–9.8.

The packet-moment checker adds 62 exact checks for PK79–PK100: noncommuting square, Haar shared-link slots, trial and small-potential constants, original graph counts, full path factors and the interacting energy-flow identity. The asymptotic escape estimate is the complete proof in Sections 9.10–9.12; no finite test is substituted for it.

The odd-observable checker adds 109 exact checks for PK101–PK131, including every sign character, Gaussian factor and radial coefficient tested. The infinite sums, actual operator limits and full higher-domain map have their written proofs in Sections 9.14–9.19. Finite checks do not certify a volume-

uniform or continuum limit.

The joint-path checker adds 82 exact checks for PK132–PK157: complete heat/Haar constants, original path and graph counts, $SU(2)$ trace signs, repeated inverse-word derivatives and the full actual ground-state product rule. The operator inequalities and asymptotic collapse are proved in Sections 9.20–9.23; finite checks are not substituted for them.

The global-moment checker adds 40 exact checks for PK158–PK177, including all chord exclusions, coincident fourth-moment contractions, the original phase scale and the complete energy-bound constants. Four larger-box numerical samples are recorded separately and are not proofs. Sections 9.24–9.26 contain the full arguments.

The phase-variational checker adds 65 exact checks for PK177a–PK206: all rooted words in the explicit configuration, original strip incidences, complete constants, amplitude factors and finite Gram identities. Three larger-box numerical checks are supplementary. The oscillatory, operator and variational conclusions have their complete proofs in Sections 9.26a–9.31.

The vacuum-matrix checker adds 64 exact checks for PK207–PK227: original face incidence/colouring, complete heat-tail constants, all path factors, polynomial derivative coefficients and non-unit-Gram complementary identities. The concentration, cutoff convergence, residual and spectral-weight arguments are the complete proofs in Sections 9.32–9.35.